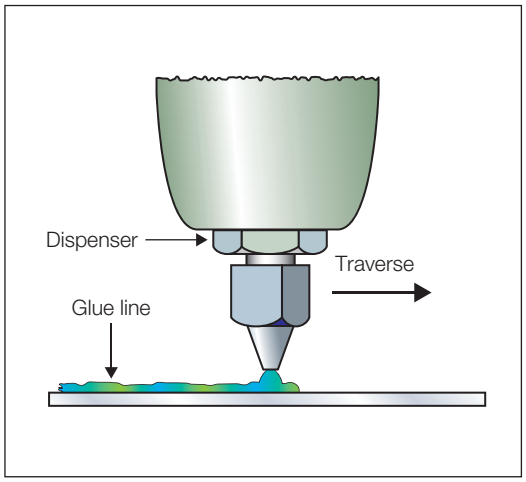
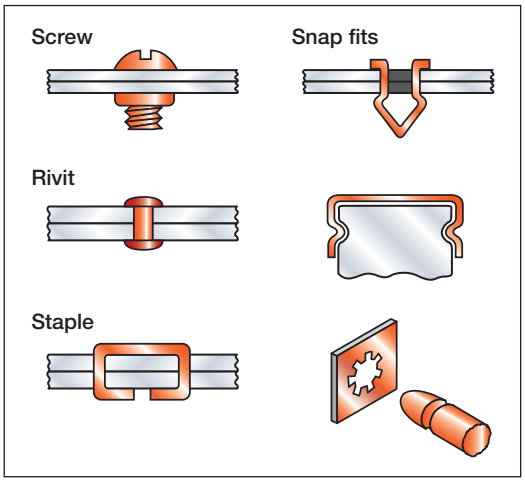
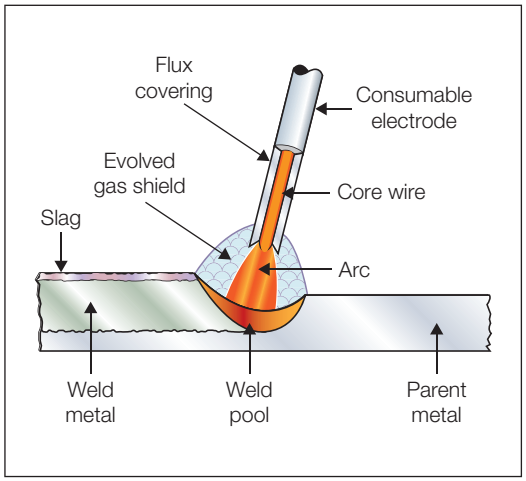


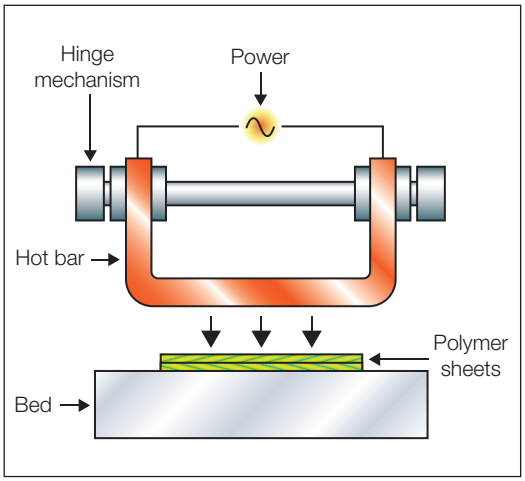
Adhesives



Fasteners



Welding: manual metal arc



Welding: hot bar polymer welding

FIGURE 13.12

Joining operations. In *adhesive bonding*, a film of adhesive is applied to one surface, which is then pressed onto the mating one. *Fastening* is achieved by bolting; riveting; stapling; push-through snap fastener; push-on snap fastener; or rod-to-sheet snap fastener. In *metal fusion welding*, metal is melted, and more is added from a filler rod to give a bond or coating. In *thermoplastic polymer welding*, heat is applied to the polymer components, which are simultaneously pressed together to form a bond.

The large-volume use of fasteners is costly; welding, crimping, or the use of adhesives can be more economical. Design for assembly (DFA) methods provide a checklist to guide minimizing assembly time.

Finishing processes

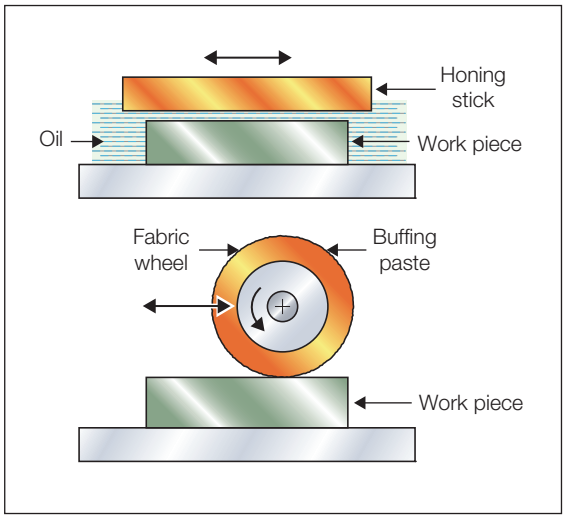
Finishing involves treatments applied to the surface of the component or assembly. Some aim to improve mechanical and other engineering properties, others to enhance appearance.

Finishing treatments to improve engineering properties (Figure 13.13) Grinding, lapping, and polishing increase precision and smoothness, particularly important for bearing surfaces. Electroplating deposits a thin metal layer onto the surface of a component to give resistance to corrosion and abrasion. Plating and painting are both made easier by a simple part shape with largely convex surfaces: Channels, crevices, and slots are difficult to reach. Anodizing, phosphating, and chromating create a thin layer of oxide, phosphate, or chromate on the surface, imparting corrosion resistance.

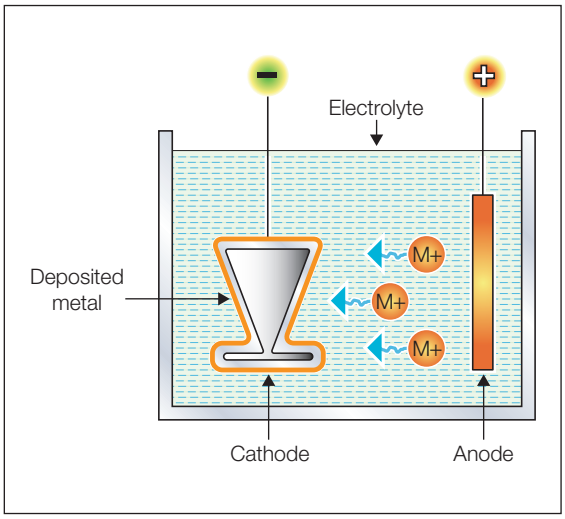
Heat treatment is a necessary part of the processing of many materials. Age-hardening alloys of aluminum, titanium, and nickel derive their strength from a precipitate produced by a controlled heat treatment: quenching from a high temperature followed by aging at a lower one. The hardness and toughness of steels are controlled in a similar way: by quenching from the "austenitizing" temperature (about 800°C) and tempering. The treatment can be applied to the entire component, as in bulk carburizing, or just to a surface layer, as in flame hardening, induction hardening, and laser surface hardening.

Quenching—very rapid cooling—is a savage procedure; the sudden thermal contraction associated with it can produce stresses large enough to distort or crack the component. The stresses are caused by a nonuniform temperature distribution, and this, in turn, is related to the geometry of the component. To avoid damaging stresses, the section thickness should be as uniform as possible, and nowhere so large that the quench rate falls below the critical value required for successful heat treatment. Stress concentrations should be avoided because they are a source of quench cracks. Materials that have been molded or deformed may contain internal stresses that can be removed, at least partially, by stress-relief anneals—another sort of heat treatment.

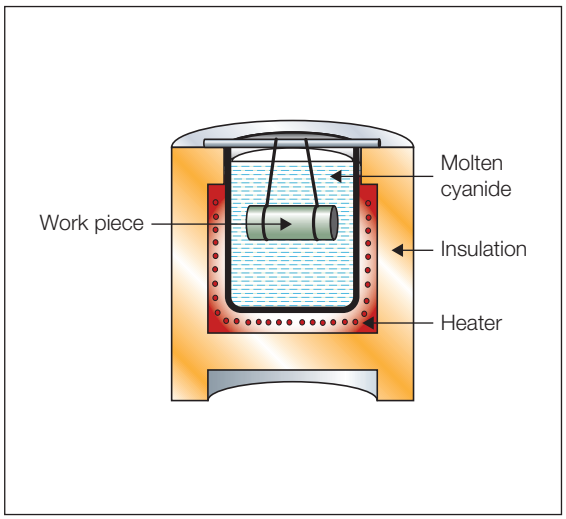
Finishing treatments that enhance aesthetics (Figure 13.14) The processes just described can be used to enhance the visual and tactile attributes of a material: Electroplating and anodizing are examples. There are many more, of which painting is the most widely used. Organic-solvent–based paints give durable coatings with high finish, but the solvent poses environmental problems. Water-based paints overcome these, but dry more slowly and the resulting paint film is less perfect. In polymer powder-coating and polymer powder-spraying a film of thermoplastic—nylon, polypropylene, or polyethylene—is deposited on the surface, giving a protective layer that can be brightly colored.



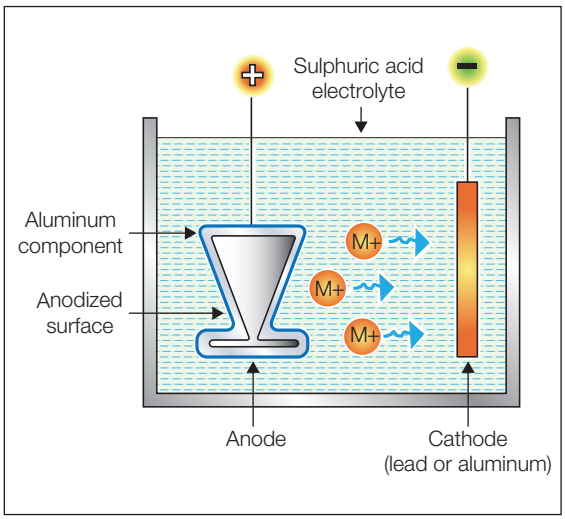
Mechanical polishing



Electroplating



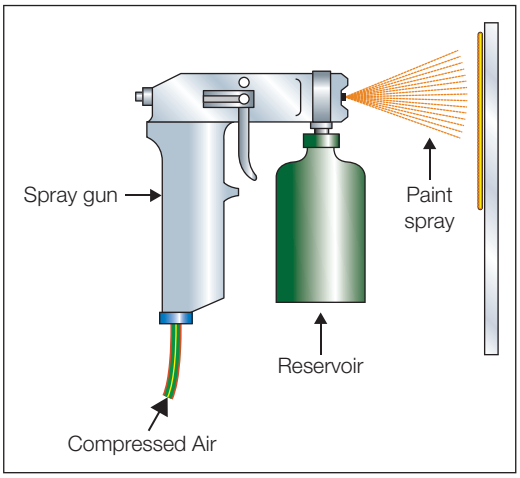
Heat treatment: carburizing



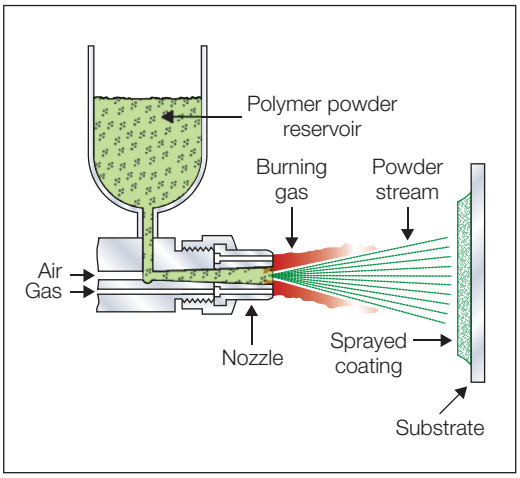
Anodizing

FIGURE 13.13

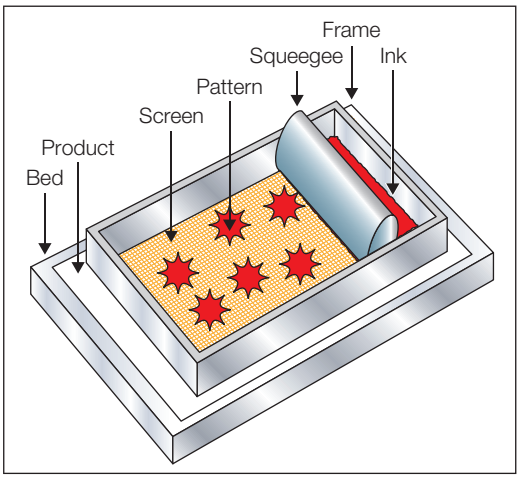
Finishing processes to protect and enhance properties. In *mechanical polishing*, the roughness of a surface is reduced and its precision increase, by material removal using finely ground abrasives. In *electroplating*, metal is plated onto a conducting work piece by electro-deposition in a plating bath. In *heat treatment*, a surface layer of the work piece is hardened and made more corrosion resistant by the inward diffusion of carbon, nitrogen, phosphorous, or aluminum from a powder bed or molten bath. In *anodizing,* a surface oxide layer is built up on the work piece (which must be aluminum, magnesium, titanium, or zinc) by a potential gradient in an oxidizing bath.



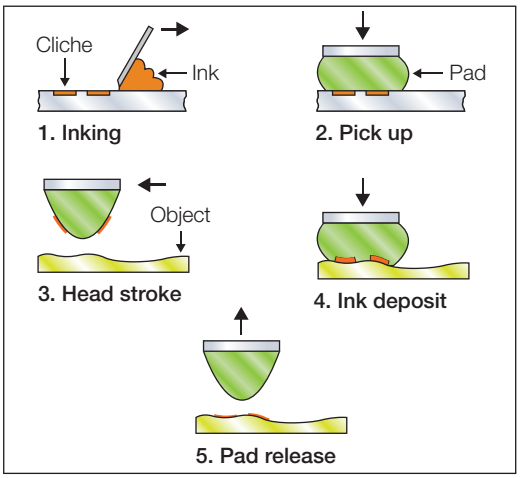
Paint spraying



Polymer powder spraying



Silk screen printing



Pad printing

FIGURE 13.14

Finishing processes to enhance appearance. In *paint spraying*, a pigment in an organic- or water-based solvent is sprayed onto the surface to be decorated. In *polymer powder-coating* a layer of thermoplastic is deposited on the surface by direct spraying in a gas flame, or by immersing the hot work piece in a bed of powder. In *silk-screen printing*, ink is wiped onto the surface through a screen onto which a blocking pattern has been deposited, allowing ink to pass in selected areas only. In *pad-printing,* an inked pattern is picked up on a rubber pad and applied to the surface, which can be curved or irregular.

In screen printing an oil-based ink is squeegeed through a mesh on which a blocking film holds back the ink where it is not wanted; full-color printing requires the successive use of up to four screens. Screen printing is widely used to apply designs to flat surfaces. Curved surfaces require the use of pad printing,

in which a pattern, etched onto a metal "cliche," is inked and picked up on a soft rubber pad. The pad is pressed onto the product, depositing the pattern on its surface; the compliant rubber conforms to the curvature of the surface.

Enough of the processes themselves; for more detail the reader will have to consult "Further reading" in Section 13.9. Now a brief look at what processing does to properties.

13.4 PROCESSING FOR PROPERTIES

The extent of the material bubbles in the property charts gives an idea of the degree to which properties can be manipulated by processing. The annotations in Figure 13.15, a copy of the modulus-strength chart, illustrates the degree to which these two properties for metals can (or cannot) be controlled by alloying, heat treatment, and cold work. The rather different

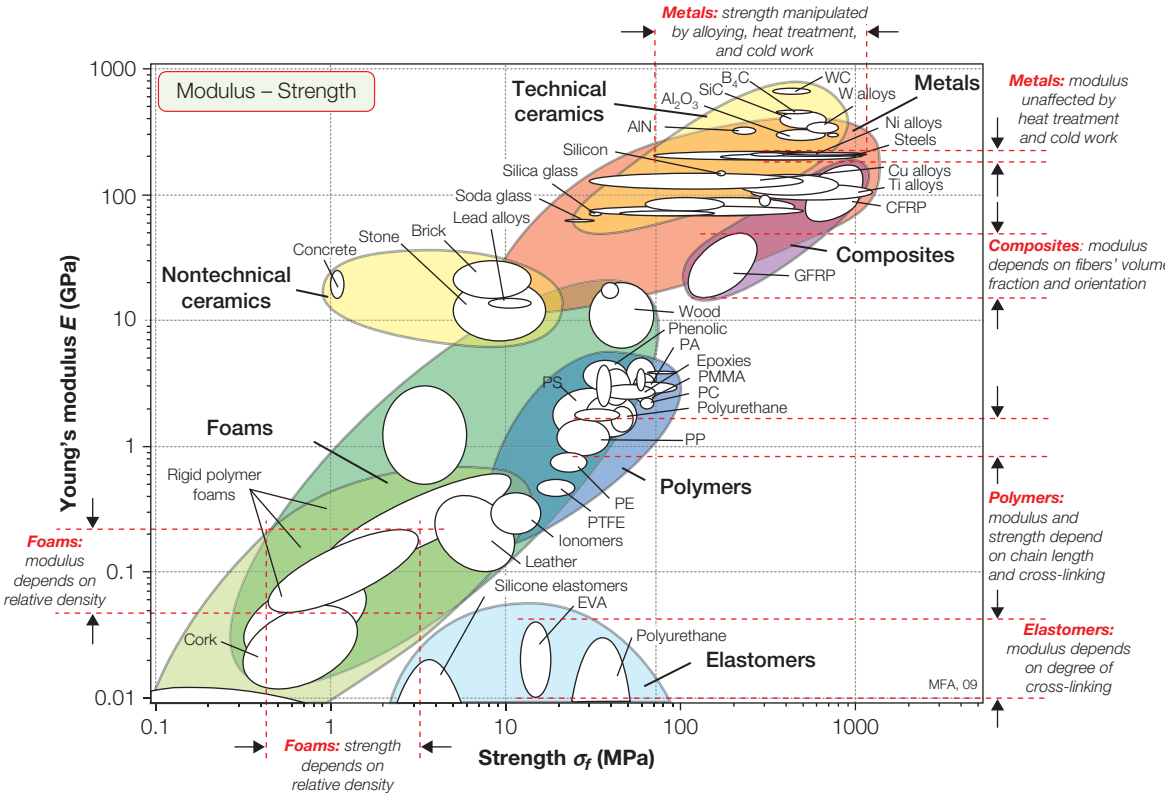


FIGURE 13.15
The extent of the material bubbles on the property charts gives an idea of the degree to which properties can be manipulated by processing.

shapes of the bubbles for composites and for foams reflect the way the properties of the first depend on fiber content and orientation; those of the second, on the extent of foaming, measured by the porosity or relative density (Chapter 11). The modulus and strength of polymers and elastomers depend on the chain length and degree of cross-linking, aspects of structure directly controlled by processing. The strength, particularly, of ceramics depends on porosity, another aspect of microstructure that is directly influenced by processing.

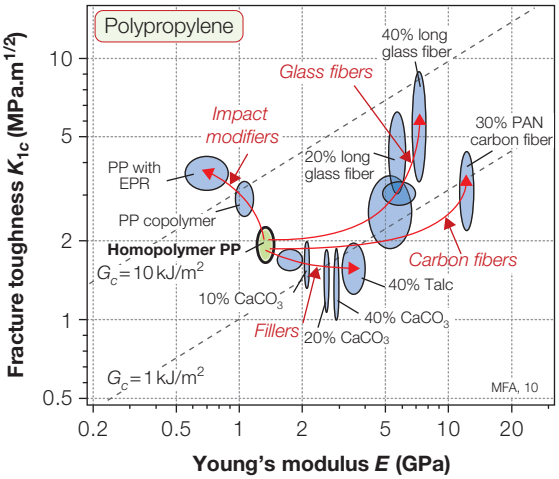


FIGURE 13.16
The strength and toughness of polypropylene, showing the effect of fillers, impact modifiers, and fibers.

We have already seen, in Figures 11.7 and 11.10, how fiber content, shape, and orientation control composite properties, and in Figure 11.24 how the foam density and connectivity dramatically influence the properties of cellular solids. The strengths of polymers span around a factor of 5 and their fracture toughnesses a factor of 20, depending on chemistry, chain length, and degree of cross-linking. More dramatic changes are possible by blending, filling, reinforcing, or plasticizing. Figure 13.16 shows how these influence the modulus E and fracture toughness K_{Ic} of polypropylene, PP. Blending or copolymerization with elastomers, such as EPR or EDPM ("impact modifiers"), reduces the modulus but increases the fracture toughness K_{Ic} and toughness G_c. Filling with inexpensive powdered glass, talc, or calcium carbonate more than doubles the modulus; however, that is at the expense of some loss of toughness. Plasticizing (blending with low-molecular-weight polymers) lowers the modulus even more dramatically. Between them, these processes can change the polymer modulus by a factor of 100 and the toughness by a factor of 10.

Of all the properties that materials scientists and engineers have sought to manipulate, the strength of metals and alloys is probably the most explored. It is easy to see why—Table 13.1 lists some of their applications. Solution hardening, precipitation hardening, and work hardening are combined to give desired combinations of strength and toughness. Figure 13.17 illustrates the large gains in strength of copper alloys that these mechanisms allow. Good things, however, have to be paid for. Here the payment required for increased strength is loss of ductility, in this case measured by the elongation to fracture ε_f.

Strength and ductility are structure-sensitive properties—they depend on composition and microstructure, and these, in turn, are controlled by

Table 13.1 Metal Alloys with Typical Applications, Indicating the Strengthening Mechanisms Used

Alloy	Typical Uses	Solution Hardening	Precipitate Hardening	Work Hardening
Pure Al	Kitchen foil			✓✓✓
Pure Cu	Wire			✓✓✓
Cast Al, Mg	Automotive parts	✓✓✓	✓	
Bronze (Cu-Sn), Brass (Cu-Zn)	Marine components	✓✓✓	✓	✓
Non-heat-treatable wrought Al	Ships, cans, structures	✓✓✓		✓✓✓
Heat-treatable wrought Al	Aircraft, structures	✓	✓✓✓	✓
Low carbon steels	Car bodies, structures, ships, cans	✓✓✓		✓✓✓
Low alloy steels	Automotive parts, tools	✓	✓✓✓	✓
Stainless steels	Pressure vessels	✓✓✓	✓	✓✓✓
Cast Ni alloys	Jet engine turbines	✓✓✓	✓✓✓	

Symbols: ✓✓✓ = routinely used; ✓ = sometimes used.

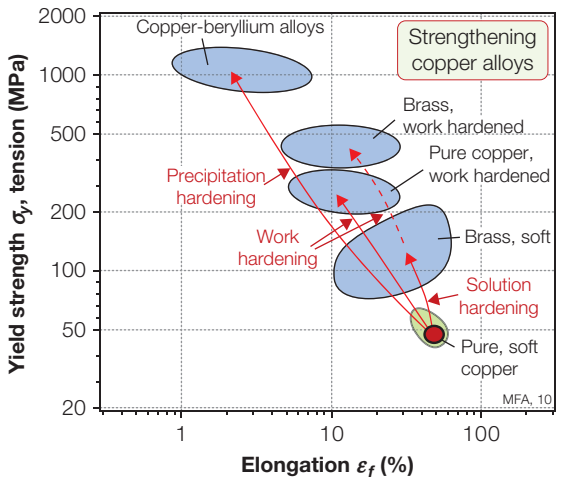


FIGURE 13.17
Strengthening mechanisms and the consequent drop in ductility, here shown for copper alloys. The mechanisms are frequently combined. The greater the strength, the lower the ductility (the elongation to fracture ε_f).

processing. Hardness, fatigue strength, fracture toughness, and thermal and electrical conductivities, too, are structure-sensitive properties. Much processing is fine-tuned to produce particular combinations of these. They are best illustrated as mini property charts, like Figures 13.16 and 13.17. Materials for heat transfer—heat exchangers, chemical engineering equipment—require

good thermal conductivity with high strength. Figure 13.18 shows how processing changes these two properties for alloys based on aluminum hardened by each of the three main mechanisms: solid solution, work hardening, and precipitation hardening. Work hardening strengthens significantly without changing the conductivity much, while solid solution and precipitation hardening introduce more scattering centers, giving a drop in conductivity. Many electrical applications—high-speed motors and power transmission, for instance—require materials with good conductivity and high strength.

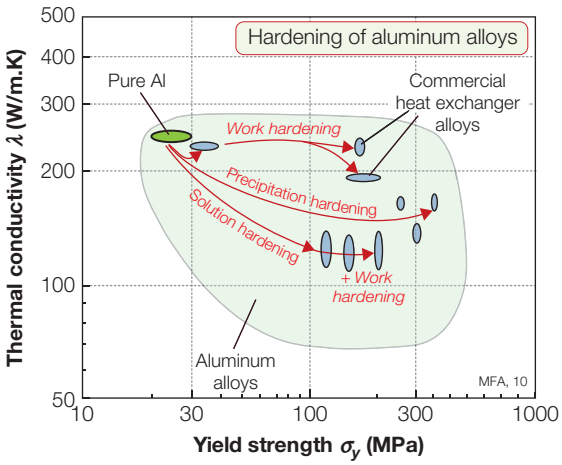


FIGURE 13.18
Thermal conductivity and strength for aluminum alloys.

Figure 13.19 illustrates this for one of the best conductors that we have: copper. Adding solute increases its strength, but the solute atoms also act as scattering centers, increasing the electrical resistivity too. Dislocations add strength (by what we called *work hardening*) and they too scatter electrons a little, though not as much as solute. Precipitates offer the greatest gain in strength with only a slight loss of conductivity. Precipitation hardening (with low residual solute) and work hardening are therefore the best ways to strengthen conductors.

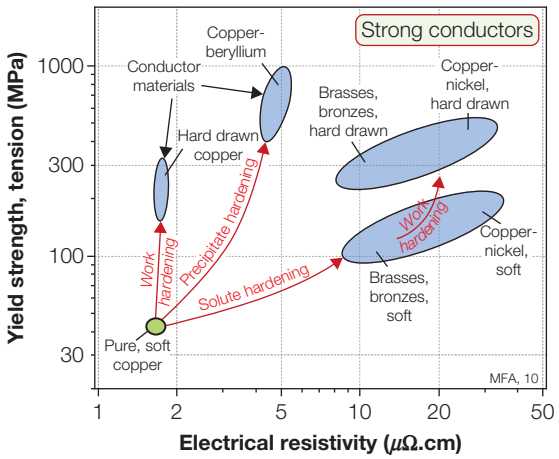


FIGURE 13.19
The best choice of material for a cable is one with high strength and low resistivity, but strengthening mechanisms increase resistivity. Work hardening and precipitation hardening do so less than solute hardening.

Nowhere is processing for properties more important than in the heat treatment of steels. Figure 13.20 shows the sequence for a medium-carbon steel. The steel is solutionized (heated into the FCC austenite field) to dissolve all of the carbon it then quenched in water or oil, which causes it to transform to hard, brittle martensite. Martensite has high hardness but is so brittle—its fracture toughness is so low that it is almost useless as a structural material. Tempering reduces the hardness and yield strength but restores toughness to a degree that depends on the tempering temperature and time. The desired property set is obtained by controlling these.

Thus processing plays a central role in manipulating material properties. Processes are chosen for their ability to create shapes and to create properties.

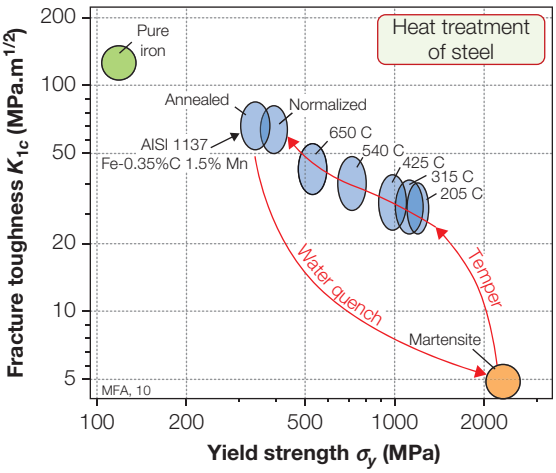


FIGURE 13.20
The changes in fracture toughness and yield strength for a plain carbon steel on heat treatment.

13.5 SYSTEMATIC PROCESS SELECTION

The selection strategy

The strategy for selecting processes parallels that for selecting materials. Figure 13.21 lists the now-familiar steps: *translation*, *screening*, *ranking*, and *documentation*.

Translation As we saw in earlier chapters, the *function* of a component dictates the initial choice of material and shape. This choice exerts constraints on the choice of processes. It is helpful to think of two types of constraint: *technical*—can the process do the job at all? And *quality*—can it do so sufficiently well? One technical constraint is always there: the compatibility of material and process. Quality constraints include achieving the desired precision, surface finish, and property profile while avoiding defects. The usual *objective* in processing is to minimize cost. The *free variables* are largely limited to choosing the process itself and its operating parameters (such as temperatures, flow rates, and so on). Table 13.2 summarizes the outcome of the translation stage.

Screening The screening step applies the constraints, eliminating processes that cannot meet them. Some process attributes are simple numeric ranges—the size or mass of component the process can handle, the precision, or the surface smoothness it can achieve. Others are nonnumeric—lists of materials to which the process can be applied, for example. Requirements such as "made of magnesium and weighing about 3 kg" are easily compared with the process attributes to eliminate those that cannot shape magnesium or cannot handle a component as large as 3 kg.

Ranking Ranking, as before, is based on one or more *objectives*, the most obvious of which is that of minimizing cost. In certain demanding applications it may be replaced by the objective of maximizing quality regardless of cost, though more usually it is a trade-off between the two that is sought.

Documentation Screening and ranking do not cope adequately with the less tractable issues of quality and productivity; they are best explored through a search for documentation—that is, design guidelines, best practice guides, case studies, and failure analyses. The most important technical expertise relates to productivity and quality. All types of processing equipment have optimum operating condition ranges under which they work best and produce products with uncompromised quality. Failure to operate within this window can lead to manufacturing defects, such as excessive porosity, cracking, or residual stress. This in turn leads to scrap and lost productivity, and, if passed on to the user, may cause premature failure. Documentation is an essential part of the selection exercise.

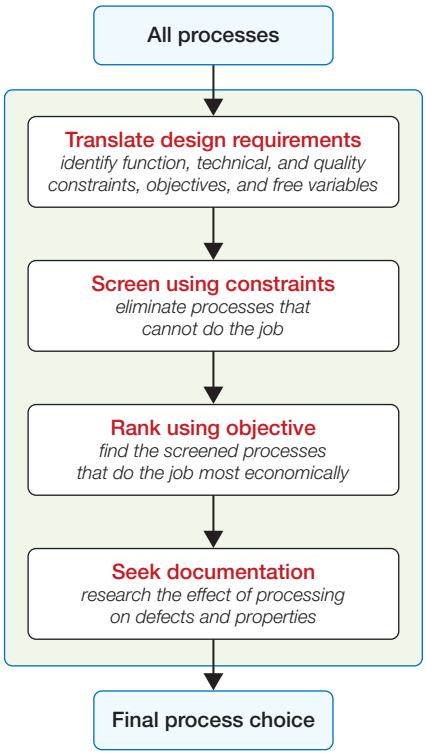


FIGURE 13.21
A flow chart of the procedure for process selection. It parallels that for material selection.

Implementing the strategy

As already explained, each process is characterized by a set of attributes. These are conveniently displayed as simple matrices and bar charts. They provide the selection tools we need for screening. The hard-copy versions, shown here, are necessarily simplified, showing only a limited number of

Table 13.2 Translation of Process Requirements

Function	What must the process do? (Shape? Join? Finish?)
Constraints	What technical limits must be met? (Material and shape compatibility) What quality limits must be met? (Precision, avoidance of defects…)
Objectives	What is to be maximized or minimized? (Cost? Time? Quality?)
Free variables	Choice of process and process-operating conditions

processes and attributes. Computer implementation allows exploration of a much larger number of both.

Material-process compatibility Figure 13.22 shows a process-material compatibility matrix. Shaping processes are at the top, with compatible combinations marked by colored dots that identify the material family. Its use for screening is straightforward: Specify the material and read off the processes or the reverse: specify the process and read off the materials. The diagonal spread of the dots in the matrix reveals that each material class—metals,

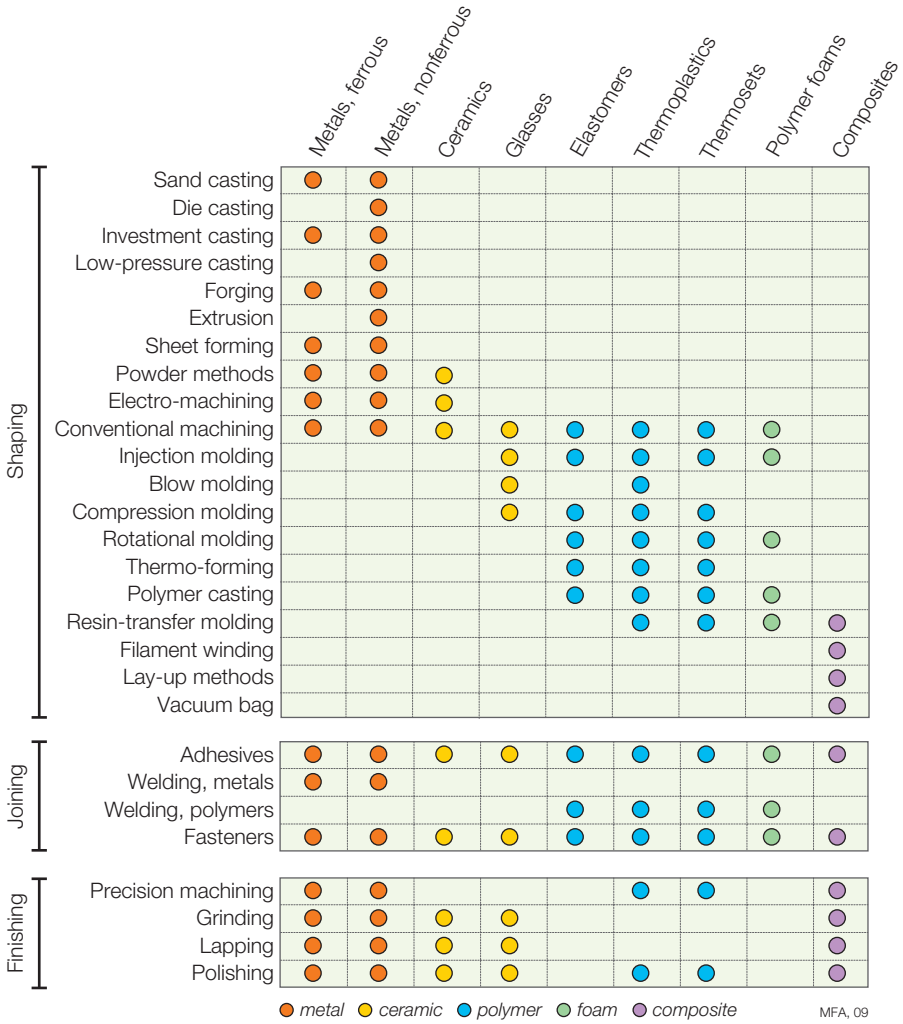


FIGURE 13.22
The process-material matrix. A colored dot indicates that the pair are compatible.

polymers, and such—has its own set of process routes. There are some overlaps —powder methods are compatible with both metals and ceramics, molding with both polymers and glasses. Machining (when used for shaping) is compatible with almost all families. Joining processes using adhesives and fasteners are very versatile and can be used with most materials, whereas welding methods are material-specific. Finishing processes are used primarily for the harder materials, particularly metals; polymers are molded to shape and rarely treated further except for decorative purposes. We will see why later.

Process-shape compatibility Shape is the most difficult attribute to characterize. Many processes involve rotation or translation of a tool or of the material, directing our thinking toward axial symmetry, translational symmetry, uniformity of section, and the like. Turning creates axisymmetric (or circular) shapes; extrusion, drawing, and rolling make prismatic shapes, both circular and noncircular. Sheet-forming processes make flat shapes (stamping) or dished shapes (drawing). Certain processes can make three-dimensional shapes, and among these some can make hollow shapes whereas others cannot. Figure 13.23 illustrates this classification scheme. The prismatic shapes shown on the left, made by rolling, extrusion, or drawing, have a special feature: They can be made in continuous lengths. The other shapes cannot—they are discrete, and the processes that make them are called batch processes. Continuous processes are well suited to long, prismatic products such as railway track or standard stock such as tube, plate, and sheet. Cylindrical rolls produce sheets. Shaped rolls make more complex profiles—rail track is one. Extrusion is a particularly versatile continuous process, since complex prismatic profiles that include internal channels and longitudinal features such as ribs and stiffeners can be manufactured in one step.

The process-shape matrix displays the links between the two. If the process cannot make the shape we want, it may be possible to combine it with a secondary process to give a process chain that adds the additional features: Casting followed by machining is an obvious example. But remember: Every additional process step adds cost.

Shaping processes: Mass and section thickness There are limits to the size of component that a process can make. Figure 13.24 shows the limits. The color coding for material compatibility has been retained, using more than one color when the process can treat more than one material family. Size can be measured by volume or by mass, but since the range of either one covers many orders of magnitude, whereas densities span only a factor of about 50, it doesn't make much difference which we use—big things are heavy, whatever they are made of. Most processes span a mass range of about a factor of 1000 or so. Note that this attribute is most discriminating at the extremes; the vast majority of components are in the 0.1 to 10 kg range, for which virtually any process will work.

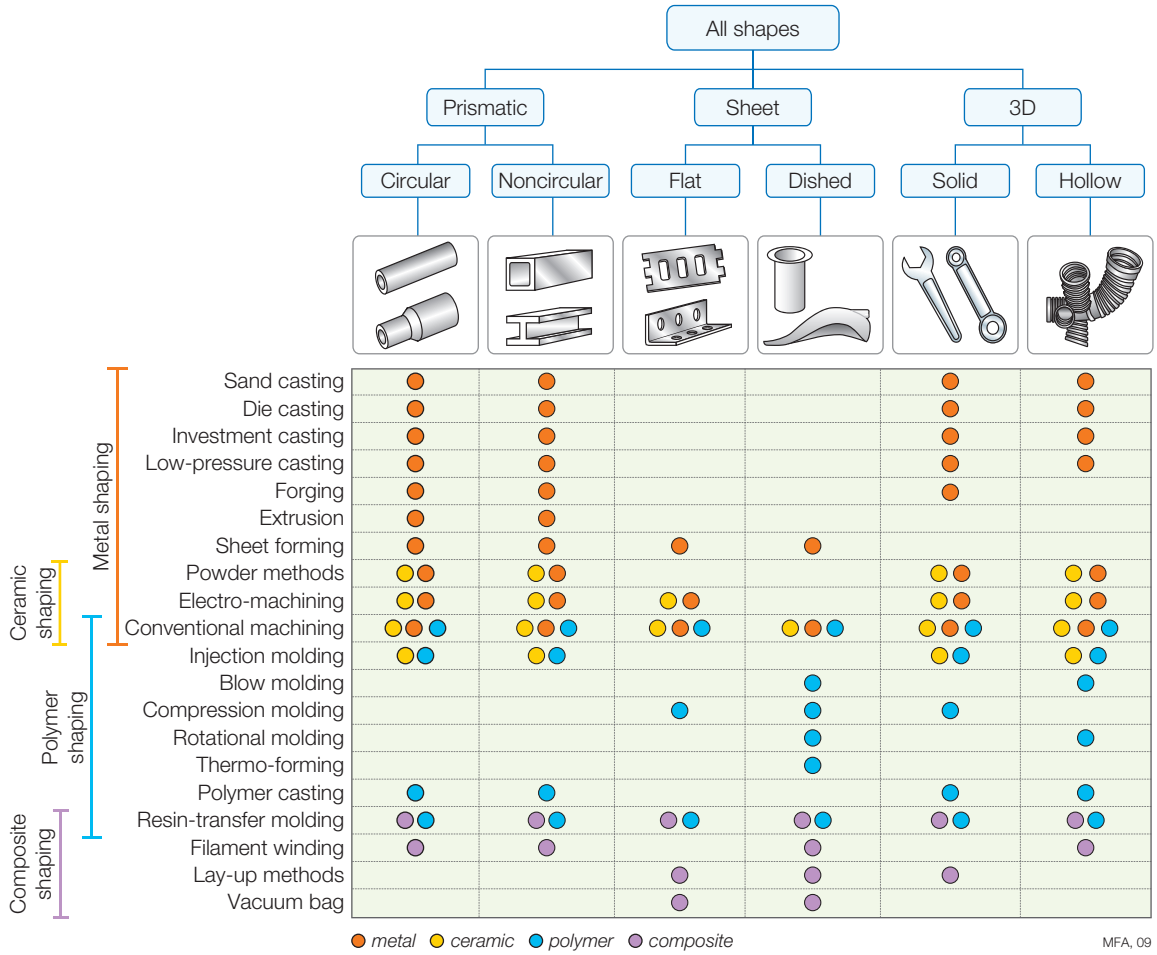


FIGURE 13.23
The process-shape matrix. Information about material compatibility is included at the extreme left.

Each bar spans the size range of which the process is capable without undue technical difficulty. All can be stretched to smaller or larger extremes but at the penalty of extra cost because the equipment is no longer standard. During screening, therefore, it is important to recognize "near misses"—processes that narrowly failed, but that could, if needed, be reconsidered and used.

Figure 13.25 shows a second bar chart: that for the ranges of section thickness of which each shaping process is capable. It is the lower end of the ranges—the minimum section thickness—where the physics of the process imposes limits. The origins of these limits are the subject of the next subsection.

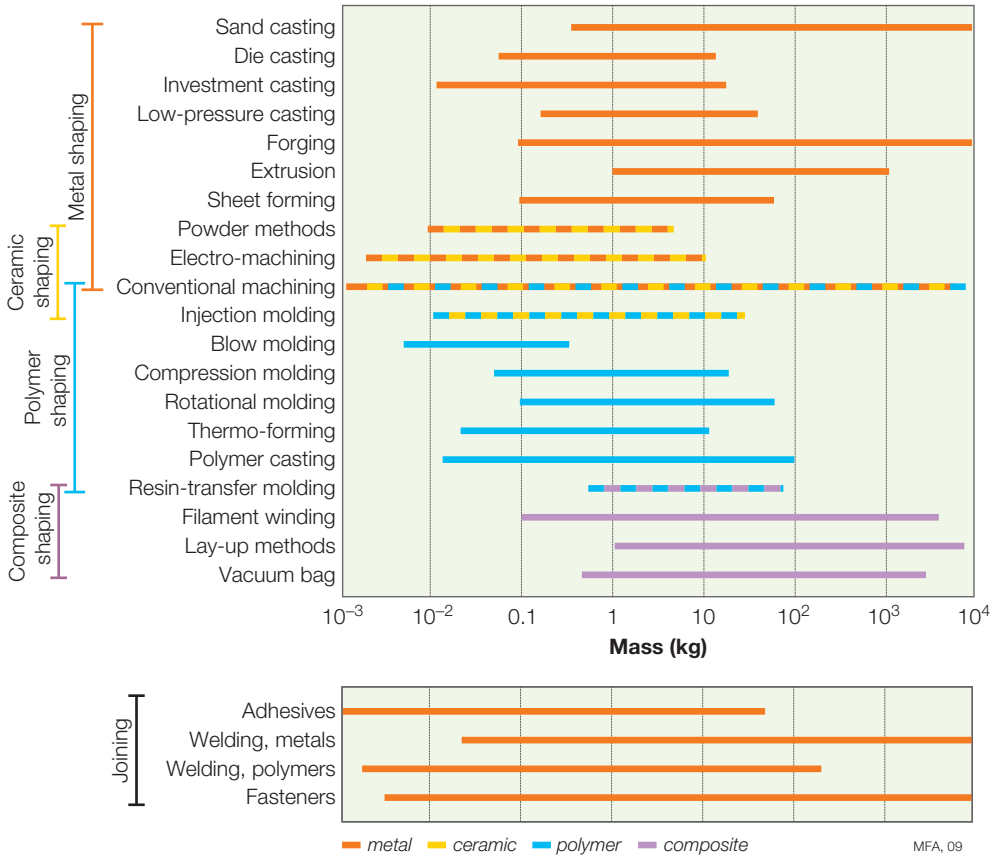


FIGURE 13.24
The process–mass-range chart. The inclusion of joining allows simple process chains to be explored.

Physical limits to size and section thickness Casting and molding both rely on material flow in the liquid or semiliquid state. Lower limits on section thickness are imposed by the physics of flow. Viscosity and surface tension oppose flow through narrow channels, and heat loss from the large surface area of thin sections cools the flowing material, raising the viscosity before the channel is filled (Figure 13.26). Pure metals solidify at a fixed temperature, with a step increase in viscosity, but for alloys solidification happens over a range of temperature, known as the "mushy zone," in which the alloy is part liquid, part solid. The width of this zone can vary from a few degrees centigrade to several hundred—so metal flow in castings depends on alloy composition. In general, higher-pressure die casting and molding methods enable thinner sections to be made, but the equipment costs more and the faster, more turbulent flow can entrap more porosity and cause damage to the molds.

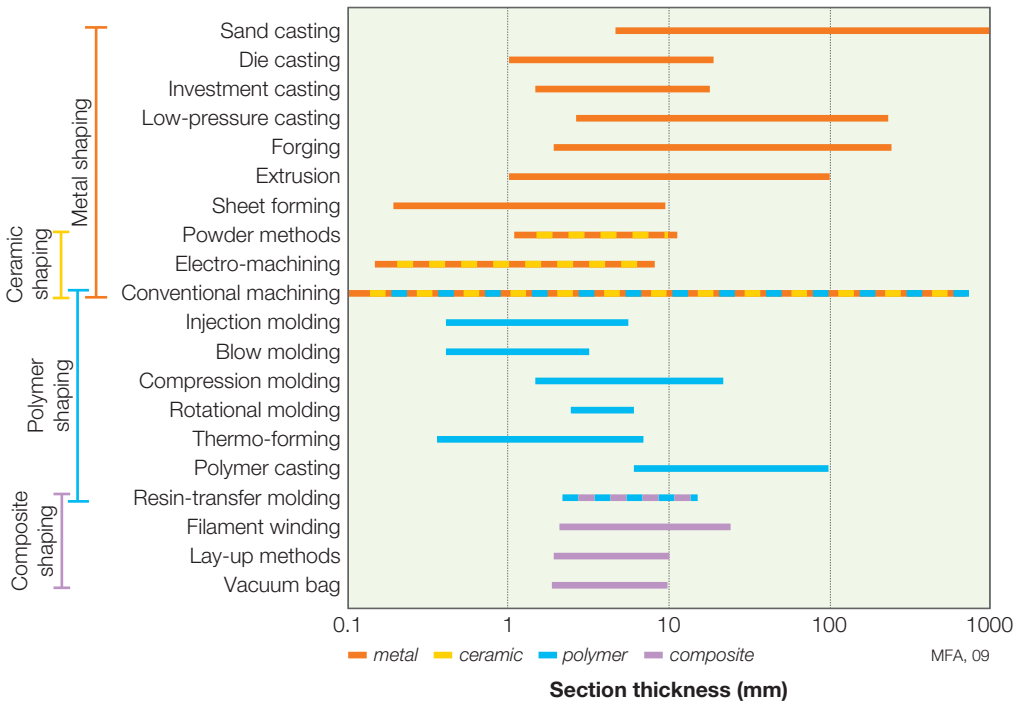


FIGURE 13.25
The process–section thickness chart.

Upper limits to size and section in casting and molding are set by problems of shrinkage. The outer layer of a casting or molding cools and solidifies first, giving it a rigid skin. When the interior subsequently solidifies, the change in volume can distort the product or crack the skin, or cause internal cavitation. Problems of this sort are most severe where there are changes of section, since the constraint introduces tensile stresses that cause *hot tearing*—cracking

Flow in thin channels

A mold for casting an intricate aluminum casting has some channel-like features with a width of only 10 μm. Will an overpressure of 1 atmosphere (0.1 MPa) be sufficient to overcome surface tension, allowing the feature to be filled? The surface tension γ of liquid aluminum is 1.1 J/m^2.

Answer

The pressure required to overcome surface tension and force the metal into a parallel-sided channel of width $2x$ is $p = \gamma/x$. Thus the narrowest channel that can be filled with an overpressure of 1 atmosphere is

$$2x = \frac{2\gamma}{p} = \frac{2.2}{0.1 \times 10^6}\, 2.2 \times 10^{-5} = 22\,\mu\text{m}$$

An overpressure of 1 atmosphere is not sufficient to fill the channel. An overpressure of 5 atmospheres would do it comfortably.

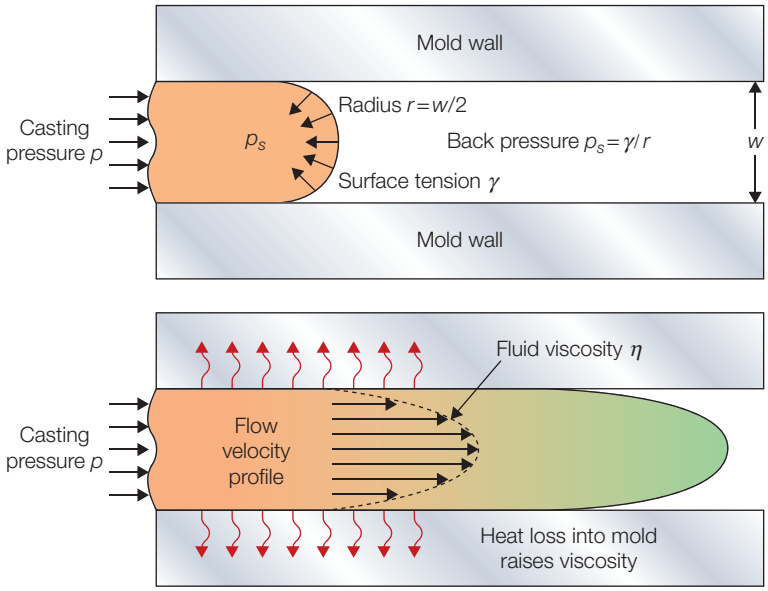


FIGURE 13.26

Flow of liquid metal or polymer into thin sections is opposed by surface tension as at (a) and by viscous forces (b). Loss of heat into the mold increases viscosity and may cause premature solidification.

caused by constrained thermal contraction. Different compositions have different susceptibilities to hot tearing—another example of coupling between material, process, and design detail.

Metal shaping by deformation—hot or cold rolling, forging, or extrusion—also involves flow. The thinness that can be forged, rolled, or extruded is limited by plastic flow in much the same way that the thinness in casting is limited by viscosity: The thinner the section, the greater the required roll pressure or forging force. Figure 13.27 illustrates the problem. Friction changes the pressure distribution on the die and under the rolls. When they are well-lubricated, as in (a), the loading is almost uniaxial and the material flows at its yield stress σ_y. With friction, as in (b), the metal shears at the die interface and the pressure ramps up because the friction resists the lateral spreading, giving a "friction hill." The area under the pressure distribution is the total forming load, so friction increases the load. The greater the aspect ratio of the section (width/thickness), the higher the maximum pressure needed to cause yielding, as in (c). This illustrates the fundamental limit of friction on section thickness—very thin sections simply stick to the tools and will not yield, even with very large pressures.

Friction also limits aspect ratio in powder processing. The externally applied pressure is diminished by die-wall friction (Figure 13.28) with the result

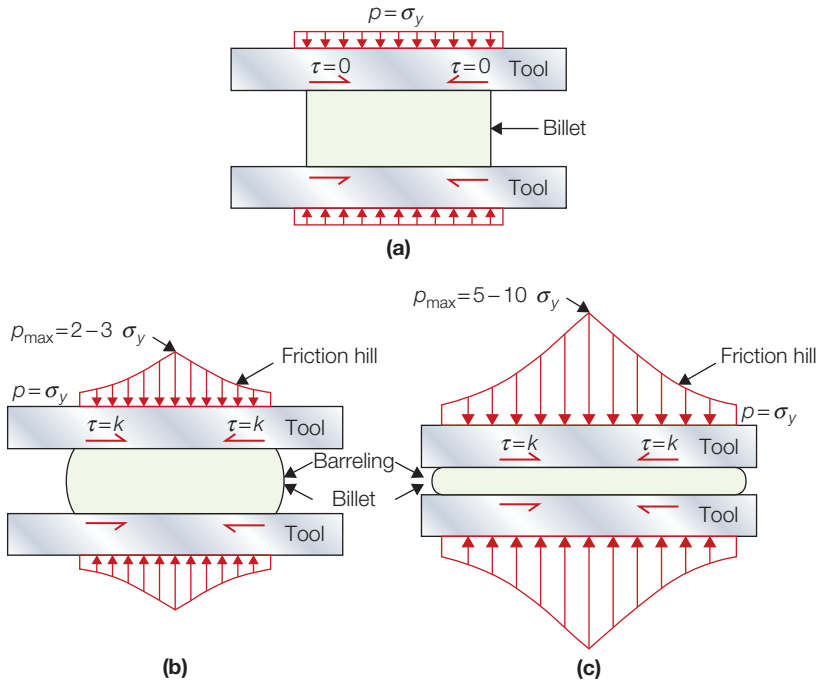


FIGURE 13.27

The influence of friction and aspect ratio on forging: (a) uniaxial compression with very low friction; (b) with sticking friction the contact pressure rises in a friction hill; (c) the greater the aspect ratio, the greater the pressure rise, ultimately limiting the thinness that can be achieved.

Limiting aspect ratio for powder pressing

How large is the pressure drop in a cylindrical powder mass like that in Figure 13.28 caused by die-wall friction if the coefficient of friction on the die wall is $\mu = 0.5$?

Answer

The frictional force opposing sliding in the green band of thickness dx is $2\pi\, r\, \mu\, p\, dx$ where p is the pressure at a distance x below the die face. Dividing this by the cross-section area of the powder pack πr^2 gives the pressure drop

$$dp = -\frac{2}{r}\mu\, p\, dx$$

Integrating from $x = 0$ where $p = p_o$ to $x = x$ where $p = p(x)$ gives

$$p(x) = p_o \exp -\frac{2\mu x}{r}$$

With a coefficient of friction of $\mu = 0.5$, the pressure drops to half its remote value p_o at a depth-to-radius ratio of only

$$\frac{x}{r} = -\frac{1}{2\mu} \ln\left(\frac{1}{2}\right) = 0.69$$

The answer is to lubricate the mold, reducing μ.

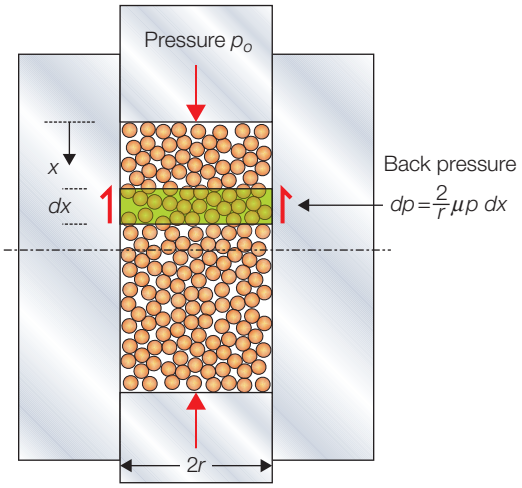


FIGURE 13.28
The height-to-width aspect ratio in powder forging is limited by die-wall friction, which causes the compaction pressure to fall exponentially with distance from the surface.

that, if the aspect ratio is too great, there is insufficient pressure to compact powder in the center of the product.

Tolerance and roughness The precision and surface finish of a component are aspects of its quality. They are measured by the *tolerance* and the *surface roughness* R. When the dimensions of a component are specified the surface quality is specified as well, though not necessarily over the entire surface. Surface quality is critical in contacting surfaces such as the faces of flanges that must mate to form a seal or sliders running in grooves. It is also important for resistance to fatigue crack initiation and for aesthetic reasons. The tolerance T on a dimension y is specified as $y = 100 \pm 0.1$ mm, or as $y = 50^{+0.01}_{-0.001}$ mm, indicating that there is more freedom to oversize than to undersize. Surface roughness is specified as an upper limit, for example, $R < 100\,\mu$m. The typical surface finish required in various products is shown in Table 13.3. The table also indicates typical processes that can achieve these levels of finish.

Surface roughness is a measure of the irregularities of the surface (Figure 13.29). It is defined as the root-mean-square (RMS) amplitude of the surface profile:

$$R^2 = \frac{1}{L}\int_0^L y^2(x)dx \qquad (13.1)$$

One way to measure it is to drag a light, sharp stylus over the surface in the x-direction while recording the vertical profile $y(x)$, like playing a gramophone record. *Optical profilometry*, which is faster and more accurate, uses laser

Table 13.3 Levels of Finish

Finish, μm	Process	Typical Application
R = 0.01	Lapping	Mirrors
R = 0.1	Precision grind or lap	High-quality bearings
R = 0.2–0.5	Precision grinding	Cylinders, pistons, cams, bearings
R = 0.5–2	Precision machining	Gears, ordinary machine parts
R = 2–10	Machining	Light-loaded bearings, noncritical components
R = 3–100	Unfinished castings	Nonbearing surfaces

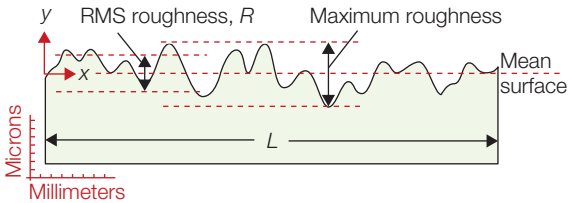


FIGURE 13.29
A section through a surface showing its irregular surface (artistically exaggerated in the vertical direction). The irregularity is measured by the RMS roughness R.

interferometry to map surface irregularity. The tolerance T is obviously greater than $2R$; indeed, since R is the root-mean-square roughness, the peak roughness, and hence the absolute lower limit for tolerance, is more like $5R$. Real processes give tolerances that range from $10R$ to $1000R$.

Figures 13.30 and 13.31 show the characteristic ranges of tolerance and roughness of which processes are capable, retaining the color coding for material families. Data for finishing processes are added below the shaping processes. Sand casting gives rough surfaces; casting into metal dies gives smoother ones. No shaping processes for metals, however, do better than T = 0.1 mm and R = 0.5 μm. Machining, capable of high dimensional accuracy and surface finish, is commonly used after casting or deformation processing to bring the tolerance or finish up to the desired level, creating a *process chain*. Metals and ceramics can be surface-ground and lapped to a high precision and smoothness: A large telescope reflector has a tolerance approaching 5 μm and a roughness of about 1/100 of this over a dimension of a meter or more. But precision and finish carry a cost: Processing costs increase exponentially as the requirements for both are made more severe. It is an expensive mistake to overspecify precision and finish.

Molded polymers inherit the finish of the molds used to shape them and thus can be very smooth; machining to improve the finish is rarely necessary. Tolerances better than $\pm$ 0.2 mm are seldom possible because internal stresses left by molding cause distortion and because polymers creep in service.

Joining: Material compatibility Processes for joining metals, polymers, ceramics, and glasses differ. A given adhesive will bond to some materials but not to others; welding methods for polymers differ from those for welding metals; and ceramics, which cannot be welded, are joined instead by diffusion bonding or glaze bonding. The process-material matrix (Figure 13.22) included four classes of joining process.

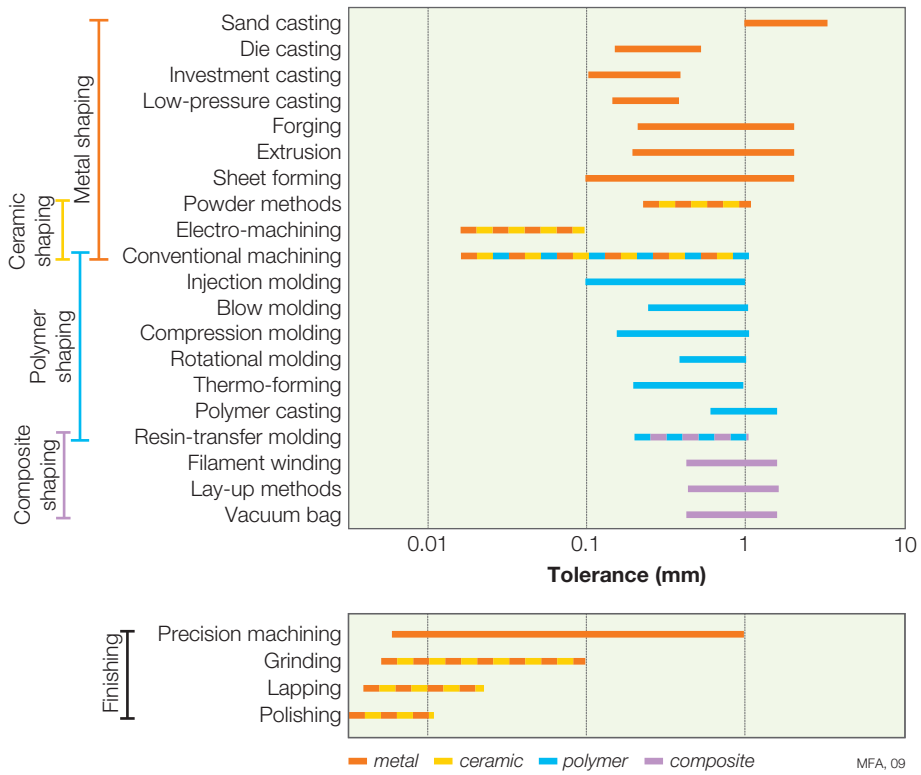


FIGURE 13.30
The process-tolerance chart. The inclusion of finishing processes allows simple process chains to be explored.

When the joint is between dissimilar materials, the process must be compatible with both. Adhesives and fasteners allow joints between different materials; many welding processes do not. If dissimilar metals are joined in such a way that they are in electrical contact, a corrosion couple appears if the joint is wet. This can be avoided by inserting an insulating layer between the surfaces. Thermal-expansion mismatch gives internal stresses in the joint if the temperature changes, with risk of distortion or damage. Identifying good practice in joining dissimilar materials is part of the documentation step.

Joint geometry and mode of loading The geometry of the joint and the way it is loaded (Figure 13.32) influence process choice. Adhesive joints support shear but are poor in peeling—think of stripping off adhesive tape. Adhesives need a large working area—lap joints work well, butt joints do not. Rivets and staples, too, are well adapted for shear loading of lap joints but are less good in tension. Welds and threaded fasteners are more adaptable, but here, too, matching choice of process to geometry and loading is important.

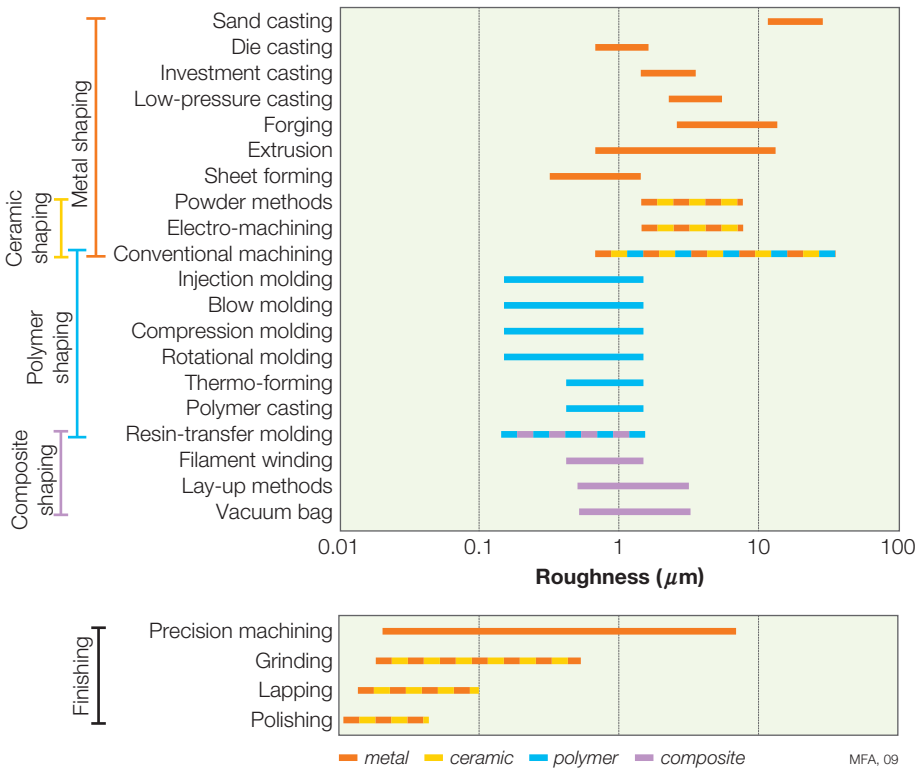


FIGURE 13.31
The process–surface roughness chart. The inclusion of finishing processes allows simple process chains to be explored.

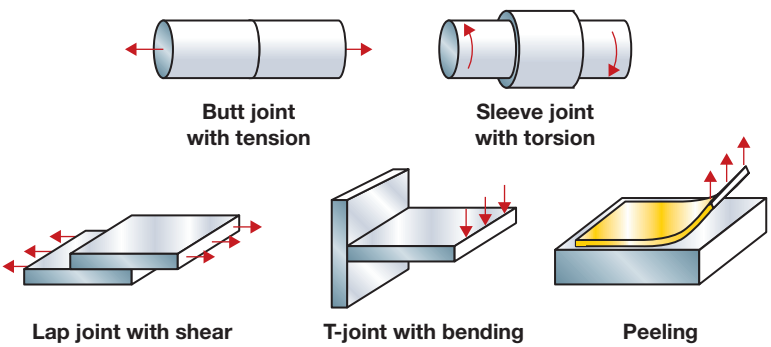


FIGURE 13.32
Joint geometries and modes of loading.

Secondary functions of joints A joint may serve as a seal, and be required to exclude gases or liquids. It may be required to conduct or insulate against the conduction of heat or electricity, or to operate at an elevated operating temperature. It may be permanent or have to be disassembled at the end of product life for recycling and reuse: Threaded fasteners and adhesives, loosened with solvents or heat, permit this.

Surface treatment: Material compatibility Material process compatibility for surface treatments is shown at the bottom of the matrix in Figure 13.22. As noted previously, surface finishing is more important for metals than for polymers.

The purpose of the surface treatment All surface treatment adds cost but the added value can be large. Table 13.4 illustrates the diversity of functions that surface treatments can provide. Some protect, some enhance performance, still others are primarily aesthetic. Protecting a component surface extends product life and increases the interval between maintenance cycles. Coatings on cutting tools enable faster cutting speeds and greater productivity. And surface-hardening processes may enable the substrate alloy to be replaced with a cheaper material—for example, a plain carbon steel with a hard carburized surface or a coating of hard titanium nitride (TiN) instead of a more expensive alloy steel.

Secondary compatibilities Some surface treatments, such as anodizing, leave the dimensions, precision, and roughness of the surface unchanged. Electro- and vapor-deposited coatings change the dimensions a little, but may still leave a perfectly smooth surface. Polymer powder coating builds up a relatively thick, smooth layer; others, such as weld deposition, create a thick layer with a rough surface, requiring refinishing. "Line-of-sight" deposition processes coat only the surface at which they are directed, leaving inaccessible areas uncoated; others, with what is called "throwing power," coat flat, curved, and reentrant surfaces equally well. Many surface treatment processes require heat. These can only be used on materials that can tolerate the rise in temperature. Some paints are applied cold, but many require a bake at up to 150°C. Heat treatments like carburizing or nitriding to give a hard surface layer require prolonged heating at temperatures up to 800°C, which can change the microstructure of the material being coated.

Table 13.4 Functions Provided by Surface Treatments

Corrosion protection, aqueous environments	Thermal insulation
Corrosion protection, gas environments	Electrical insulation
Wear resistance	Magnetic response
Friction control	Decoration
Fatigue resistance	Color
Thermal conduction	Reflectivity

13.6 RANKING: PROCESS COST

Part of the cost of any component is the expense of the material from which it is made. The rest is the cost of manufacture—that is, forming it to a shape and joining and finishing it. Before turning to details, there are four common-sense rules for minimizing cost that the designer should bear in mind, as follows.

Keep things standard If someone already makes the part you want, it will almost certainly be cheaper to buy it than to make it. If nobody does, then it is cheaper to design it to be made from standard stock (sheet, rod, tube) than from nonstandard shapes or from special castings or forgings. Try to use standard materials, and as few of them as possible: This will reduce inventory costs and the range of tooling the manufacturer needs, and it can help with recycling.

Keep things simple If a part has to be machined, it will have to be clamped; the cost increases with the number of times it will have to be rejigged or reoriented, especially if special tools are necessary. If a part is to be welded or brazed, the welder must be able to reach it with his torch and still see what he is doing. If it is to be cast or molded or forged, it should be remembered that high (and expensive) pressures are required to make fluids flow into narrow channels, and that reentrant shapes greatly complicate mold and die design. Think of making the part yourself: Will it be awkward? Could slight redesign make it less awkward?

Make the parts easy to assemble Assembly takes time, and time is money. If the overhead rate is a mere $60 per hour, every minute of assembly time adds another $1 to the cost. Design for assembly (DFA) addresses this problem with a set of common-sense criteria and rules. Briefly, there are three:

1. Minimize part count.
2. Design parts to be self-aligning on assembly.
3. Use joining methods that are fast; snap fits and spot welds are faster than threaded fasteners or, usually, adhesives.

Do not specify more performance than is needed Performance must be paid for. High-strength metals are more heavily alloyed with expensive additions; high-performance polymers are chemically more complex; high-performance ceramics require greater quality control in their manufacture. All of these increase material costs. In addition, high-strength materials are hard to fabricate. The forming pressures (whether for a metal or a polymer) are higher; tool wear is greater; ductility is usually less so that deformation processing can be difficult or impossible. This can mean that new processing routes must be used: investment casting or powder forming instead of conventional casting and mechanical working; more expensive molding equipment operating at higher temperatures and pressures; and so on. The better performance of the high-strength material must be paid for, not only in greater material cost but also in the higher cost of processing. Finally, there

are the questions of tolerance and roughness. Cost rises exponentially with demands for precision and surface finish. The message is clear. Performance costs money. Do not overspecify it.

To make further progress, we must examine the contributions to process costs, and their origins.

Economic criteria for selection If you have to sharpen a pencil, you can do it with a knife. If, instead, you had to sharpen a thousand pencils, it would pay to buy an electric sharpener. And if you had to sharpen a million, you might wish to equip yourself with an automatic feeding, gripping, and sharpening system. To cope with pencils of different length and diameter, you could go further and devise a microprocessor-controlled system with sensors to measure pencil dimensions, sharpening pressure, and so on —an "intelligent" system that can recognize and adapt to pencil size. The choice of process, then, depends on the number of pencils you wish to sharpen, that is, on the *batch size*. The best choice is the one that costs the least per pencil sharpened.

Figure 13.33 is a schematic of how the cost of sharpening a pencil might vary with batch size. A knife does not cost much but it is slow, so the labor

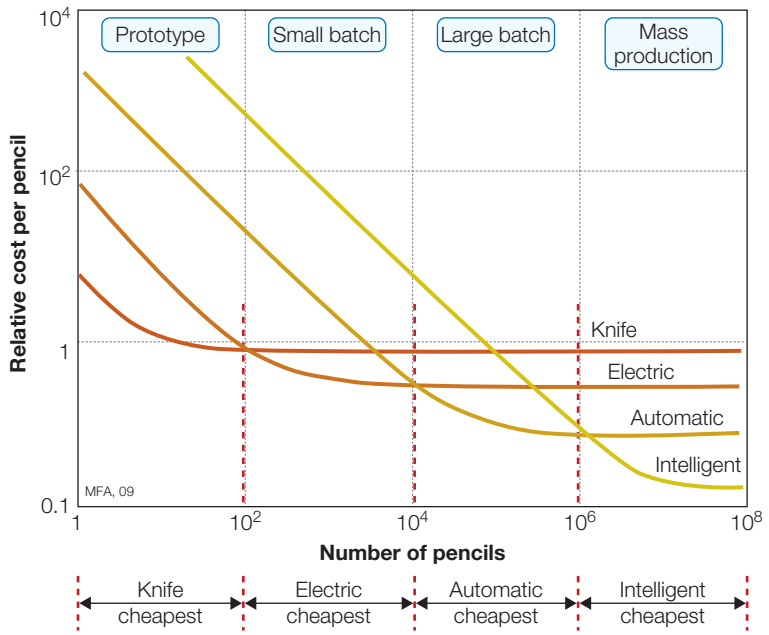


FIGURE 13.33
The cost of sharpening a pencil plotted against batch size for four processes. The curves all have the form of Equation (13.5).

cost is high. The other processes involve progressively greater capital investment but do the job more quickly, reducing labor costs. The balance between capital cost and rate gives the shape of the curves. In this figure the best choice is the lowest curve—a knife for up to 100 pencils; an electric sharpener for 10^2 to 10^4, an automatic system for 10^4 to 10^6, and so on. Each process has an *economic batch size.*

Economic batch size Process cost depends on a large number of independent variables, not all within the control of the modeler. Cost modeling is described in the next section, but—given the disheartening implications of the last sentence—it is comforting to have an alternative, if approximate, way out. The economic batch size provides it. Values for the processes described in this chapter are shown in Figure 13.34. A process with an economic batch size with the range B_1–B_2 is one that is found by experience to be competitive in cost when the output lies in that range, just as the electric sharpener was economic in the range 10^2 to 10^4. The economic batch size is commonly cited for processes. The easy way to introduce economy into the selection is to rank candidate processes by economic batch size and retain those that are economic in the range you want. But do not harbor false illusions: Many variables cannot be rolled into one without loss of discrimination. A cost model gives deeper insight.

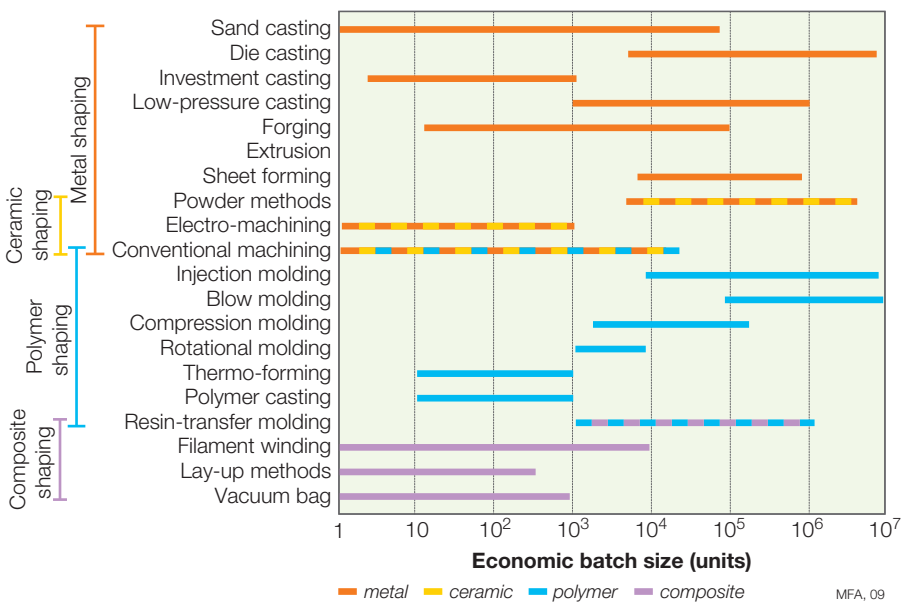


FIGURE 13.34
The economic batch size chart.

Cost modeling

The manufacture of a component consumes resources (Figure 13.35), each of which has an associated cost. The final cost is the sum of expenses of all of the resources it consumes (detailed in Table 13.5). Thus the cost of producing a component of mass m entails the cost Cm (\$/kg) of the materials and feedstocks from which it is made. It involves the cost of dedicated tooling Ct (\$) and that of the capital equipment Cc (\$) in which the tooling will be used. It requires time, chargeable at an overhead rate $\dot{C}_{oh}$ (thus with units of \$/hr), in which we include the cost of labor, administration, and general plant costs. It requires energy, which is sometimes charged against a process step if it is very energy intense but more commonly is treated as part of the overhead and lumped into $\dot{C}_{oh}$, as we shall do here. Finally there is the cost of information, meaning research and development, royalty or license fees; this, too, we view as a cost per unit time and lump it into the overhead.

Think now of the manufacture of a component (the "unit of output") weighing m kg, made of a material costing C_m \$/kg. The first contribution to the unit cost is that of the material mC_m magnified by the factor $1/(1-f)$ where f is the scrap fraction—the fraction of the starting material that ends up as sprues, risers, turnings, rejects, or waste:

$$C_1 = \frac{mC_m}{(1-f)} \qquad (13.2)$$

The cost C_t of a set of tooling—dies, molds, fixtures, and jigs—is what is called a *dedicated cost*: one that must be wholly assigned to the production run of this single component. It is written off against the numerical size n of the production

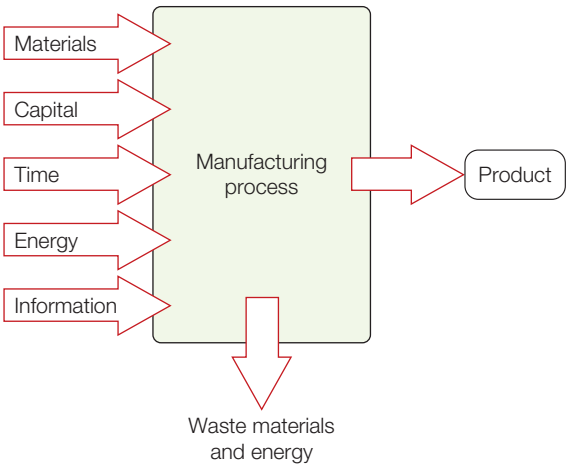


FIGURE 13.35
The inputs to a cost model.

Table 13.5 Symbols, Definitions, and Units

Resource	Symbol	Unit
Materials: including consumables	C_m	\$/kg
Capital: cost of tooling	C_t	\$
cost of equipment	C_c	\$
Time: overhead rate, including labor, administration, rent …	$\dot{C}_{oh}$	\$/hr
Energy: cost of energy	$\dot{C}_e$	\$/hr
Information: R & D or royalty payments	$\dot{C}_i$	\$/year

run. Tooling wears out. If the run is a long one, replacement will be necessary. Thus tooling cost per unit takes the form

$$C_2 = \frac{C_t}{n}\left\{Int\left(\frac{n}{n_t} + 0.51\right)\right\} \tag{13.3}$$

where n_t is the number of units that a set of tooling can make before it has to be replaced, and *Int* is the integer function. The term in curly brackets simply increments the tooling cost by that of one tool set every time n exceeds n_t.

The capital cost of equipment C_c, by contrast, is rarely dedicated. A given piece of equipment—a powder press, for example—can be used to make many different components by installing different die-sets or tooling. It is usual to convert the capital cost of *nondedicated* equipment, and the cost of borrowing the capital itself, into an overhead by dividing it by a capital write-off time t_{wo} (five years, say) over which it is to be recovered. The quantity C_c/t_{wo} is then a cost per hour—provided the equipment is used continuously. That is rarely the case, so the term is modified by dividing it by a load factor L—the fraction of time for which the equipment is productive. The cost per unit is then this hourly cost divided by the rate $\dot{n}$ at which units are produced:

$$C_3 = \frac{1}{\dot{n}}\left(\frac{C_c}{Lt_{wo}}\right) \tag{13.4}$$

Finally there is the overhead rate $\dot{C}_{oh}$. It becomes a cost per unit when divided by the production rate $\dot{n}$ units per hour (averaged over the production run to allow for downtime):

$$C_4 = \frac{\dot{C}_{oh}}{\dot{n}} \tag{13.5}$$

The total shaping cost per part is the sum of these four terms, taking the form

$$C = \frac{mC_m}{(1-f)} + \frac{C_t}{n}\left\{Int\left(\frac{n}{n_t} + 0.51\right)\right\} + \frac{1}{\dot{n}}\left(\frac{C_c}{Lt_{wo}} + \dot{C}_{oh}\right) \tag{13.6}$$

The equation says: The cost has three essential contributions—a material cost per unit of production that is independent of batch size and rate, a dedicated cost per unit of production that varies as the reciprocal of the production volume $(1/n)$, and a gross overhead per unit of production that varies as the reciprocal of the production rate $(1/\dot{n})$. The equation describes a set of curves relating cost C to batch size n, one for each process. Each has the shape of the pencil-sharpening curves of Figure 13.33.

The use of the model is illustrated more fully in the case studies of Chapter 14.

Technical cost modeling Equation (13.6) is the first step in modeling cost. Greater predictive power is possible with technical cost models that exploit understanding of the way in which the design, the process, and cost interact. The capital cost of equipment depends on size and degree of automation. Tooling cost and production rate depend on complexity. These and many other dependencies can be captured in theoretical or empirical formulae or look-up tables that can be built into the cost model, giving more resolution in ranking competing processes. For more advanced analyses the reader is referred to the literature listed in "Further reading," at the end of this chapter.

13.7 COMPUTER-AIDED PROCESS SELECTION

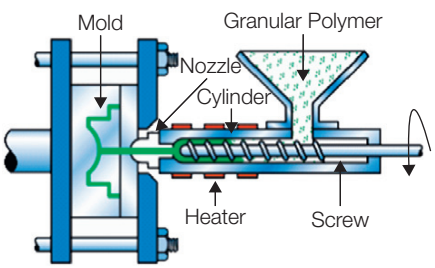
Screening

If process attributes are stored in a database with an appropriate user interface, selection charts can be created and selection boxes manipulated with much greater freedom. The CES platform, mentioned earlier, is an example of such a system. The database contains records, each describing the attributes of a single process. Example 13.1 shows part of a typical record: that for injection molding. A schematic indicates how the process works; it is supported by a short description. This is followed by a list of attributes: the shapes it can make, the attributes relating to shape and physical characteristics, and those that describe economic parameters; ending with brief documentation in the form of guidelines, technical notes, and typical uses. The numeric attributes are stored as ranges, indicating the range of capability of the process. Each record is linked to records for the materials with which it is compatible, allowing choice of material to be used as a screening criterion, like the material compatibility matrix of Figure 13.22 but with greater resolution. A short list of candidates is extracted in two steps: screening to eliminate processes that cannot meet the design specification, and ranking to order the survivors by economic criteria.

EXAMPLE 13.1

Injection Molding

The Process The most widely used process for shaping thermoplastics is the reciprocating screw injection molding machine, shown in the diagram on the right. Polymer granules are fed into a spiral press where they are heated, mixed, and softened to a dough-like consistency that is forced through one or more channels ("sprues") into the die. The polymer solidifies under pressure and the component is then ejected.



Thermoplastics, thermosets, and elastomers can also be injection molded. Co-injection allows molding of components with different materials, colors, and features. Injection foam molding allows economical production of large molded components by using inert gas or chemical blowing agents to make components that have a solid skin and a cellular inner structure.

Shapes	
Circular prismatic	✓
Noncircular prismatic	✓
Solid 3D	✓
Hollow 3D	✓
Physical Attributes	
Mass range	0.01–25 kg
Range of section thickness	0.4–6.3 mm
Tolerance	0.2–1 mm
Roughness	0.2–1.6 μm
Surface roughness (A = v. smooth)	A
Economic Attributes	
Economic batch size (units)	10^4–10^6
Cost Modeling	
Capital cost	3×10^4–7×10^5 USD
Material utilization fraction	0.6–0.9
Production rate (units)	60–1000/hr
Tooling cost	3000–30,000 USD
Tool life (units)	10^4–10^6

Documentation

Design Guidelines Injection molding is the best way to mass-produce small, precise polymer components with complex shapes. The surface finish is good; texture and pattern can be easily altered in the tool; and fine detail reproduces well. Decorative labels can be molded onto the surface of the component. The only finishing operation is the removal of the sprue.

Technical Notes Most thermoplastics can be injection molded, although those with high melting temperatures (e.g., PTFE) are difficult. Thermoplastic-based composites (short-fiber and particulate filled) can be processed providing the filler-loading is not too large. Large changes in section area are not recommended. Small reentrant angles and complex shapes are possible, though some features (e.g., undercuts, screw threads, inserts) may result in increased tooling costs.

Typical Uses Housings, containers, covers, knobs, tool handles, plumbing fittings, lenses, toys, and so on.

To enable this, the cost model described in Section 13.6 is implemented in the CES software. The records contain approximate data for the ranges of capital and tooling costs (C_c and C_t) and for the rate of production ($\dot{n}$). Equation (13.6) contains other parameters not listed in the record because they are not attributes of the process itself but depend on the design, or the material, or the economics (and thus the location) of the plant in which the processing will be done. The user must provide this information, conveniently

entered through a dialog box. The output is a plot of cost against batch size, like that shown earlier in Figure 13.33.

More information about computer-aided selection can be found in the sources listed under Granta Design (2010) in "Further reading."

13.8 SUMMARY AND CONCLUSIONS

A wide range of shaping, joining, and finishing processes is available to the design engineer. Each has certain characteristics, which, taken together, suit it to the processing of certain materials in certain shapes, but disqualify it for others. Faced with the choice, the designer in the past, relied on locally available expertise or on common practice. Neither of these leads to innovation, nor are they well matched to current design methods. The structured, systematic approach of this chapter provides a way forward. It ensures that potentially interesting processes are not overlooked, and guides the user quickly to processes capable of achieving the desired requirements.

The method parallels that for selection of material, using process selection matrices and charts to implement the procedure. A component design dictates a certain, known combination of process attributes. These design requirements are plotted onto the charts, identifying a subset of possible processes. The method lends itself to computer implementation, allowing selection from a large portfolio of processes by screening on attributes and ranking by economic criteria.

There is, of course, much more to process selection than this. It is to be seen, rather, as a first systematic step, replacing a total reliance on local experience and past practice. The narrowing of choice helps considerably: It is now much easier to identify the right source for more expert knowledge and to ask it the right questions. But the final choice still depends on local economic and organizational factors that can only be decided on a case-by-case basis.

13.9 FURTHER READING

ASM Handbook Series (1971–2004). Heat treatment, vol. 4; Surface engineering, vol. 5; Welding, brazing and soldering, vol. 6; Powder metal technologies, vol. 7; Forming and forging, vol. 14; Casting, vol. 15; Machining, vol. 16. ASM International.
A comprehensive set of handbooks on processing, occasionally updated, and now available online at http://products.asminternational.org/hbk/index.jsp

Bralla, J.G. (1998). *Design for manufacturability handbook* (2nd ed.). McGraw-Hill, ISBN 0-07-007139-X.
Turgid reading, but a rich mine of information about manufacturing processes.

Bréchet, Y., Bassetti, D., Landru, D., & Salvo, L. (2001). Challenges in materials and process selection. *Prog. Mat. Sci., 46*, 407–428.
An exploration of knowledge-based methods for capturing material and process attributes.

Budinski, K.G., & Budinski, M.K. (2010). *Engineering materials, properties and selection* (9th ed.). Prentice Hall, ISBN 978-0-13-712842-6.
A well-established text on the processing and use of engineering materials, now in its 9th edition.

Campbell, J. (1991). *Casting*. Butterworth-Heinemann, ISBN 0-7506-1696-2.
The fundamental science and technology of casting processes.

Clark, J.P., & Field, F.R. III (1997). Techno-economic issues in materials selection. In: *ASM Metals Handbook, 20*. American Society for Metals.
A paper outlining the principles of technical cost modeling and its use in the automobile industry.

Dieter, G.E. (1991). *Engineering design, a materials and processing approach* (2nd ed.). McGraw-Hill, ISBN 0-07-100829-2.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

Dieter, G.E., & Schmidt, L.C. (2009). *Engineering design* (4th ed.). McGraw-Hill, ISBN 978-0-07-283703-2.
Professor Dieter is a pioneer in presenting design from a materials perspective. The book contains a notable chapter on conceptualization.

Esawi, A., & Ashby, M.F. (1998). Computer-based selection of manufacturing processes: Methods, software and case studies.. *Proc. Inst. Mech. Eng., 212*, 595–610.
A paper describing the development and use of the CES database for process selection.

Grainger, S., & Blunt, J. (1998). *Engineering coatings, design and application*. Abington Publishing, ISBN 1-85573-369-2.
A handbook of surface treatment processes to improve surface durability—generally meaning surface hardness.

Granta Design (2010). *The CES Edu system* and other teaching resources, accessed from *www.grantadesign.com/education/*

Houldcroft, P. (1990). *Which process?*. Abington Publishing, ISBN 1-85573-008-1.
The title of this useful book is misleading—it deals only with a subset of the joining process: the welding of steels. But here it is good, matching the process to the design requirements.

Kalpakjian, S., & Schmidt, S.R. (2008). *Manufacturing processes for engineering materials* (5th ed.). Prentice Hall, ISBN 978-0-13-227271-1.
A comprehensive and widely used text on material processing.

Kalpakjian, S., & Schmidt, S.R. (2010). *Manufacturing engineering and technology* (6th ed.). Prentice Hall, ISBN 978-0-13-608168-5.
A comprehensive and widely used text on material processing.

Lascoe, O.D. (1988). *Handbook of fabrication processes*. ASM International, ISBN 0-87170-302-5.
A reference source for fabrication processes.

Shackelford, J.F. (2009). *Introduction to materials science for engineers* (7th ed.). Prentice Hall, ISBN 978-0-13-601260-3.
A long-established text on materials from an engineering perspective.

Swift, K.G., & Booker, J.D. (1997). *Process selection, from design to manufacture*. Arnold, ISBN 0-340-69249-9.
Details of 48 processes in a standard format, structured to guide process selection.

Wise, R.J. (1999). *Thermal welding of polymers*. Abington Publishing, ISBN 1-85573-495-8.
An introduction to the thermal welding of thermoplastics.

CHAPTER 14

Case Studies: Process Selection



Aluminum die castings. (Image courtesy of Aluminum Recovery Technologies Kendallville, Indiana.)

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14.1 INTRODUCTION AND SYNOPSIS

The last chapter described a systematic procedure for process selection. The inputs are design requirements; the output is a shortlist of processes capable of meeting them. Where processes compete, a ranking is made possible by a cost model. The case studies in this chapter illustrate the method. The first four make use of hard-copy charts; the last two show how computer-based selection works.

The case studies follow a standard pattern. First, we list the *design requirements*: material, shape, size, minimum section, precision, and finish. Then we plot these requirements onto the process matrices, identifying search areas. The processes that lie within the search areas are capable of making the component to its design specification: They are the candidates. If no one process meets all the design requirements, then processes have to be "stacked": casting followed by machining (to meet the tolerance specification on one surface, for instance); or powder methods followed by grinding.

More details for the most promising are then sought, starting with the data sources listed under "Further reading" and in the more comprehensive compilation of Appendix D. The final choice evolves from this subset, taking into account local factors, often specific to a particular company, geographical area, or country.

14.2 CASTING AN ALUMINUM CON-ROD

Connecting rods link oscillatory motion to rotary motion in IC engines, and rotary to oscillatory motion in pumps. Here we explore competing processes for casting a small aluminum alloy connecting rod (Figure 14.1), using the cost model of Chapter 13 to distinguish between them.

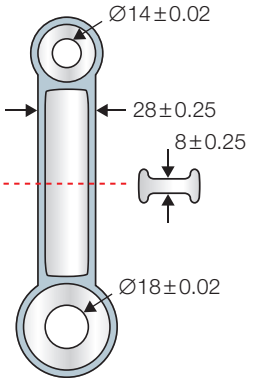


FIGURE 14.1
A connecting rod. The precision of the bores and bore facings is much higher than that of the rest of the body, requiring subsequent machining.

The design requirements The con-rod is a solid 3D shape to be made of a nonferrous alloy. The dimensions of the con-rod are such that its mass is about 0.3 kg and its minimum section 8 mm. The precision and tolerance of the casting are not critical since the bores and their faces will have to be machined to give the required precision. A batch size of 100,000 is envisaged. Table 14.1 summarizes the requirements.

Table 14.1 Requirements for the Con-rod

Function	Connecting rod	
Constraints	Material: aluminum casting alloy	– *Technical constraints*
	Shape: solid, 3D	
	Estimated mass: 0.3–0.4 kg	
	Minimum section: 8 mm	
	Tolerance: < 1.0 mm (surfaces), < 0.02 mm (bores)	– *Quality constraints*
	Roughness: not critical	
	Batch size: 100,000	– *Economic constraint*
Objective	Minimize cost	
Free variables	Choice of process	
	Process-operating conditions	

Table 14.2 Shortlist for Shaping the Con-rod

Process	Comment
Sand casting plus machining Low-pressure casting Die casting	The process–tolerance chart, Figure 14.8, reveals that none of these can meet the requirement on bore tolerance. All will require a subsequent machining operation.

The selection The constraints are plotted on the process matrices and charts in Figures 14.4 through 14.10 in the chapter. The material, the shape, and the required batch size eliminate most of the processes listed on them, leaving three casting processes and machining from solid as options. Machining is rejected on the grounds of material wastage; retained are sand casting, die casting, and low-pressure casting (Table 14.2).

More insight is given by examining their relative cost, using the model of Equation (13.6). The relevant data are assembled in Table 14.3, in which all costs all are normalized by the material cost, the term $mC_m/(1-f)$. The results are plotted in Figure 14.2 as unit cost C versus batch size n curves, following the example in Figure 13.33. At small batch sizes the unit cost is dominated by the "fixed" costs of tooling (the second term on the right of Equation (13.6)). As the batch size n increases, this contribution falls (provided, of course, that the tooling has a life that is greater than n) until it flattens out at a value that is dominated by the "variable" costs of material, labor, and other overheads.

Competing processes usually differ in tooling cost C_t and production rate $\dot{n}$, causing their $C - n$ curves to intersect, as they do here. Sand-casting equipment is inexpensive but the process is slow. The cost of molds for low-pressure casting is greater than for sand casting, and the process is a little faster. Die-casting equipment costs much more but it is also much

Table 14.3 Data for the Cost Equation for the Processes in Table 14.2

Relative Cost*	Sand Casting	Die Casting	Low-pressure Casting	Comment
Material, $mC_m/(1-f)$	1	1	1	*Process-independent parameters*
Basic overhead $\dot{C}_{oh}$ (hr^{-1})	10	10	10	
Capital write-off time *two* (yrs)	5	5	5	
Load factor	0.5	0.5	0.5	
Dedicated tool cost, C_t	210	16,000	2,000	*Process-dependent parameters*
Capital cost, C_c	1800	30,000	8,000	
Batch rate, $\dot{n}$ (hr^{-1})	3	50	10	
Tool life, n_t (number of units)	200,000	1,000,000	500,000	

* *All costs normalized to the material cost.*

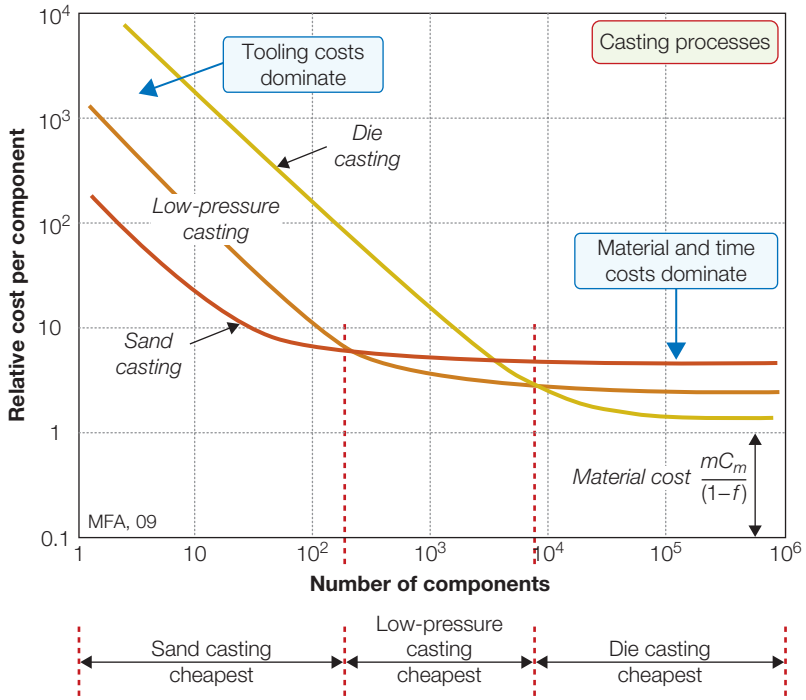


FIGURE 14.2
The relative cost of casting the con-rod as a function of the production run. The costs are normalized by the cost of the material.

faster. The material cost, the labor cost per hour, and the capital write-off time are taken to be the same for all.

The curves for sand, low-pressure, and die casting intersect at a batch size of 200: Below this, sand casting is the more economical. Low-pressure casting becomes marginally less expensive than the others for batches of between 200 and 8000, above which die casting becomes the most economic choice. The best choice for a batch size of 100,000 is die casting. Note that, for small batches, the component cost is dominated by that of the tooling—the material cost hardly matters. But as the batch size grows, the contribution of the second term in the cost equation diminishes; and if the process is fast, the unit cost falls until it is typically about three times that of the material of which the component is made.

Postscript There are other issues of quality besides precision and smoothness that enter into the choice of process. Sand castings tend to trap bubbles and inclusions that act as starting points for fatigue cracks in a cyclically loaded component like a con-rod. Some low-pressure–casting techniques smooth the flow of liquid metal into the die, reducing defect content. Die casting uses higher pressures, and generally gives the highest-quality casting. Considerations such as these can shift the economic switch-points, expanding the economic batch range of the process that offers the highest quality.

14.3 FORMING A FAN

Fans for vacuum cleaners (Figure 14.3) are designed to be cheap, quiet, and efficient, probably in that order. The key to minimizing process costs is to form the fan to its final shape in a single operation, leaving only the central hub to be machined to fit the shaft with which it mates. This means the selection of a single process that can meet the specifications for precision and tolerance, avoiding the need for machining or finishing of the disk or blades.

The design requirements Nylon is the material of choice for the fan. The pumping rate of a fan is determined by its rate of revolution and its size. The designer calculates the need for a fan of radius 60 mm, with 12 profiled blades of average thickness 4 mm. The volume of material in the fan is, roughly, its surface area times its thickness—about 10^{-4} m^3—giving (when multiplied by the density of nylon, 1100 kg/m^3) a weight in the range 0.1 to 0.15 kg. The fan has a fairly complex shape, although its high symmetry simplifies it somewhat. We classify it as 3D solid.

In the designer's view, balance and surface smoothness are what really matter. They (and the geometry) determine the pumping efficiency of the fan and influence the noise it makes. The designer specifies a tolerance of $\pm$ 0.5 mm and a surface roughness of $\leq$ 1 μm. A production run of 10,000 fans is planned.

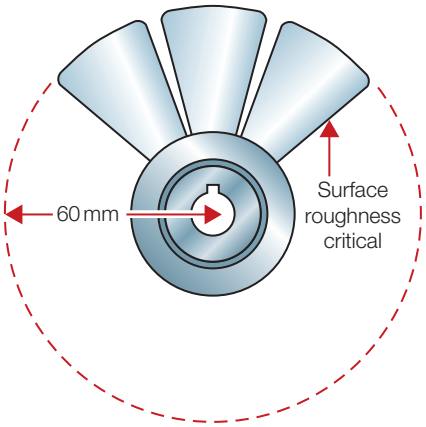


FIGURE 14.3
A fan. It is to be made of nylon, requires low roughness and a certain precision, and is to be produced in large numbers.

Table 14.4 Process Requirements for the Fan

Function	*Fan*	
Constraints	Material: nylon Shape: solid, 3D Estimated mass: 0.1–0.15 kg Minimum section: 4 mm	– *Technical constraints*
	Tolerance: ± 0.5 mm Roughness: < 1 μm	– *Quality constraints*
	Batch size: 100,000	– *Economic constraint*
Objective	Minimize cost	
Free variables	Choice of process Process-operating conditions	

The design requirements are summarized in Table 14.4. What processes can meet them?

The selection We turn first to the material-process and shape-process matrices (Figures 14.4 and 14.5) on which selection boxes have been drawn. The intersection of the two leaves five classes of shaping process—those boxed in the second figure. Screening on mass and section thickness (Figures 14.6 and 14.7) eliminates polymer casting and RTM, leaving the other three.

The constraints on tolerance and roughness are upper limits only (Figures 14.8 and 14.9); all three process classes survive. The planned batch size of 10,000 plotted on the economic batch-size chart (Figure 14.10) eliminates machining from solid. The surviving processes are listed in Table 14.5.

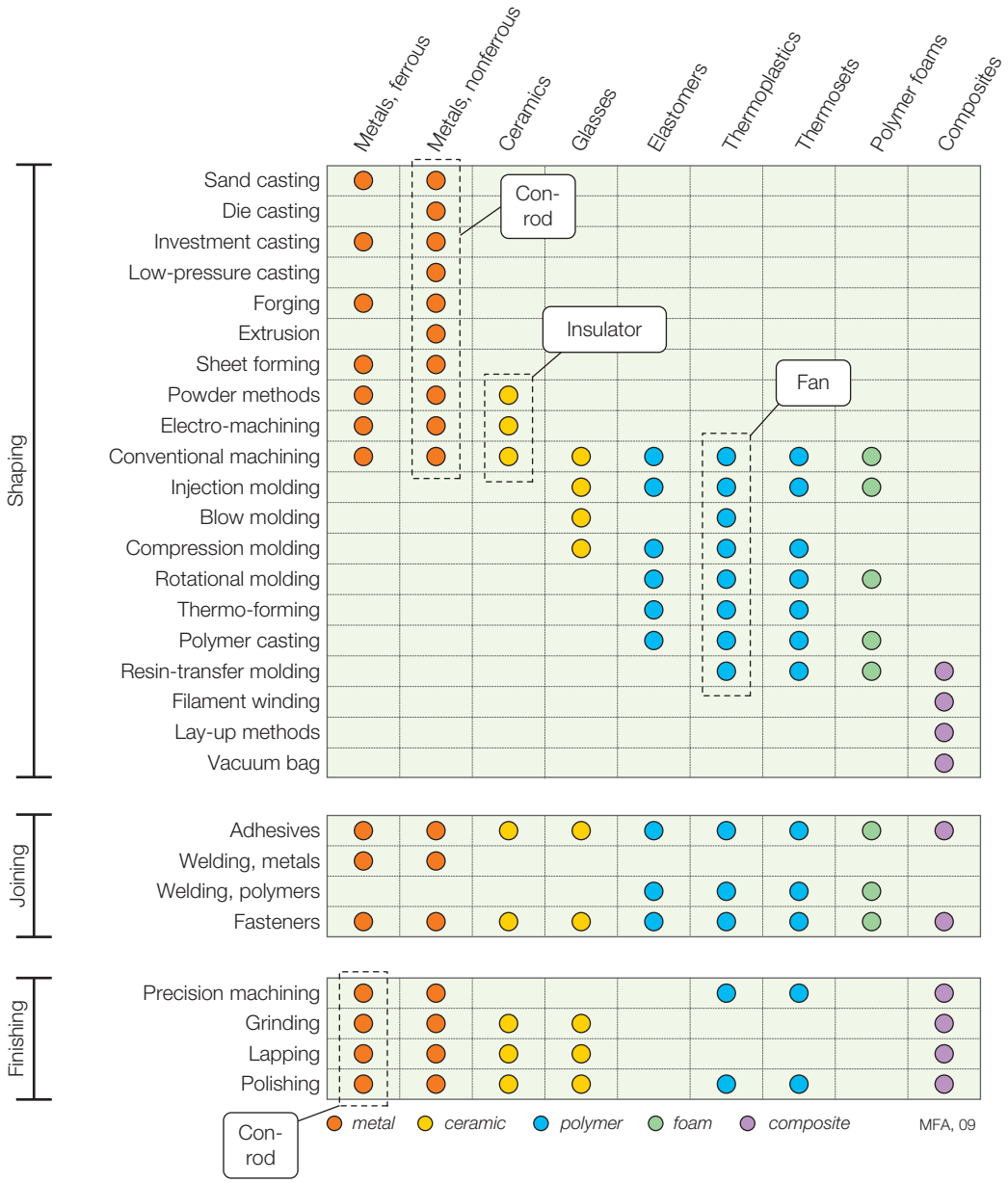


FIGURE 14.4

The process–material compatibility matrix, showing the requirements of the case studies. By including the joining and finishing processes it becomes possible to check that the more restrictive requirements can be met by combining processes.

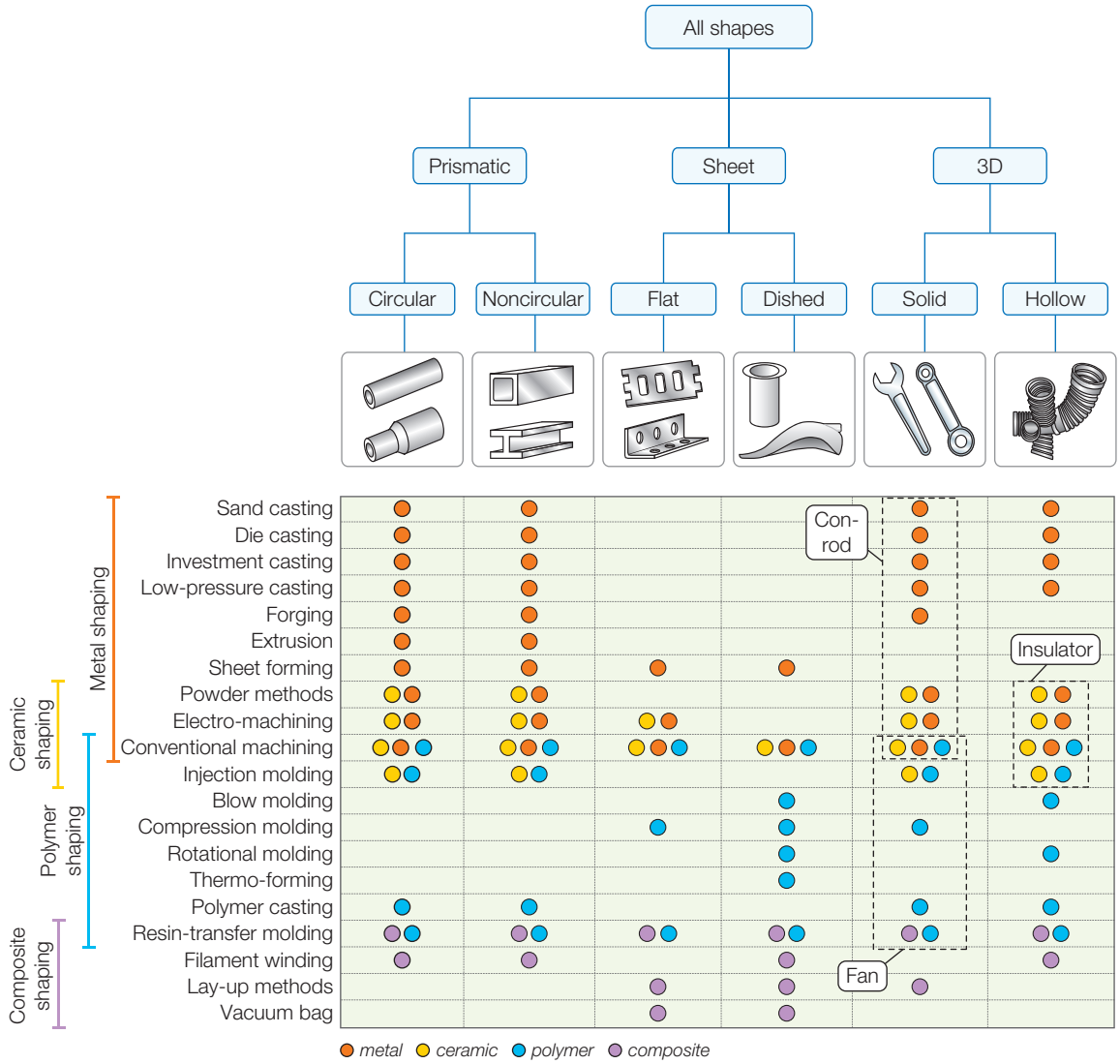


FIGURE 14.5

The process–shape compatibility matrix, showing the requirements of the case studies. A summary of the material compatibility appears at the left. The intersection of this selection stage and the last one narrows the choice.

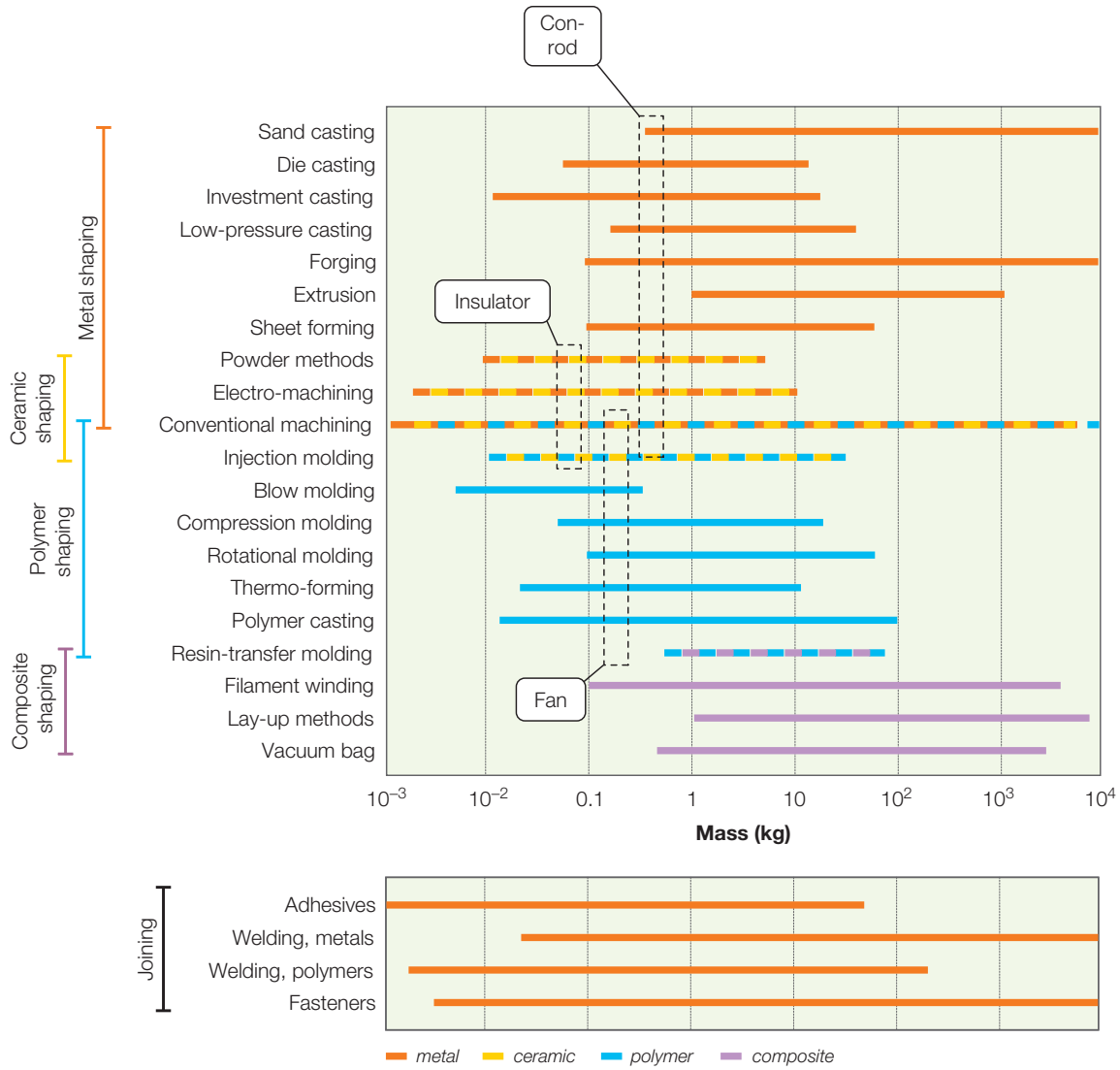


FIGURE 14.6

The process–mass range chart, showing the requirements of the three case studies. The inclusion of joining processes allows the possibility of fabrication of large structures to be explored.

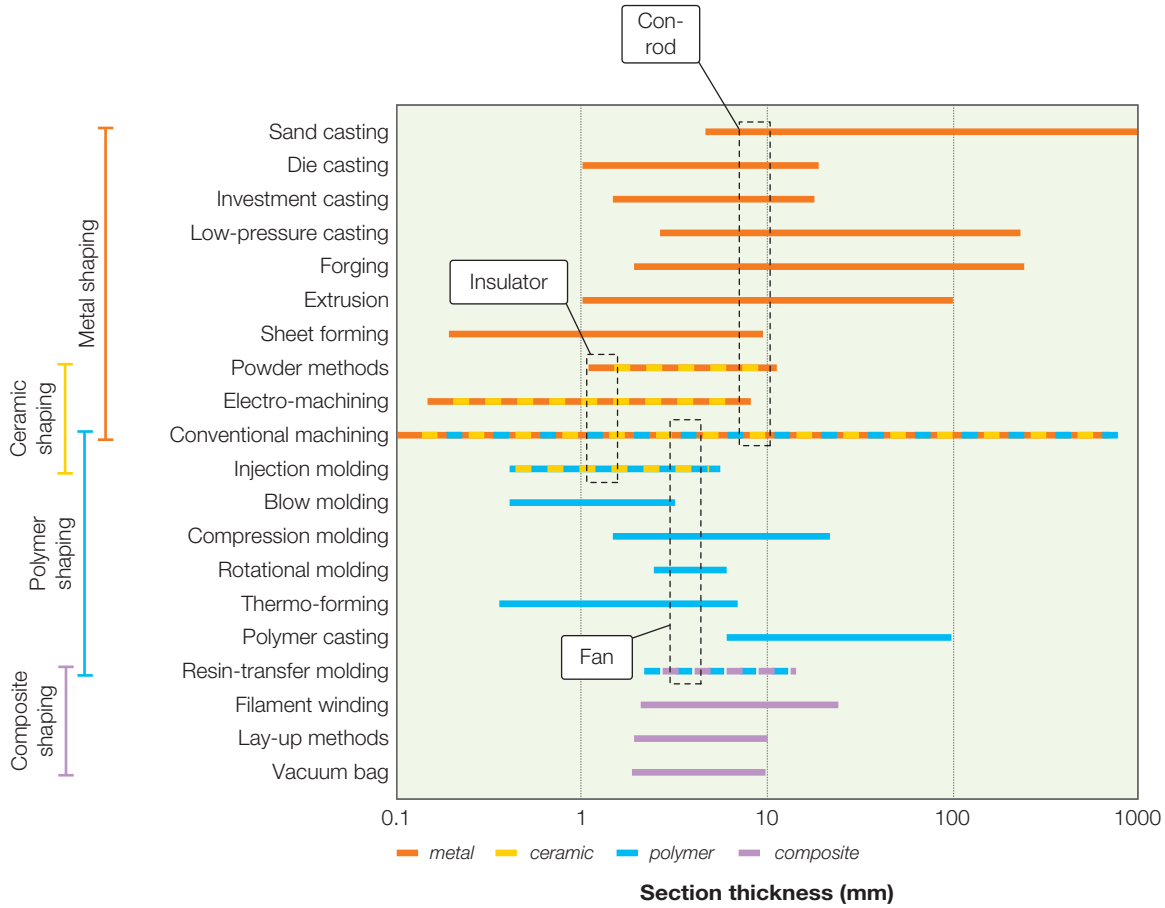


FIGURE 14.7
The process–section thickness chart, showing the requirements of the case studies.

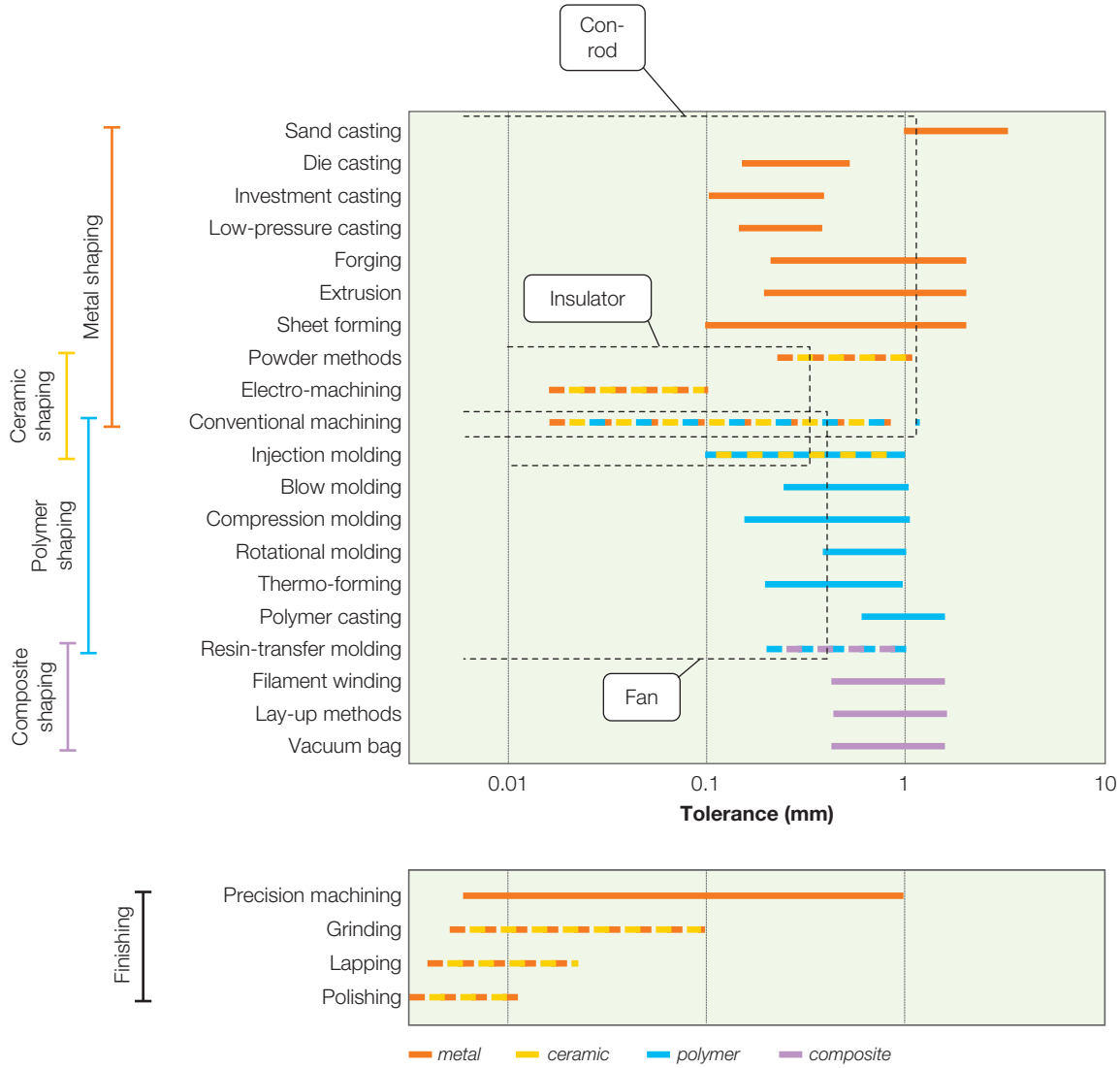


FIGURE 14.8

The process–tolerance chart, showing the requirements of the case studies. The inclusion of joining and finishing processes allows the possibility of fabrication of large structures to be explored. Tolerance and surface roughness are specified as an upper limit only, so the selection boxes (*left*) are open-ended.

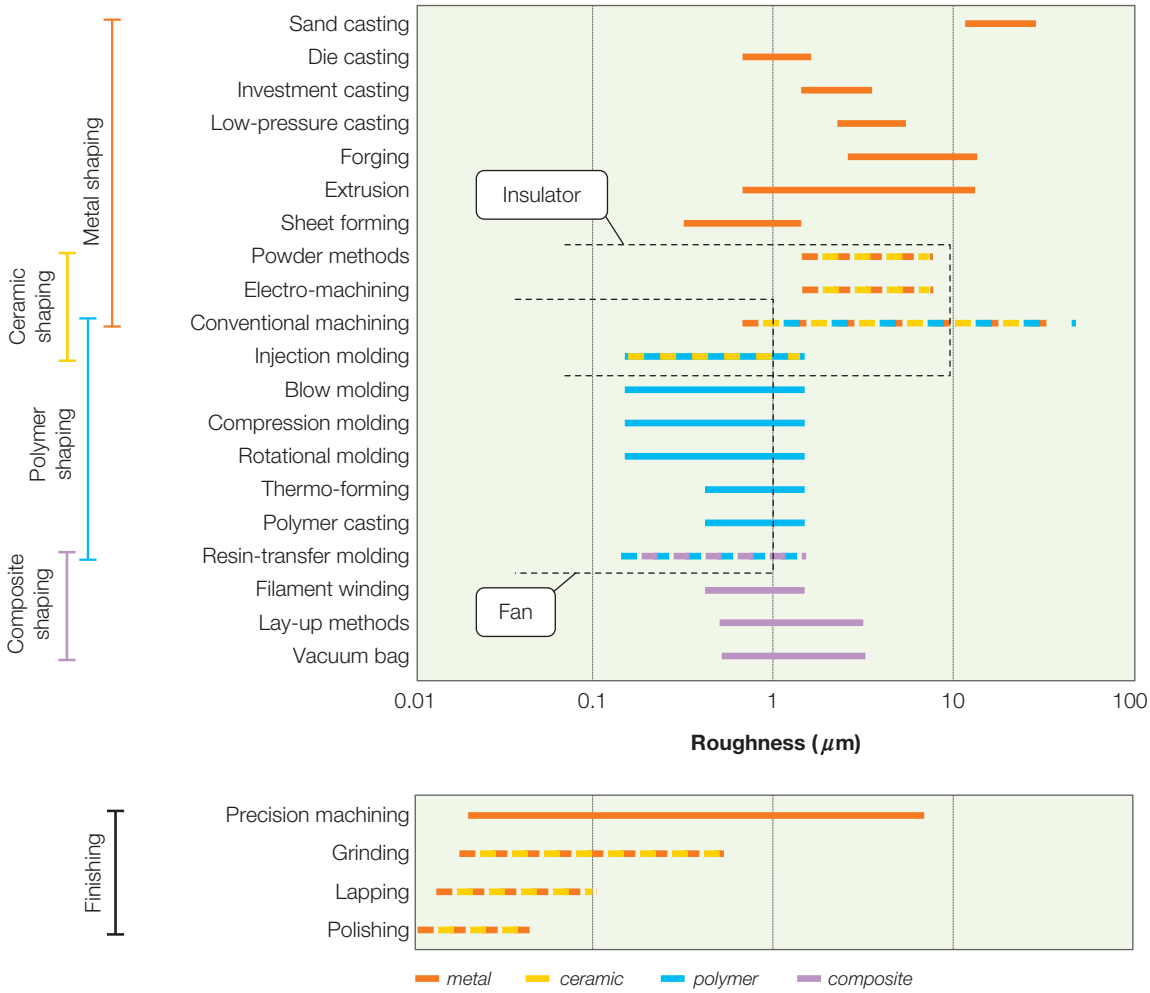


FIGURE 14.9
The process–surface roughness chart. Only one case study—the fan—imposes restrictions on this.

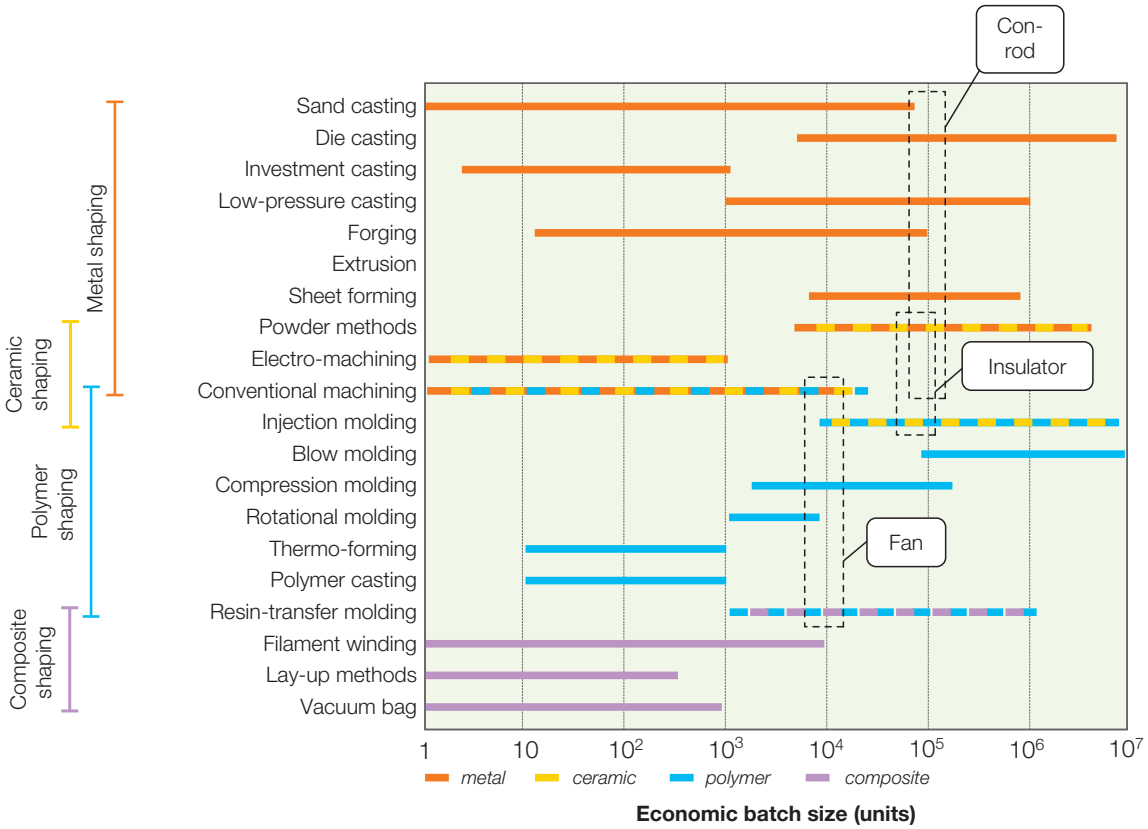


FIGURE 14.10

The process–economic batch-size chart, showing the requirements of the three case studies. The box for the optical table spans the range of possible production volumes listed in the requirements.

Table 14.5 Processes for Forming the Fan

Process	Comment
Injection molding Compression molding	Injection molding meets all the design requirements; compression molding may require further finishing operations

Table 14.6 Data for the Cost Equation for the Processes in Table 14.5

Relative Cost	Compression Molding	Injection Molding	Comment
Material, $mC_m/(1-f)$	1	1	*Process-independent parameters*
Basic overhead $\dot{C}_{oh}$ (hr^{-1})	20	20	
Capital write-off time *two* (yrs)	5	5	
Load factor	0.5	0.5	
Dedicated tool cost, C_t	2000	10,000	*Process-dependent parameters*
Capital cost, C_c	20,000	100,00	
Batch rate, $\dot{n}$ (hr^{-1})	30	150	
Tool life, n_t (number of units)	100,000	200,000	

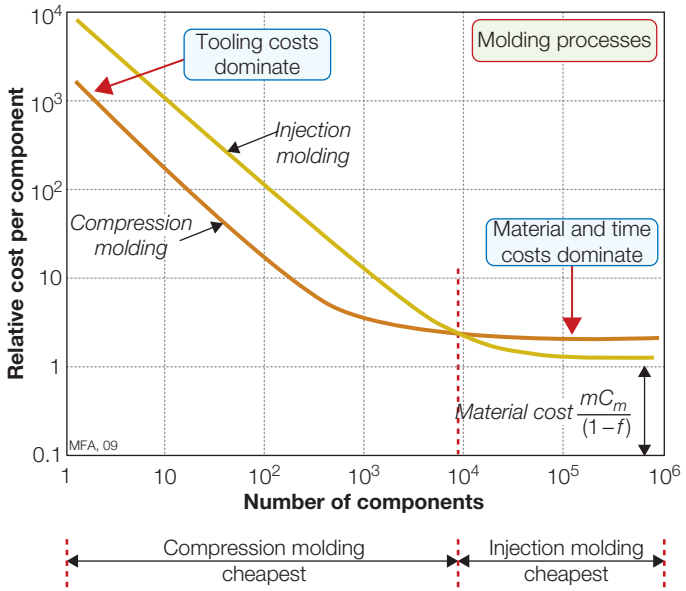


FIGURE 14.11
The relative cost of molding the fan as a function of the production run. The costs are normalized by the cost of the material.

To get further we need the cost model. The data required to implement it[1] are gathered in Table 14.6. Figure 14.11 shows the resulting cost curves. The tooling for compression molding is less expensive than that for injection molding, but it is slower because, by using multiple die cavities, injection

[1] Data from the records in the CES software for these processes.

molding can produce several units in one shot. This makes the two curves intersect at about 10,000 units: Below, compression molding is the less expensive; above, it is injection molding.

Postscript There are (as always) other considerations—the questions of capital investment, local skills, overhead rate, and so forth. The charts cannot answer these. But the procedure has been helpful in narrowing the choice, suggesting alternatives, and providing a background against which a final selection can be made.

Related case studies

6.6 "Materials for flywheels"

14.2 "Casting an aluminum con-rod"

14.4 SPARK PLUG INSULATORS

The difficulties of using hard-copy charts for process selection will, by now, be obvious: The charts have limited resolution and are clumsy to use. They give a helpful overview but they are not the way to get a definitive selection. Computer-based methods overcome both problems.

The CES system, which builds on the methods in Chapter 5, was described earlier. It allows limits to be set on material, shape, mass, section, tolerance, and surface roughness, delivering the subset of processes that meet all the limits. The economics are then examined by plotting the desired batch size on a bar chart of economic batch size, or by implementing the cost model that is built into the software. If the requirements are very demanding, no single process can meet them all. Then the procedure is to relax the most demanding of them (often tolerance and surface roughness), seeking processes that can meet the others. A second process is then sought to add the desired refinement.

The next two case studies show how the method works.

The design requirements The anatomy of a spark plug is sketched in Figure 14.12. It is an assembly of components, one of which is the insulator. This is to be made of a ceramic, *alumina*, with the shape shown in the figure: an axisymmetric, hollow, stepped shape. It weighs about 0.05 kg and has a minimum section of 1.2 mm.

Precision is important, since the insulator is part of an assembly; the design specifies a precision of $\pm$ 0.3 mm and a surface finish of better than 10 μm. The insulators are to be made in large numbers: The projected batch size is 100,000. Cost should be as low as possible. Table 14.7 summarizes the requirements.

FIGURE 14.12
A spark plug. We seek a process to make the insulator.

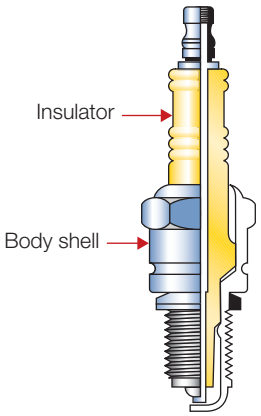


Table 14.7 Process Requirements for the Spark Plug Insulator

Function	*Insulator*	
Constraints	Material: alumina	– *Technical constraints*
	Shape: 3D hollow	
	Mass: 0.04–0.06 kg	
	Minimum section: 1.5 mm	
	Tolerance: $< \pm 0.3$ mm	– *Quality constraints*
	Surface roughness: $< 10\ \mu$m	
	Batch size: 100,000	– *Economic constraint*
Objective	Minimize cost	
Free variables	Choice of process	
	Process operating conditions	

The selection The constraints are plotted on the compatibility matrices and bar charts in Figures 14.4 through 14.10. Only one process family survives: powder methods. The required batch size of 100,000 lies within its economic range (refer to Figure 14.6).

The family of powder methods is a large one, including die pressing and sintering, hot pressing, powder extrusion, powder injection molding, hot isostatic pressing, and spray deposition. These (and others) could be added to the charts, but to do so makes them cumbersome. It is better to manage the information in a computer-based environment. The CES system does this.

Applying the constraints listed in Table 14.7 eliminates all but two of the processes in its database: die pressing and sintering and powder injection molding (PIM: the shaping of a powder by mixing it with a polymer binder, molding, then burning off the binder during the subsequent sintering). The same system implements the cost model that was described in Chapter 13.

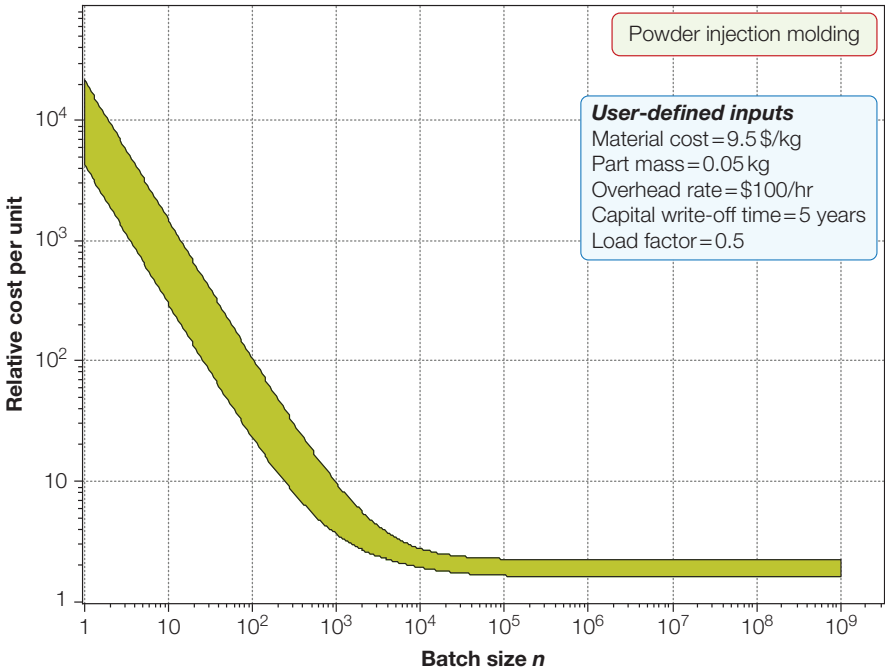


FIGURE 14.13
The unit cost as a function batch size for powder-injection molding of a small insulator. Cost is normalized by the cost of the material.

Its output, showing the cost, in units of material cost for PIM, is shown in Figure 14.13. The user-defined inputs are listed at the top right. The unit cost, high at low batch sizes, falls to about twice that of the material at batch sizes above about 10,000.

Postscript Insulators are made commercially by PIM. Economics, for a mass-produced product like this, are critical—pennies count. Technical cost modeling, described before in Chapter 13, can guide the choice of the best equipment and optimum operating conditions to minimize cost.

Related case studies

6.18 "Heat sinks for hot microchips"
14.5 "A manifold jacket"

14.5 A MANIFOLD JACKET

The design requirements The manifold jacket shown in Figure 14.14 is part of the propulsion system of a space vehicle. It is to be made of nickel. It is large, weighing about 7 kg, and complex, having an unsymmetrical 3D-hollow

FIGURE 14.14
A manifold jacket. (Redrawn from Bralla, 1986.)

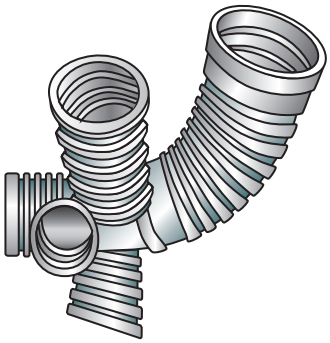


Table 14.8 Process Requirements for the Manifold Jacket

Function	Manifold jacket	
Constraints	Material: nickel	– *Technical constraints*
	Shape: 3D hollow	
	Mass: 7 kg	
	Minimum section: 2–5 mm	
	Tolerance: $< \pm 0.1$ mm	– *Quality constraints*
	Surface roughness: $< 20\ \mu$m	
	Batch size: 20	– *Economic constraint*
Objective	Minimize cost	
Free variables	Choice of process	
	Process-operating conditions	

shape. The minimum section thickness is between 2 and 5 mm. The requirements for precision and surface finish are severe (tolerance $< \pm 0.1$ mm, roughness $< 20\ \mu$m). Because of its limited application, only 20 units are to be made. Table 14.8 lists the requirements.

The charts cannot give much guidance in selecting processes to make such a complex component. We turn instead to computer-aided selection.

The selection The constraints are applied by entering them in a limit-selection dialog box like that in Figure 5.12 (*left*). The material constraint (nickel) and the shape constraint (3D hollow) eliminate a large number of processes. Figure 14.15 shows 15 that survive. The subsequent constraints on mass and section thickness eliminate more, leaving only the six that remain in color in the figure. Applying the severe constraints on tolerance and surface roughness eliminates all but one: electroforming. The required batch size lies within its range.

Postscript A search for further information in the sources listed in Appendix D reveals that electroforming of nickel is an established practice and that components as large as 20 kg are routinely made by this process. It looks like a good choice.

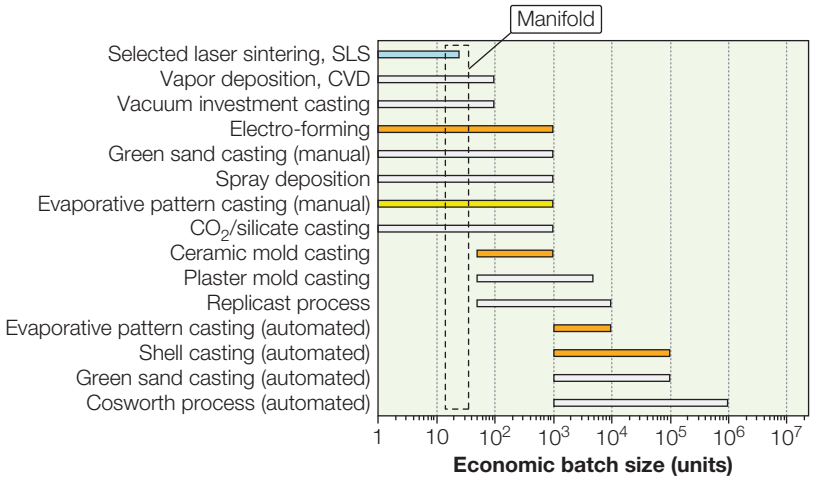


FIGURE 14.15

Output of the computer-based selection of processes to make the manifold. The grayed-out processes have failed one or more constraints.

Related case study

14.4 "Spark plug insulators"

14.6 JOINING A STEEL RADIATOR

Figure 14.16 shows a section through a domestic radiator made from corrugated pressed sheet steel. The task is to choose a joining process for the seams between the sheets.

The design requirements The process must be compatible with the material, here low carbon steel sheet of thickness 1.5 mm. The lap joints carry only low loads in service but handling during installation may impose tension and shear. They must conduct heat, be watertight, and able to tolerate temperatures up to 100°C. There is no need for the joints to be disassembled for recycling at the end of life since the whole thing is steel. Table 14.9 summarizes the translation.

The selection The CES system has records and attributes for 52 joining processes. Applying the material constraint leaves 32 of them—those eliminated are polymer specific or composite specific. Further screening on joint geometry, mode of loading, and section thickness reduces the list to 20. The requirement to conduct heat reduces this further to 12. Water resistance and operating temperature do not change the shortlist further. The processes passing the screening stage are listed in Table 14.10.

FIGURE 14.16
A section through a domestic radiator. The three pressed steel sections are joined by lap joints.

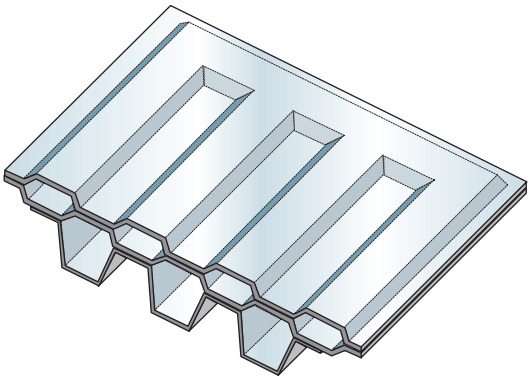


Table 14.9 Translation for Joining a Steel Radiator

Function	Domestic radiator	
Objective	Minimize cost	
Constraints	Material compatibility: low carbon steel Joint geometry: lap joint Mode of loading: tension, shear (moderate) Sheet thickness: 1–2 mm Joint must conduct heat Joint must be watertight Service temperature > 100°C Disassembly not required	} – *Technical constraints*
Free variables	Choice of shaping process Process operating conditions	

Table 14.10 Shortlist of Processes for Joining a Steel Radiator

Process	Comment
Soldering Brazing	Risk of corrosion arising from dissimilar metals in electrical contact
Laser beam welding Electron beam welding Explosive welding	Expensive, specialized processes
Metal inert gas arc welding (MIG) Tungsten inert gas arc welding (TIG) Manual metal arc welding (MMA) Oxyacetylene welding Projection welding Seam welding	Well-established, conventional welding processes

Postscript At this point we seek documentation for the processes. This reveals that explosive welding requires special facilities and permits (hardly a surprise). Electron and laser welding require expensive equipment, so use of a shared facility might be necessary to make them economic. Resistance spot welding is screened out because it failed the requirement to be watertight, although that requirement only applies to the edge seams.

Quality constraints are not severe, but distortion and residual stress will make fit-up of adjacent joints difficult. Low carbon steels are readily weldable, so it is unlikely that welding will cause cracking or embrittlement—but this can be checked by seeking documentation.

14.7 SURFACE-HARDENING A BALL-BEARING RACE

The balls of ball races run along grooved tracks (Figure 14.17). The life of a ball race is limited by wear and fatigue. Both are suppressed by using hard materials. Hard materials, however, are not tough, incurring the risk that shock loading or mishandling might cause the race to fracture.

The design requirements The solution is to use an alloy steel, which has excellent bulk properties, and to apply a separate surface treatment to increase hardness where it matters. We therefore seek processes to surface-harden alloy steels for wear and fatigue resistance. The precision of both balls and race is critical, so the process must not compromise the dimensions or the surface smoothness. Table 14.11 summarizes the translation.

The selection The CES system contains records for 46 surface treatment processes. Many are compatible with alloy steels. More discriminating is the purpose of the treatment—to impart fatigue and wear resistance—reducing the list to 8. Imposing the requirement for a very smooth surface knocks out processes that coat or deform the surface because these compromise the finish. Adding the further constraint that the curved surface coverage must be *good* or *very good* leaves just five, listed in Table 14.12. The records for these describe and illustrate the process, and summarize their typical uses.

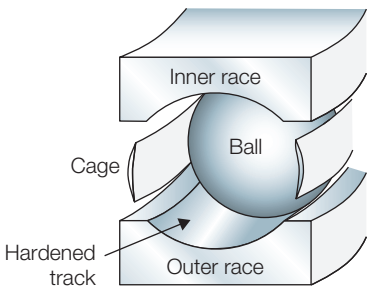


FIGURE 14.17
A section through a ball race. The surfaces of the race are to be hardened to resist wear and fatigue crack nucleation.

Table 14.11 Translation for Surface-hardening a Ball-bearing Race

Function	Ball-bearing race	
Objective	Minimize cost	
Constraints	Material: alloy steel	– *Technical constraints*
	Function of treatment: hardening for fatigue and wear resistance	
	Curved surface coverage: good	
	Precision and surface finish do not compromise	– *Quality constraint*
Free variables	Choice of surface treatment process Process operating conditions	

Table 14.12 Shortlist of Processes for Hardening an Alloy Steel Ball Race

Process	Comment
Carburizing Carbo-nitriding Nitriding Aluminizing Boriding Chromizing	All are high-temperature, diffusion-controlled processes, and so are slow. Documentation in the records gives details.

Postscript To get further we seek documentation for these processes. The hardness of the surface and the depth of the hardened layer depend on process variables: the time and temperature of the treatment and the composition of the steel in ways that are tabulated. And economics, of course, enters the equation. Ball races are made in enormous numbers and while their sizes vary, their geometry does not. This is where dedicated equipment, even if very expensive, is viable.

14.8 SUMMARY AND CONCLUSIONS

Process selection, at first sight, looks like a black art: The initiated know; the rest of the world cannot even guess how they do it. But this—as the chapter demonstrates—is not really so. The systematic approach, developed in Chapter 13, and illustrated here, identifies a subset of viable processes using design information only: size, shape, complexity, precision, roughness, and material—itself chosen by the systematic methods in Chapter 5. It does not identify the single, best choice; that depends on too many case-specific considerations. But, by identifying candidates, it directs the user to data sources (starting with those listed in Appendix D) that provide the details needed to make a final selection.

CHAPTER 15

Materials and the Environment



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15.1 INTRODUCTION AND SYNOPSIS

All human activity has some impact on the environment. The environment has some capacity to cope with this so that a certain level of impact can be absorbed without lasting damage. But it is clear that current human activities exceed this threshold with increasing frequency, diminishing the quality of the world in which we now live and threatening the well-being of future generations. The manufacture and use of products, with their associated consumption of materials and energy, are among the culprits. The position is dramatized by the following statement: At a global growth rate of 3% per year we will mine, process, and dispose of more "stuff" in the next 25 years than in the entire industrial history of mankind.

Design for the environment is generally interpreted as the effort to adjust our present design methods to correct known, measurable, environmental degradation; the time scale of this thinking is 10 years or so, an average product's expected life. *Design for sustainability* is the longer view: that of adaptation to a lifestyle that meets present needs without compromising the needs of future generations. The time scale here is less clear—it is measured in decades or centuries—and the adaptation required is much greater. This chapter focuses on the role of materials and processes in achieving design for the environment. Sustainability requires social and political changes that are beyond the scope of this book.

The ideas in this chapter are developed further in the text that appears as the first entry in "Further reading."

15.2 THE MATERIAL LIFE-CYCLE

The nature of the problem is brought into focus by examining the material life-cycle shown in the sketch in Figure 15.1. Ore and feedstock, most of them nonrenewable, are processed to give materials; these are manufactured into products that are used and, at the end of their lives, disposed of, a fraction perhaps entering a recycling loop, the rest committed to incineration or landfill. Energy and materials are consumed at each point in this cycle (we shall call them "phases"), with an associated penalty of CO_2 and other emissions—heat and gaseous, liquid, and solid waste.

The problem, crudely put, is that the sum of these unwanted by-products now often exceeds the capacity of the environment to absorb them. Some of the injury is local in scale and its origins can be traced and remedial action taken. For some the scale is national, for others it is global, and here remedial action has wider social and organizational prerequisites. Much present environmental legislation aims at modest reductions in the damaging

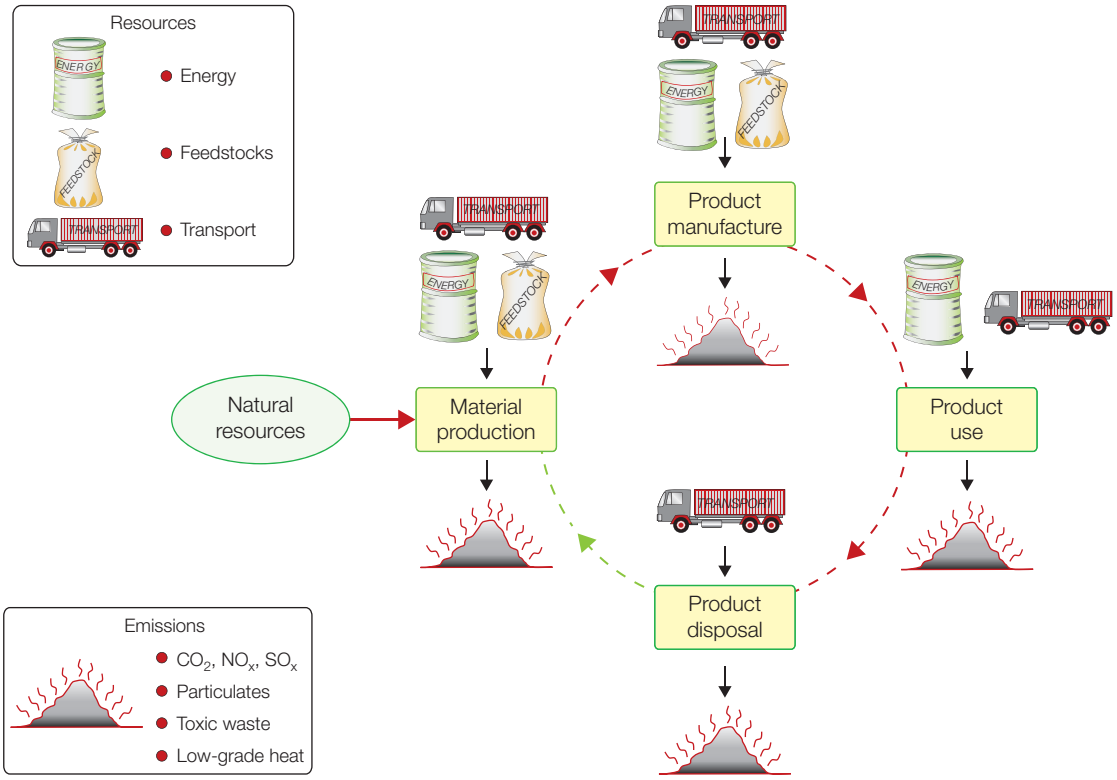


FIGURE 15.1

The material life-cycle. Ore and feedstock are mined and processed to yield a material. These are manufactured into a product that is used and, at the end of its life, discarded or recycled. Energy and materials are consumed in each phase, generating waste heat and solid, liquid, and gaseous emissions.

activity; a regulation requiring a 20% reduction in—say—the average gasoline consumption of passenger cars is seen by car makers as a major challenge.

Sustainability requires solutions of a completely different kind. Even conservative estimates of the adjustment needed to restore long-term equilibrium with the environment envisage a reduction in the flows of Figure 15.1 by a factor of four or more. Population growth and the growth of the expectations of this population more than cancel any modest savings that the developed nations may achieve. It is here that the challenge is greatest, requiring difficult adaptation; it is one for which no generally agreed-on solutions yet exist. But it remains the long-term driver of eco-design, to be retained as background to any creative thinking.

15.3 MATERIAL AND ENERGY-CONSUMING SYSTEMS

It would seem that the obvious ways to conserve materials are to make products smaller, make them last longer, and recycle them when they finally reach the end of their lives. But the seemingly obvious can sometimes be deceptive. Materials and energy form part of a complex and highly interactive system, as illustrated in Figure 15.2. Here primary catalysts of consumption such as *new technology*, *planned obsolescence*, *increasing wealth*, *education*, and *population growth* influence aspects of product use and through these the consumption of materials and energy and the by-products that these produce. The connecting lines indicate influences; a green line suggests positive, broadly desirable influence; a red line suggests negative, undesirable influence; and red-green suggests that the driver has the capacity for both positive and negative influence.

The diagram brings out the complexity. Follow, for instance the lines of influence of *new technology* and its consequences. It offers more material and energy-efficient products, but by also offering new functionality it creates obsolescence and the desire to replace a product that has useful life left in it. Electronic products are prime examples of this: Most are discarded while

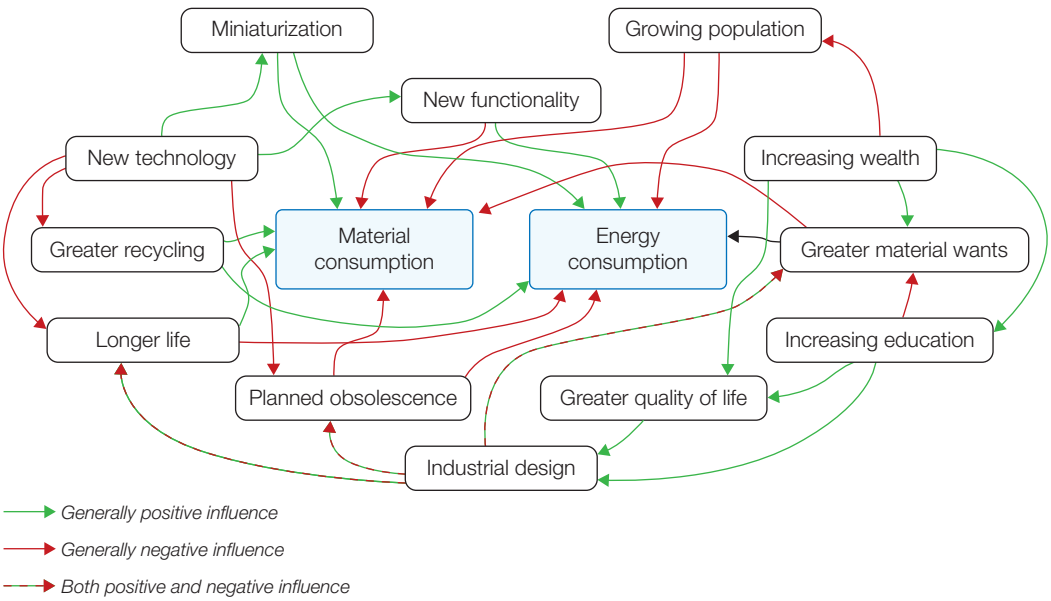


FIGURE 15.2

The influences on consumption of materials and energy. It is essential to see eco-design as a systems problem, not solved by simply choosing "good" and avoiding "bad" materials but rather by matching the material to the system requirements.

still functional. And observe, even at this simple level, the consequences of *longer life*—a seemingly obvious measure. It can certainly help to conserve materials (a positive influence) but, in an era in which new technology delivers more energy-efficient products (particularly cars, electronics, and household appliances today), extending the life of old products can have a negative influence on energy consumption.

As a final example, consider the bivalent influence of industrial design—the subject of Chapter 16. The lasting designs of the past are evidence of industry's ability to create products that are treasured and conserved. But today industrial design is frequently used as a potent tool to stimulate consumption by deliberate obsolescence, creating the perception that "new" is desirable and that even slightly "old" is unappealing.

The use patterns of products Table 15.1 suggests a matrix of product use patterns. Those in the first row require energy to perform their primary function. Those in the second could function without energy but, for reasons of comfort, convenience, or safety, consume energy to provide a secondary function. Those in the third row provide their primary function without any need for energy other than human effort. The load factor across the top is an approximate indicator of the intensity of use—one that will, of course, vary widely.

The choice of materials and processes influences all the phases of Figure 15.1: *production*, through the drainage of resources and the undesired by-products of refinement; *manufacture*, through the level of efficiency and cleanness of the shaping, joining, and finishing processes; *use*, through the ability to conserve energy through lightweight design, higher thermal efficiency, and lower energy consumption; and finally *disposal* through a greater ability to allow reuse, disassembly, and recycling.

Table 15.1 Use Matrix of Product Classes

	High Load Factor	Modest Load *P* Factor	Low Load Factor
Primary power-consuming	Family car Train set Aircraft	Television Freezer	Coffee maker Vacuum cleaner Washing machine
Secondary power-consuming	Housing (heat, light)	Parking lot (light)	Household dishes Clothing (washing)
Nonpower-consuming	Bridges Roads	Furniture Bicycle	Canoe Tent

Energy intense ↕ *Material intense*

High impact ↔ *Low impact*

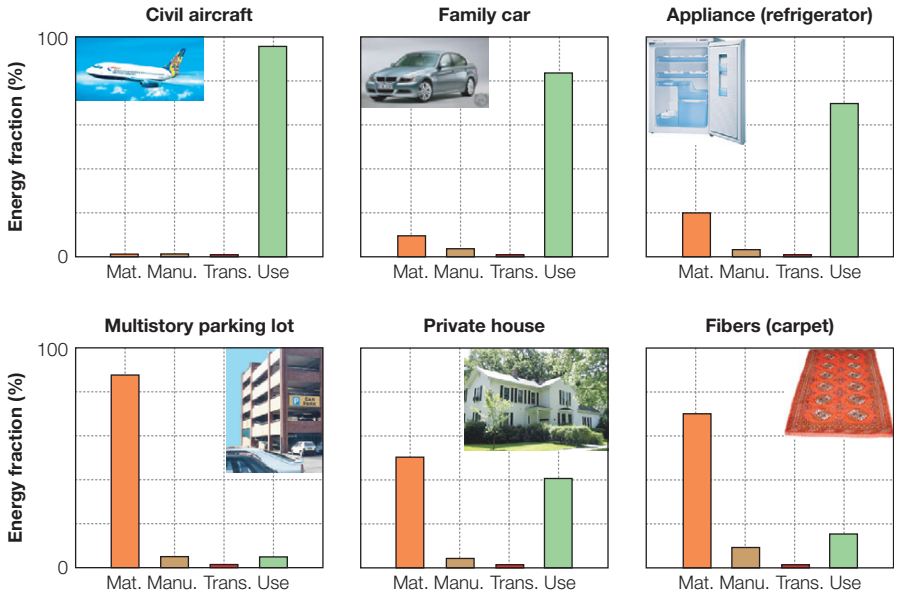


FIGURE 15.3
Approximate values for the energy consumed at each phase of Figure 15.1 for a range of products.

It is generally true that one of the four phases of Figure 15.1 dominates the picture. Simplifying for a moment, let us take *energy consumption* as a measure of both the inputs and undesired by-products of each phase and use it for a character appraisal of use sectors. Figure 15.3 presents the evidence, using this measure. It has two significant features, with important implications. The first feature is that one phase almost always dominates, accounting for 80% or more of the energy, sometimes much more. If large changes are to be achieved, it is this phase that must be the target; a reduction by a factor of 2, even of 10, in any other makes little significant difference to the total. The second feature: When differences are as great as those of Figure 15.3, precision is not the issue—an error of a factor of 2 changes very little. It is the nature of people who *measure* things to wish to do so with precision, and precise data must be the ultimate goal. But it is possible to move forward without precision: Precise judgments can be drawn from imprecise data. This is an important consideration: Much information about eco-attributes is imprecise.

15.4 THE ECO-ATTRIBUTES OF MATERIALS

Material production: Energy and emissions

Most of the energy consumed in the four phases of Figure 15.1 is derived from fossil fuels. Some is consumed in that state—as gas, oil, coal, or coke. Much is first converted to electricity at a European average conversion

efficiency of about 38%. Not all electricity is generated from fossil fuels—there are contributions from hydroelectric, nuclear, and wind/wave sources. But with the exception of Norway (70% hydro) and France (80% nuclear), the predominant energy sources are fossil fuels; and since the national grids of European countries are linked, with power flowing from one to another as needed, it is adequate here to think of a European average fossil-fuel energy per kilowatt of delivered electrical power.

The fossil-fuel energy consumed in making one kilogram of material is called its *embodied energy*. Some of the energy is stored in the created material and can be reused, in one sense or another, at the end of life. Polymers made from oil (as most are) contain energy in another sense—that of the oil that enters the production as a primary feedstock. Natural materials such as wood similarly contain "intrinsic" or "contained" energy, this time derived from solar radiation absorbed during growth. Views differ on whether intrinsic energy should be included in embodied energy or not. There is a sense in which not only polymers and woods but also metals carry intrinsic energy that could—by chemical reaction or by burning the metal in the form of finely divided powder—be recovered, so omitting it when reporting production energy for polymers but including it for metals seems inconsistent. For this reason we will include intrinsic energy from nonrenewable resources in reporting embodied energies, which generally lie in the range of 25 to 250 MJ/kg, though a few are much higher. The existence of intrinsic energy has another consequence: The energy to recycle a material is sometimes much less than that required for its first production, because the intrinsic energy is retained. Typical values lie in the range 10 to 100 MJ/kg.

The production of 1 kilogram of material is associated with *undesired gas emissions*, among which CO_2, NO_x, SO_x, and CH_4 cause general concern (global warming, acidification, ozone-layer depletion). The quantities can be large—each kilogram of virgin aluminum produced with energy from fossil fuels creates some 9 kilograms of CO_2, 40 grams of NO_x, and 90 grams of SO_x (Table 15.2). Material production is generally associated with other undesirable outputs, particularly toxic wastes and particulates, but these can, in principle, be dealt with at the source.

Approximate data for embodied energy and CO_2 burden of materials are listed in Appendix A.

Estimates for material processing energies (at an overall conversion efficiency of 15%)

Many processes depend on casting, evaporation, or deformation. It is helpful to have a feel for the approximate magnitudes of energies required by these. The use of primary energy (thought of as the equivalent quantity of oil) for

Table 15.2 The Eco-attributes of One Grade of Wrought 1,000-grade Aluminum Alloy

Material Production: Energy and Emissions	
Embodied energy	195–210 MJ/kg
Carbon dioxide	9–10 kg/kg
Nitrogen oxides	72–79 g/kg
Sulphur oxides	120–140 g/kg
Material Processing Energy	
Casting energy	2.5–2.8 MJ/kg
Minimum energy to vaporization	17–19 MJ/kg
Minimum energy to 90% deformation	2.55–2.8 MJ/kg
End of Life	
Downcycle	Yes
Recycle	Yes
Biodegrade	No
Incinerate	No
Landfill	Acceptable
Recycling energy	17–20 MJ/kg
Recycle fraction	52–58%

processing materials involves several energy conversion steps, each with a conversion efficiency that is less than 1. Many processes use electric energy, generated from primary energy with a conversion efficiency on the order of 38%. The use of this for electroheating or for electroprocessing (like electroforming) involves further losses. Scrap, discarded material, and other wastes carry with them an energy investment that does not appear in the "good" casting, moldings, or forgings that are the output of the processes.

Other process routes involve other energy conversion sequences, all with their losses. It is possible, as we do below, to estimate the energy to melt, mold, vaporize, or deform a material, but to express this in terms of primary energy it must be divided by the product of the conversion efficiencies of each conversion step. For the sake of example we set this overall efficiency at a (realistic) 15%.

Melting To melt a material, it must first be heated to its melting point, requiring a minimum input of the heat $C_p\,(T_m - T_0)$, and then caused to melt, requiring the latent heat of melting L_m:

$$H_{\min} = C_p\,(T_m - T_0) + L_m \tag{15.1}$$

where $H_{\min}$ is the minimum energy per kilogram for melting, C_p is the specific heat, T_m is the melting point, and T_0 is the ambient temperature. A close correlation exists between L_m and $C_p\,T_m$:

$$L_m \approx 0.4\, C_p T_m \tag{15.2}$$

and for metals and alloys $T_m >> T_0$, giving

$$H_{\min} \approx 1.4\, C_p T_m \tag{15.3}$$

Assuming efficiency of 15%, the estimated energy to melt 1 kilogram H_m^* is

$$H_m^* \approx 8.4\, C_p T_m \tag{15.4}$$

the asterisk recalling that it is an estimate. For metals and alloys, the quantity H_m^* lies in the range 1 to 8 MJ/kg.

Vaporization As a rule of thumb, the latent heat of vaporization L_v is larger than that for melting, L_m, by a factor of 24 ± 5, and the boiling point T_b is larger than the melting point T_m by a factor 2.1 ± 0.5. Using the same assumptions as before, we find an estimate for the energy to evaporate 1 kg of material (as in PVD processing) to be

$$H_v^* \approx 76\, C_p T_m \tag{15.5}$$

again assuming an efficiency of 15%. For metals and alloys, the quantity H_v^* lies in the range 6 to 60 MJ/kg.

Deformation Deformation processes like rolling or forging generally involve large strains. Assuming an average flow strength of $(\sigma_y + \sigma_{uts})/2$, a strain of order 1, and an efficiency factor of 15%, we find, the work of deformation per kg, to be

$$W_D^* \approx 3(\sigma_y + \sigma_{uts}) \tag{15.6}$$

where σ_y is the yield strength and σ_{uts} is the tensile strength. For metals and alloys, the quantity W_D^* lies in the range 0.05 to 2 MJ/kg.

We conclude that casting or deformation requires processing energies that are small compared to the production energy of the material being processed, but the larger energies required for vapor-phase processing may become comparable with those for material production.

End of life

Figure 15.4 introduces the options: landfill, combustion for heat recovery, recycling, reengineering, and reuse.

Landfill Much of what we now reject is committed to landfill. Already there is a problem—the land available to "fill" in this way is already, in some European countries, almost full. Administrations react by charging a landfill tax—currently somewhere near €50 per tonne and rising, seeking to divert waste into the other channels of Figure 15.4.

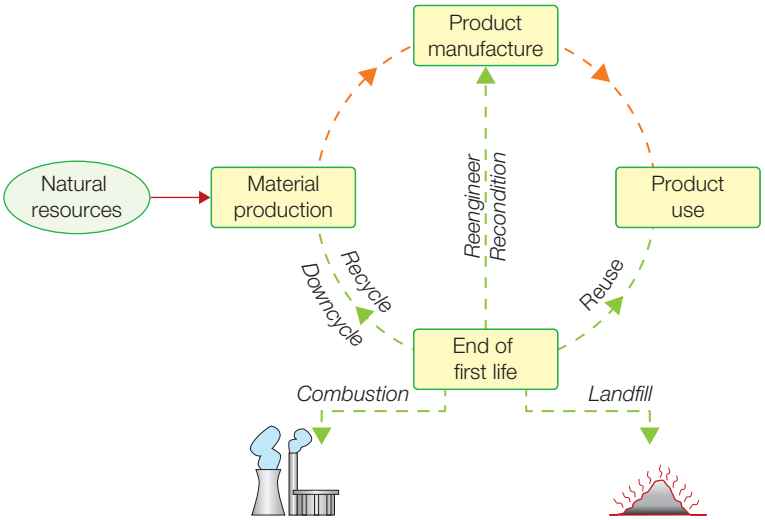


FIGURE 15.4
End of life options: landfill, combustion, recycling, refurbishment or upgrading, and reuse.

Combustion for heat recovery Materials, we know, contain energy. Rather than throw them away it would seem better to retrieve and reuse some of their energy by controlled combustion, capturing the heat. But this is not as easy as it sounds. First there is the need for a primary sorting to separate combustibles from noncombustible material. Then the combustion must be carried out under conditions that do not generate toxic fumes or residues, requiring high temperatures, sophisticated control, and expensive equipment. Energy recovery is imperfect partly because it is incomplete and partly because the incoming waste carries a moisture content that has to be boiled off.

Recycling Recycling requires energy, and this energy carries its burden of gases. But the *recycle energy* is generally small compared to the initial production energy, making recycling—when it is possible at all—an energy-efficient proposition. It may not, however, be one that is cost-efficient; that depends on the degree to which the material has become dispersed. In-house scrap, generated at the point of production or manufacture, is localized and is already recycled efficiently (near 100% recovery). Widely distributed "scrap"—material contained in discarded products—is much more expensive to collect, separate, and clean. Many materials cannot be recycled, although they may still be reused in a lower-grade activity; continuous-fiber composites, for instance, cannot be reseparated economically into fiber and polymer in order to recycle them, though they can be chopped and used as fillers. Most other materials require an input of virgin material to avoid build-up of uncontrollable impurities. Thus the

fraction of a material's production that can ultimately reenter the cycle of Figure 15.4 depends both on the material itself and on the product into which it has been incorporated.

Reengineering Reengineering is the refurbishment or upgrading of the product or of its recoverable components. Certain criteria must be met to make it practical. One is that the design of the product is fixed or that the technology on which it is based is evolving so slowly that there remains a market for the restored product. Some examples are housing, office space, and road and rail infrastructure; these are sectors with enormous appetites for materials.

Reuse Reuse is the redistribution of the product to a consumer sector that is willing to accept it in its used state, perhaps to reuse it for its original purpose (e.g., a second-hand car), perhaps to adapt it to another (converting a car into a hot rod or a bus into a mobile home). Charity stores pass on clothing, books, and objects acquired from those for whom they had become waste; the stores sell them to others who perceive them to have value. Reuse is the most benign of the end-of-life scenarios.

15.5 ECO-SELECTION

To select materials to minimize impact on the environment we must first ask—as we did in Section 15.2—which phase of the life-cycle of the product under consideration makes the largest contribution. The answer guides the choice of strategy to ameliorate that contribution (Figure 15.5). The strategies are described in the following sections.

The material production phase If material production is the dominant phase of life, this becomes the first target. Drink containers (Figure 15.6) provide an example: They consume materials and energy during material extraction and production, but apart from transport and refrigeration, which are small, not thereafter. We use the energy consumed in extracting and refining the material (the embodied energy noted in Table 15.2) as the measure; CO_2, NO_x, and SO_x emissions are related to it, although not in a simple way. The energy associated with the production of 1 kilogram of a material is H_p, that per unit volume is $H_p\rho$, where ρ is the density of the material.

The bar charts of Figures 15.7 and 15.8 show these two quantities for ceramics, metals, polymers, and composites (hybrids). On a "per kg" basis (upper chart), glass, the material of the first container, carry the lowest penalty. Steel is slightly higher. Polymers carry a much higher energy penalty than steel. Aluminum and the other light alloys carry the highest penalty of all. But if these same materials are compared on a "per m^3" basis (lower chart) the conclusions change: Glass is still the lowest, but now commodity polymers, such

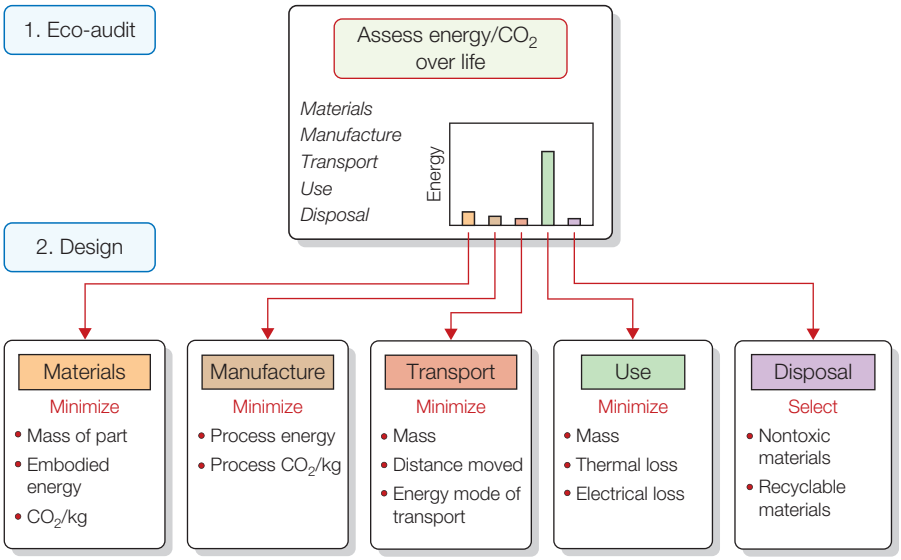


FIGURE 15.5
Rational design for the environment starts with an analysis of the phase of life to be targeted. This decision then guides the method of selection to minimize the impact of the phase on the environment.



FIGURE 15.6
Liquid containers: glass, polyethylene, PET, aluminum, and steel. All can be recycled. Which carries the lowest production energy penalty?

as PE and PP, carry a *lower* burden than steel; the composite GFRP is only a little higher. But is comparison "per kg" or "per m^3" the right way to do it? Rarely. To deal with environmental impact at the production phase properly we must seek to minimize the energy, the CO_2 burden, or the eco-indicator value *per unit of function.*

Performance indices that include energy content are derived in the same way as those for weight or cost (Chapter 5). As an example, consider the

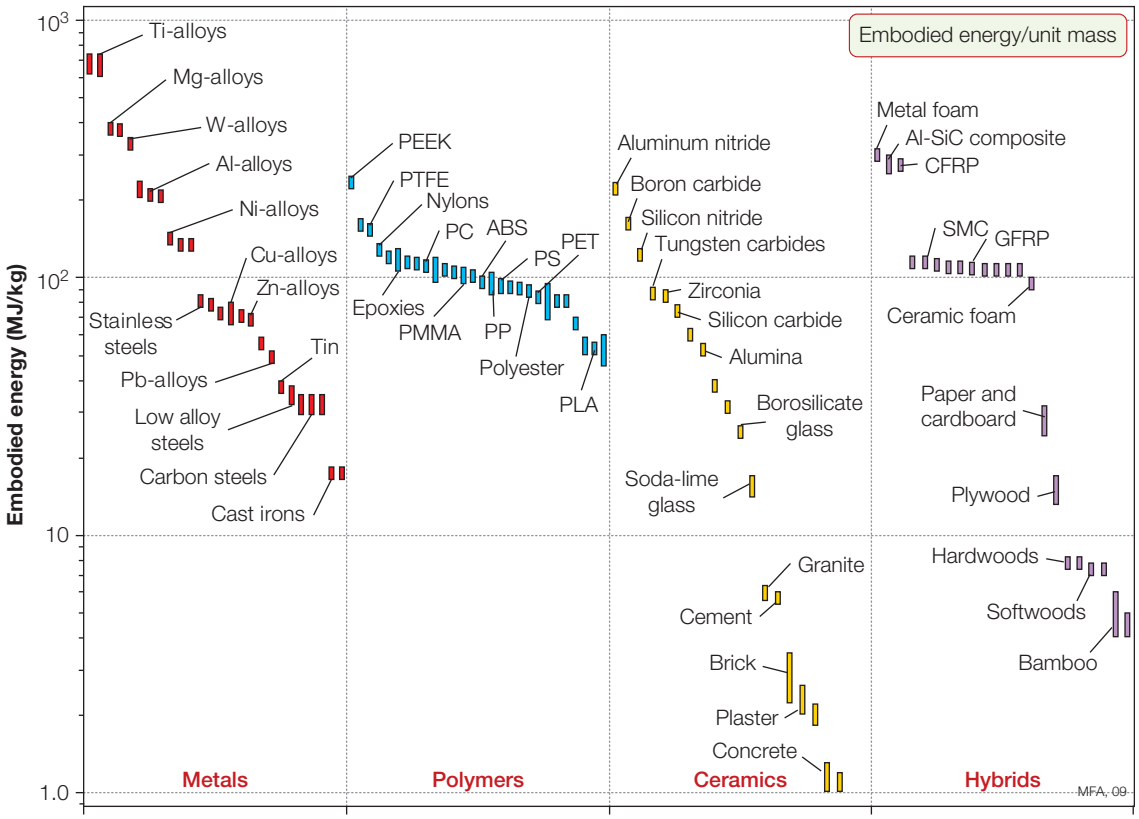


FIGURE 15.7
The embodied energy per unit mass of materials (data in Appendix A).

selection of a material for a beam that must meet a stiffness constraint at minimum energy content. Repeating the derivations of Chapter 5 but with the objective of minimizing embodied energy rather than mass leads to performance equations and material indices that are simply those of Chapter 5 with ρ replaced by $H_p\rho$. Thus the best materials to minimize the embodied energy of a beam of specified stiffness and length are those with large values of the index

$$M_1 = \frac{E^{1/2}}{H_p\rho} \tag{15.7}$$

where E is the modulus of the material of the beam. The stiff tie of minimum energy content is best made of a material of high $E/H_p\rho$; the stiff plate, of a material with high $E^{1/3}/H_p\rho$, and so on.

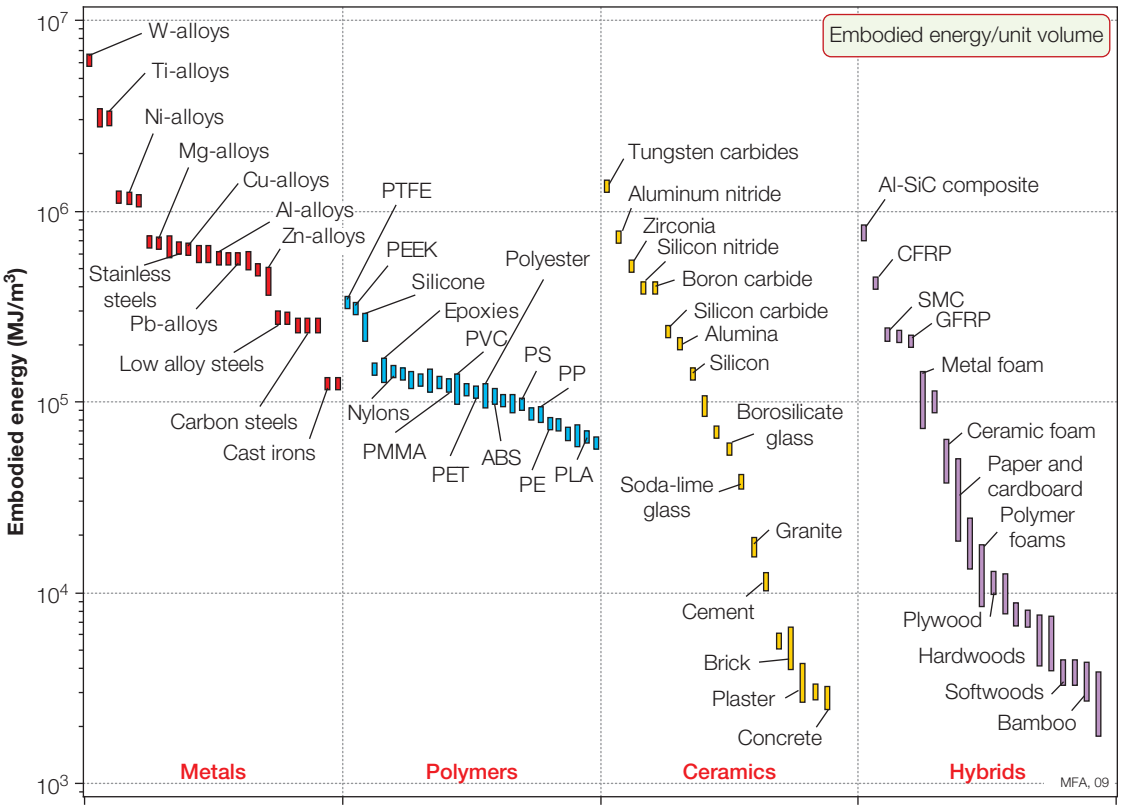


FIGURE 15.8
The embodied energy per unit volume of materials.

Strength works in the same way. The best materials for a beam of specified bending strength and minimum embodied energy are those with large values of

$$M_3 = \frac{\sigma_f^{2/3}}{H_p \rho} \tag{15.8}$$

where σ_f is the failure strength of the beam material. Other indices follow in a similar way.

Figures 15.9 and 15.10 are a pair of charts for selection to minimize embodied energy H_p per unit of function (similar charts for CO_2 burden can be made using the CES Edu software). The first chart shows modulus E plotted against $H_p\rho$; the guide lines give the slopes for three of the commonest performance indices. The second shows strength σ_f (defined in

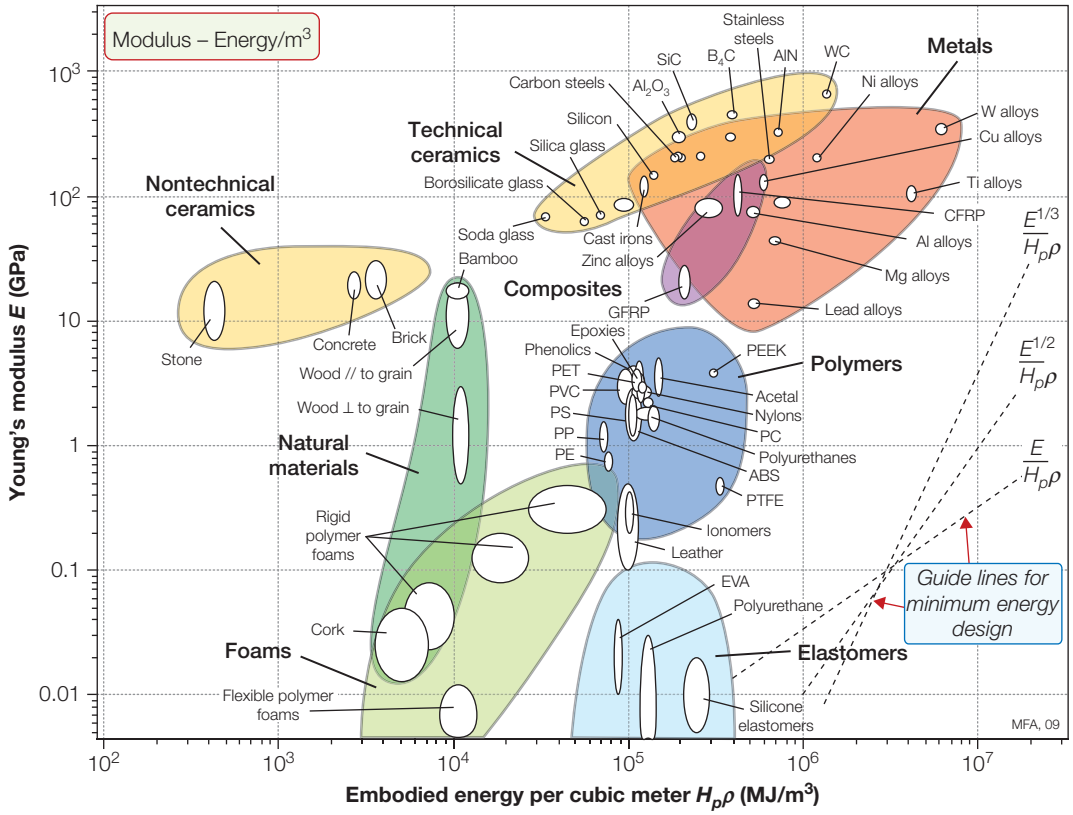


FIGURE 15.9
A selection chart for stiffness with minimum production energy. It is used in the same way as Figure 4.3.

Chapter 4) plotted against $H_p\rho$; again, guide lines give the slopes. The two charts give survey data for minimum energy design. They are used in exactly the same way as the $E - \rho$ and $\sigma_f - \rho$ charts for minimum mass design.

Most polymers are derived from oil. This leads to statements that they are energy-intense, with implications for their future. The two charts in Figures 5.9 and 5.10 show that, per unit of function in bending (the commonest mode of loading), most polymers carry a lower energy penalty than primary aluminum, magnesium, or titanium, and several are competitive with steel. Most of the energy consumed in the production of metals such as steel, aluminum, a magnesium is used to reduce the ore to the elemental metal, so that these materials, when recycled, require much less energy. Efficient collection and recycling makes important contributions to energy saving.

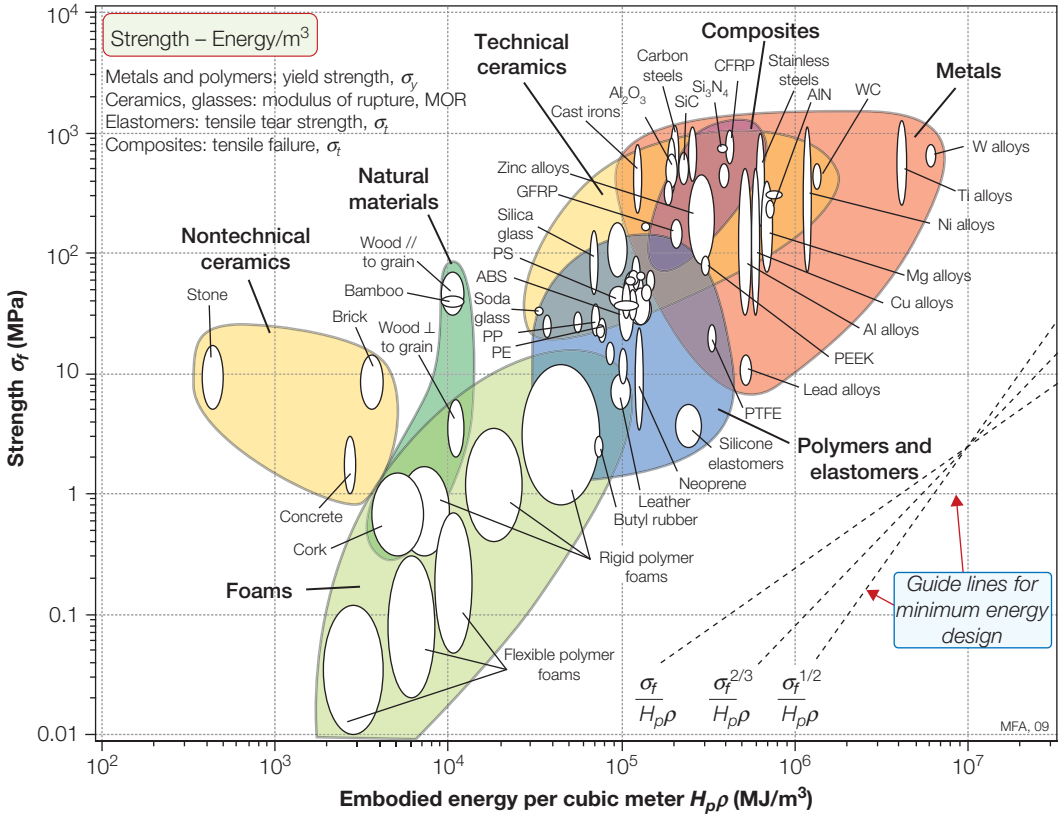


FIGURE 15.10
A selection chart for strength with minimum production energy. It is used in the same way as Figure 4.4.

The product manufacture phase Shaping materials, as we discussed earlier, requires energy. Certainly it is important to save energy in production. But higher priority often attaches to the *local* impact of emissions and toxic waste during manufacture, and this depends crucially on local circumstances. Paper making (to take an example) uses very large quantities of water. Historically the wastewater was heavily polluted with alkalis and particulates, devastating the river systems into which it was dumped. Today, the best paper mills discharge water that is as clean and pure as it was when it entered. Production sites of the former communist-bloc countries are terminally polluted; those producing the same materials elsewhere, using best-practice methods, have no such problems. Clean manufacture is the issue here.

The use phase The eco-impact of the use phase of energy-consuming products has nothing to do with the energy content of the materials

themselves—indeed, minimizing this may frequently have the opposite effect on use energy. Use energy depends on mechanical, thermal, and electrical efficiencies; it is minimized by maximizing these. Fuel efficiency in transport systems (measured, say, by MJ/km) correlates closely with the mass of the vehicle itself; the objective then becomes minimizing mass. Energy efficiency in refrigeration or heating systems is achieved by minimizing the heat flux into or out of the system; the objective is then minimizing thermal conductivity or thermal inertia. Energy efficiency in electrical generation, transmission, and conversion is maximized by minimizing the ohmic losses in the conductor; here the objective is minimizing electrical resistance while meeting necessary constraints on strength, cost, and so on. Selection to meet these objectives is exactly what the previous chapters of this book were about.

The product disposal phase The environmental consequences of the final phase of product life has many aspects. Increasingly, legislation dictates disposal procedures, take-back, and recycle requirements, and—through landfill taxes and subsidized recycling—deploys market forces to determine the end-of-life choice.

15.6 CASE STUDIES: DRINK CONTAINERS AND CRASH BARRIERS

The energy content of containers

The containers shown earlier in Figure 15.6 are examples of products for which the first and second phases of life—material production and product manufacture—consume the most energy and generate the most emissions. Thus material selection to minimize energy and consequent gas and particle emissions focuses on these. Table 15.3 summarizes the requirements.

The masses of five competing container types, the material of which they are made, and the specific energy content of each are listed in Tables 15.4 and 15.5. Their production involves molding or deformation; approximate energies for each are listed. All five materials can be recycled. Which container type carries the lowest overall energy penalty per unit of fluid contained?

Table 15.3 Design Requirements for the Containers

Function	Container for cold drink
Constraint	Must be recyclable
Objective	Minimize embodied energy per unit capacity
Free variable	Choice of material

Table 15.4 Details of the Containers

Container Type	Material	Mass, g	Mass/ Liter, g	Energy/ Liter, MJ/Liter
PET 400-ml bottle	PET	25	62	5.4
PE 1-liter milk bottle	High-density PE	38	38	3.2
Glass 750-ml bottle	Soda glass	325	433	8.2
Al 440-ml can	5000 series Al alloy	20	45	9.0
Steel 440-ml can	Plain carbon steel	45	102	2.4

Table 15.5 Data for the Materials of the Containers (from Appendix A)

Material	Embodied Energy, MJ/kg	Forming Method	Forming Energy, MJ/kg
PET	84	Molding	3.1
PE	81	Molding	3.1
Soda glass	15.5	Molding	4.9
5000 series Al alloy	210	Deep drawing	0.13
Plain carbon steel	32	Deep drawing	0.15

The method and results A comparison of the energies in Table 15.5 shows that the energy to shape the container is always less than that to produce the material in the first place. Only in the case of glass is the forming energy significant. The dominant phase is material production. Summing the two energies for each material and multiplying by the container mass per liter of capacity gives the ranking shown in the second to last column of Table 15.4. Steel can carry the lowest energy penalty, glass and aluminum the highest.

Crash barriers

Barriers to protect the driver and passengers in road vehicles are of two types: those that are static—the central divider of a freeway, for instance—and those that move—the bumper of the vehicle itself (Figure 15.11). The static types line tens of thousands of miles of road. Once in place they consume no energy, create no CO_2, and last a long time. The dominant phases of their life in the sense of the life-cycle shown in Figure 15.1 are material production and manufacture. The bumper, by contrast, is part of the vehicle; it adds to its weight and thus to its fuel consumption.

The dominant phase here is use. This means that, if eco-design is the objective, the criteria for selecting materials for the two sorts of barrier will differ. We take as a criterion for the first that of maximizing the energy that the

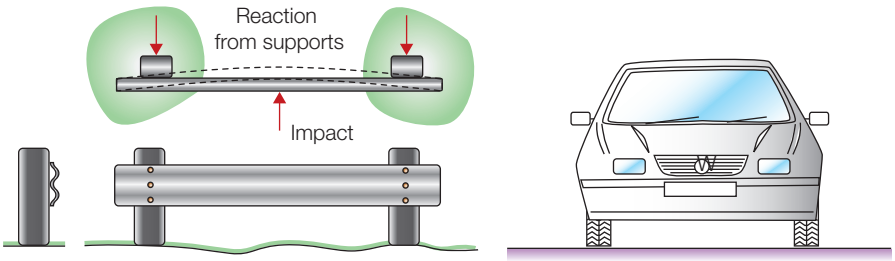


FIGURE 15.11
Two crash barriers: one static, the other—the bumper—attached to something that moves. Different eco-criteria are needed to select materials for each.

Table 15.6 Design Requirements for the Crash Barriers

Function	Energy-absorbing crash barriers
Constraint	Must be recyclable
Objectives	Maximize energy absorbed per unit production energy, or Maximize energy absorbed per unit mass
Free variable	Choice of material

barrier can absorb per unit of production energy; for the second we take absorbed energy per unit mass. Table 15.6 summarizes.

In an impact, the barrier is loaded in bending (Figure 15.11). Its function is to transfer load from the point of impact to the support structure, where reaction from the foundation or from crush elements in the vehicle support or absorb it. To do this the material of the barrier must have high strength, σ_f, be adequately tough and able to be recycled. That of the static barrier must meet these constraints with *minimum embodied energy* as the objective, since this will reduce the overall life energy most effectively. We know from Section 15.5 that this means materials with large values of the index

$$M_3 = \frac{\sigma_f^{2/3}}{H_p \rho} \tag{15.9}$$

where σ_f is the yield strength, ρ the density, and H_p the embodied energy per kg of material. For the car bumper it is mass, not embodied energy, that is the problem. If we change the objective to *minimum mass*, we require materials with high values of the index

$$M_4 = \frac{\sigma_f^{2/3}}{\rho} \tag{15.10}$$

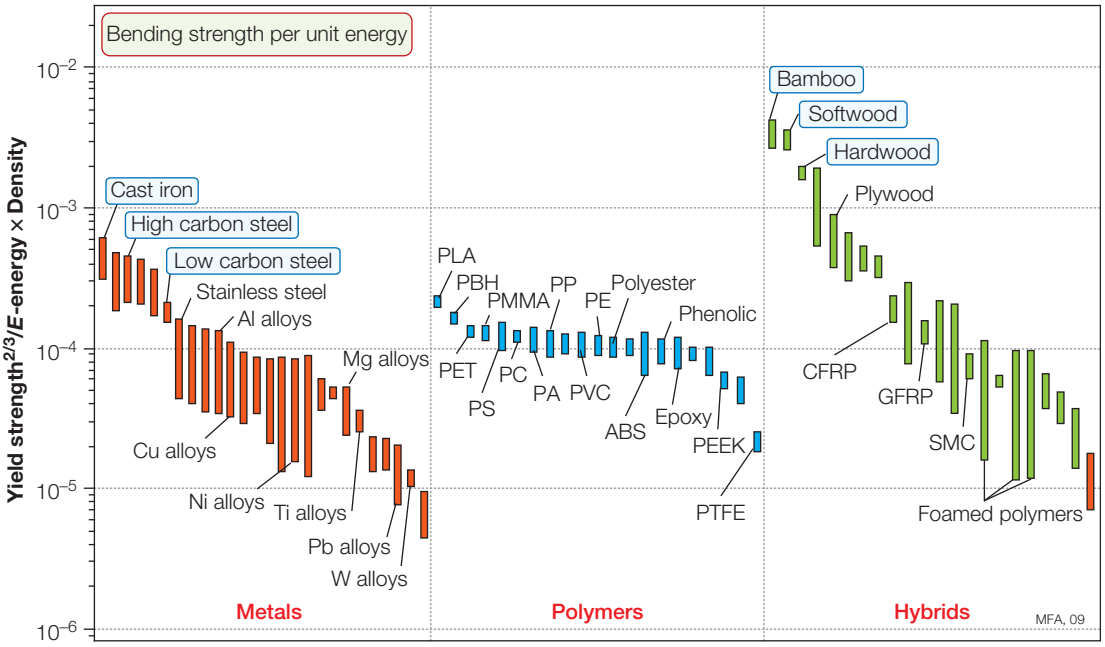


FIGURE 15.12

Material choice for the static barrier; the units are $(MPa)^{2/3}/(MJ/m^3)$. Cast irons, carbon steels, low alloy steels, and wood are the best choices.

These indices can be plotted on the charts shown in Figure 15.10 (see also Figure 4.4), enabling a selection. We leave that as one of the exercises in order to show here an alternative: simply plotting the index itself as a bar chart. Figures 15.12 and 15.13 show the result for metals, polymers, and polymer-matrix composites. The first guides the selection for static barriers. It shows that embodied energy (for a given load-bearing capacity) is minimized by making the barrier out of carbon steel or cast iron or wood; nothing else comes close. The second figure guides selection for the mobile barrier. Here CFRP (continuous fiber carbon epoxy, for instance) excels in its strength per unit weight, but it is not recyclable. Heavier but recyclable are alloys of magnesium, titanium, and aluminum. Polymers, which rank poorly in the first figure, now become candidates—even without reinforcement, they can be as good as steel.

Postscript Metal crash barriers have a profile like that shown in Figure 15.11 (*left*). The curvature increases the second moment of area of the cross-section and through this the bending stiffness and strength. This is an example of combining material choice and section shape (Section 9.5) to optimize a design.

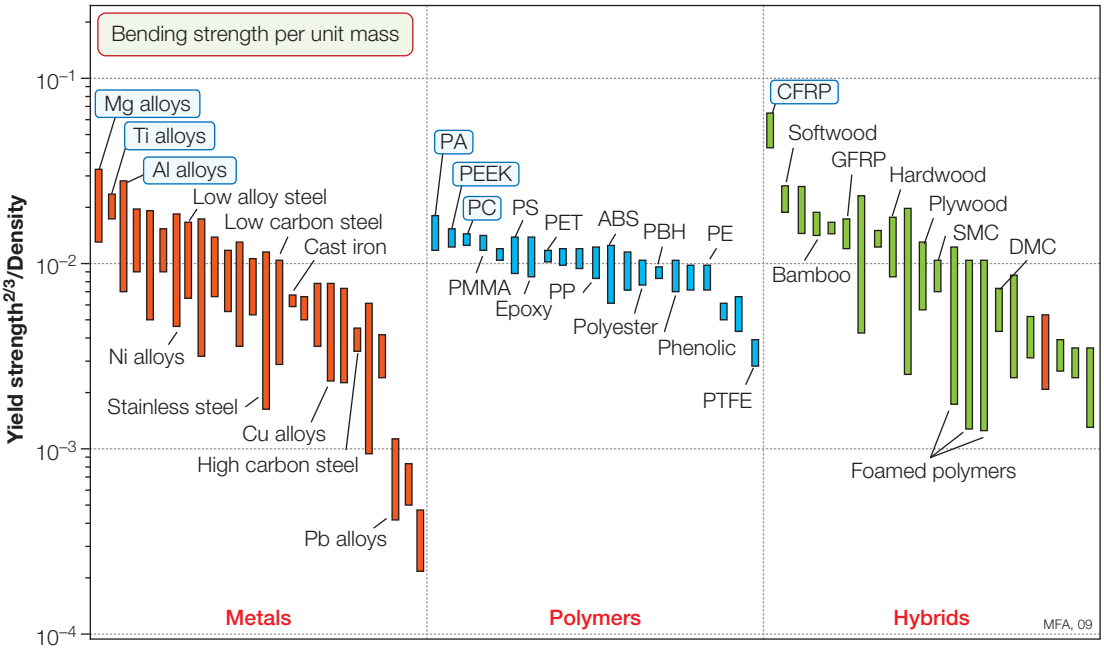


FIGURE 15.13
Material choice for the mobile barrier; the units are $(MPa)^{2/3}/(kg/m^3)$. CFRP and light alloys offer the best performance; nylon and polycarbonate perform as well as steel.

15.7 SUMMARY AND CONCLUSIONS

Rational selection of materials to meet environmental objectives starts by identifying the phase of product life that causes the greatest concern: production, manufacture, use, or disposal. Dealing with all of these requires data not only for the obvious eco-attributes (e.g., energy, CO_2 and other emissions, toxicity, and the ability to be recycled) but also for mechanical, thermal, electrical, and chemical properties. Thus if material production is the phase of concern, selection is based on minimizing production energy or the associated emissions (e.g., CO_2 production). But if it is the use phase that is of concern, selection is based instead on light weight and excellence as a thermal insulator or as an electrical conductor (while meeting other constraints on stiffness, strength, cost, etc.).

This chapter developed methods to deal with these issues. The methods are most effective when implemented in software. The CES Edu system, described in Chapters 5, 6, 13, and 14 contains eco-data for materials and includes an eco-audit tool to analyze product life in the manner of Figure 15.3.

If you found this chapter interesting and would like to read more, you will find the ideas it contains developed more fully in the first book listed in the following section.

15.8 FURTHER READING

Ashby, M.F. (2009). *Materials and the environment*. Butterworth-Heinemann, ISBN 978-1-85617-608-8.
A teaching text that provides the resources—background, methods, data—to enable environmental issues relating to materials to be explored in depth.

CES Edu (2010). *The Cambridge Engineering Selector*. Granta Design.
The material selection platform now has an optional eco-design module. www.grantadesign.com.

Fuad-Luke, A. (2002). *The eco-design handbook*. Thames and Hudson, ISBN 0-500-28343-5.
A remarkable sourcebook of examples, ideas, and materials of eco-design.

Goedkoop, M.J., Demmers, M., & Collignon, M.X. (1995). *Eco-indicator '95 manual*. PRé Consultants and The Netherlands Agency for Energy and the Environment, ISBN 90-72130-80-4.

Goedkoop, M., Effting, S., & Collignon, M. (2000). *The Eco-indicator 99: A damage oriented method for life cycle impact assessment*. Manual for Designers.
PRé Consultants market a leading life-cycle analysis tool and are proponents of the eco-indicator method. http://www.pre.nl.

ISO 14001 (1996) and ISO 14040 (1997, 1998, 1999). Environmental management system-specification with guidance for use. International Organization for Standardization (ISO), Geneva.

Kyoto Protocol (1997). Environmental management—life cycle assessment (and subsections). Framework Convention on Climate Change. Document FCCC/CP1997/7/ADD.1. United Nations, Geneva.
An agreement among the developed nations to limit their greenhouse gas emissions relative to the levels emitted in 1990. The United States agreed to reduce emissions from 1990 levels by 7% during the period 2008–2012; European nations adopted a more stringent agreement.

Lovins, L.H., von Weizsäcker, E., & Lovins, A.B. (1998). *Factor four: Doubling wealth, halving resource use*. Earthscan, ISBN 1-853834-068.
An influential book arguing that resource productivity can and should grow fourfold—that is, the wealth extracted from one unit of natural resources can quadruple, allowing the world to live twice as well yet use half as much.

MacKay, D.J.C. (2008). *Sustainable energy—without the hot air*. Cambridge University: Department of Physics, *www.withouthotair.com/*.
MacKay brings a welcome dose of common sense into the discussion of energy sources and use. Fresh air replacing hot air.

Mackenzie, D. (1997). *Green design: Design for the environment* (2nd ed.). Lawrence King Publishing, ISBN 1-85669-096-2.
A lavishly produced compilation of case studies of eco-design in architecture, packaging, and product design.

Meadows, D.H., Meadows, D.L., Randers, J., & Behrens, W.W. (1972). *The limits to growth—1st report of the club of Rome*. Universe Books, ISBN 0-87663-165-0.
A pivotal publication, alerting the world to the possibility of resource depletion, undermined by the questionable quality of the data used for the analysis; despite that, it was the catalyst for subsequent studies and for views that are now more widely accepted.

Schmidt-Bleek, F. (1997). *How much environment does the human being need—factor 10—the measure for an ecological economy*. Deutscher Taschenbuchverlag.
The author argues that true sustainability requires a reduction in energy and resource consumption by the developed nations by a factor of 10.

Wenzel, H., Hauschild, M., & Alting, L. (1997). *Environmental assessment of products, Vol. 1*. Chapman and Hall, ISBN 0-412-80800-5.
Professor Alting leads Danish research in eco-design.

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CHAPTER 16

Materials and Industrial Design





Two cars that are about the same in power and price but very different in style, comfort, and sophistication. How is it that they can compete in the same market? It has to do with aesthetics, association, and perception.

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16.1 INTRODUCTION AND SYNOPSIS

Good design works. Excellent design also gives pleasure.

Pleasure derives from form, color, texture, feel, and the associations that these invoke. Pleasing design says something about itself; generally speaking, honest statements are more satisfying than deception, though eccentric or humorous designs can be appealing too.

Materials play a central role in this. A major reason for introducing new materials is the greater freedom of design that they allow. Metals, during the past century, allowed the building of structures that could not have been built before: cast iron, the Crystal Palace; wrought iron, the Eiffel Tower; drawn steel cable, the Golden Gate Bridge—all undeniably beautiful. Polymers lend themselves to bright colors, satisfying textures, and great freedom of form; they have opened new styles of design, of which some of the best examples are found in household appliances: Food mixers, hair dryers, mobile phones, MP3 players, and vacuum cleaners make extensive and imaginative use of materials to allow styling, weight, feel, and form that give pleasure.

Those who concern themselves with this aesthetic dimension of engineering are known, rather confusingly, as "industrial designers." This chapter introduces some of the ideas of industrial design, emphasizing the role of materials. It ends with case studies. But first a word of caution.

Previous chapters dealt with systematic ways of choosing material and processes. "Systematic" means that if *you* do it and *I* do it, following the same procedure, we will get the same result, and that result, next year, will be the same as it is today. Industrial design is not, in this sense, systematic. Succes, here involves sensitivity to fashion, custom, and educational background, and is influenced (even manipulated) by advertising and association. The views in this chapter are partly those of writers who seem to me to say sensible things,

and partly my own. You may not agree with them, but if they make you think about designing to give pleasure, the chapter has done what it should.

The ideas in this chapter are developed more fully in the first text listed in "Further reading."

16.2 THE REQUIREMENTS PYRAMID

The pen with which I am writing this chapter cost $5 (Figure 16.1, *upper image*). If you go to the right shop, you can find a pen that costs well over $1000 (*lower image*). Does it write 200 times better than mine? Unlikely; mine writes perfectly well. Yet there is a market for expensive pens. Why?

A product has a *cost*—the outlay in manufacture and marketing it. It has a *price*—the sum at which it is offered to the consumer. And it has a *value*—a measure of what the consumer thinks it is worth. The expensive pens command the price they do because the consumer perceives their value to justify it. What determines value? Three things.

Functionality, provided by sound technical design, clearly plays a role. The requirements pyramid in Figure 16.2 has this as its base: The product must work properly and be safe and economical. Functionality alone is not enough: The product must be easy to understand and operate, and these are questions of *usability*, the second tier in the figure. The third, completing the pyramid, is the requirement that the product give *satisfaction*: that it enhances the life of those who own or use it.



FIGURE 16.1
Pens, inexpensive and expensive. The chosen material—acrylic in the upper two; gold, silver, and enamel in the lower two—creates the aesthetics and the associations of the pens. (*Bottom* image courtesy of David Nishimura, *Vintagepens.com*.)

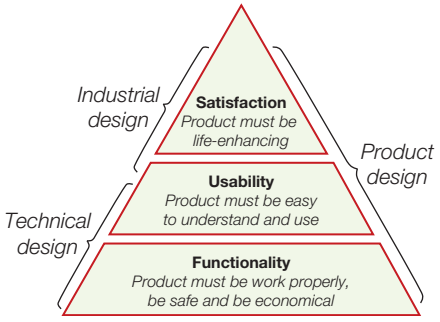


FIGURE 16.2
The requirements pyramid. The lower part of the pyramid tends to be called "technical design," the upper part, "industrial design," suggesting that they are separate activities. It is better to think of all three tiers as part of a single process that we shall call "product design."

The value of a product is a measure of the degree to which it meets (or exceeds) the expectation of the consumer for all three of these—functionality, usability, and satisfaction. Think of this as the product. It is very like human character. An admirable character is one that functions well, interacts effectively, and is rewarding company. An unappealing character is one that does none of these. An odious character is one that behaves in such an unattractive way that you cannot bear to be near them.

Products are the same. All the pens in Figure 16.1 function well and are easy to use. The huge difference in price implies that the lower two provide a degree of satisfaction not offered by the upper two. The most obvious difference between them is the materials of which they are made—the upper pair of molded acrylic, the lower pair of gold, silver, and enamel. Acrylic is the material of toothbrush handles, something you throw away after use. Gold and silver are the materials of precious jewelry; they have associations of craftsmanship, of heirlooms passed from one generation to the next. Well, that's part of the difference, but there is more. To find it we need to address a question—what creates product character?

16.3 PRODUCT CHARACTER

Figure 16.3 shows a way of dissecting product character. It is a map of the ideas we are going to explore; like all maps there is a lot of detail, but we need it to find our way. In the center is information about the *product* itself: the basic design requirements, its function, its features. The way these are thought through and developed is conditioned by the *context*, shown in the circle above the product. The context is set by the answers to the questions in the top box: *Who? Where? When? Why?* Consider the first of these: *Who?* A designer seeking to create a product attractive to women will make choices that differ from those for a product intended for children, or for elderly people, or for sportsmen. *Where?* A product for use in the home requires a different choice of material and form than one to be used—say—in a school or hospital. *When?* One intended for occasional use is designed in a different way from one used all the time; one for formal occasions differs from one for informal use. *Why?* A product that is primarily utilitarian involves different design decisions from one that is largely a lifestyle statement. The context influences and conditions all the decisions that the designer makes in finding a solution. It sets the *mood*.[1]

On the left of the *product* circle of Figure 16.3 lie packages of information about the *materials* and the *processes* used to shape, join, and finish it. Each

[1] Many designers working on a project assemble a *mood board* with images of the sort of people for whom the product is intended, the surroundings in which they suppose it will be used, and other products that the intended user group might own, seeking to capture the flavor of their lifestyle.

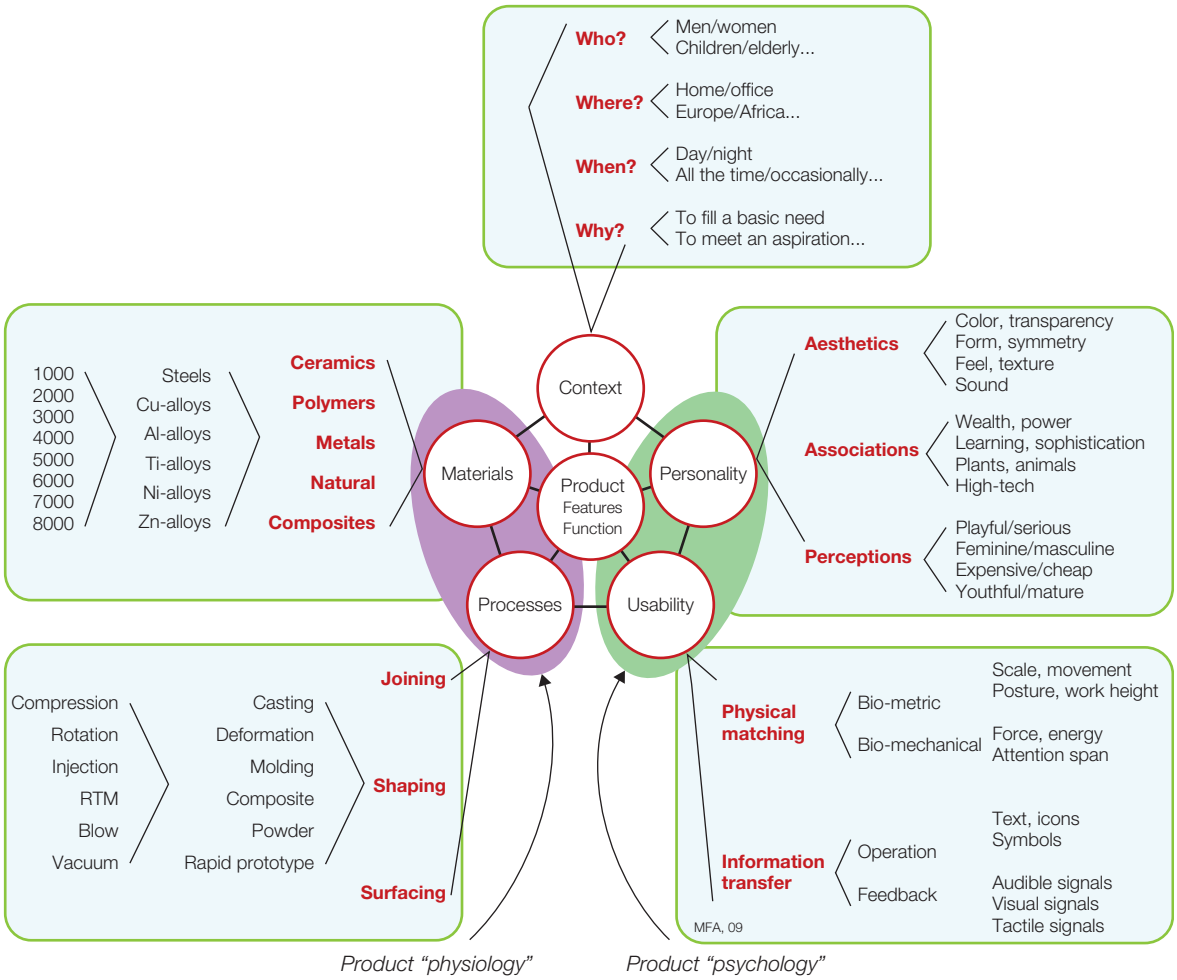


FIGURE 16.3

The breakdown of product character. *Context* defines the intentions or "mood"; *materials* and *processes* create the flesh and bones; the user interface determines *usability*; and the product's aesthetics, associations, and perceptions create its *personality*.

illustrates the library, so to speak, from which the choices can be made. They are the families, classes, and members that we first met in Chapter 3. Choosing these to provide functionality—the bottom of the pyramid—has been the subject of this book so far. Material and process give the product its tangible form, its flesh and bones so to speak; they create the *product physiology*.

On the right in Figure 16.3 are two more packages of information. The lower one—*usability*—characterizes the ways in which the product communicates

with the user: the interaction with their sensory, cognitive, and motor functions. Product success requires a mode of operation that, as far as possible, is intuitive and does not require taxing effort, and an interface that communicates the state of the product and its response to user action by visible, acoustic, or tactile response. It is remarkable how many products fail in this and, by doing so, exclude many of their potential users. There is today an awareness of this, giving rise to research on *inclusive design*: design to make products that can be used by a larger spectrum of the population.

One package remains: the one labeled *personality*. Product personality derives from *aesthetics*, *associations*, and *perceptions*, three words that need explanation.

Anaesthetics dull the senses. Aesthetics do the opposite: They stimulate the five senses: sight, hearing, touch, taste, smell, and via them, the brain. The first row of the personality box elaborates: We are concerned here with color, form, texture, feel, smell, and sound. Think of a new car: its styling, its smell, the sound its doors make on closing. They are no accident. Car makers spend millions to make them as they are.

A products also has associations—the second row of the box. Associations are the things the product reminds you of, the things it suggests. The Land Rover and other SUVs have forms and (often) colors that mimic those of military vehicles. The streamlining of American cars of the 1960s and 1970s carried associations with aerospace. It may be an accident that the VW Beetle has a form that suggests the insect, but the others are no accident; they were deliberately chosen by the designer to appeal to the consumer group (the *Who?*) at which the product was aimed.

Finally, the most abstract quality of all, perceptions. Perceptions are the reactions the product induces in an observer, the way it makes you *feel*.

Table 16.1 Some Perceived Attributes of Products

Perceptions (with Opposites)	
Aggressive—Passive	Extravagant—Restrained
Cheap—Expensive	Feminine—Masculine
Classic—Trendy	Formal—Informal
Clinical—Friendly	Handmade—Mass-produced
Clever—Silly	Honest—Deceptive
Common—Exclusive	Humorous—Serious
Decorated—Plain	Informal—Formal
Delicate—Rugged	Irritating—Lovable
Disposable—Lasting	Lasting—Disposable
Dull—Sexy	Mature—Youthful
Elegant—Clumsy	Retro—Futuristic

Here there is room for disagreement; the perceptions of a product change with time and depend on the culture and background of the observer. Yet in the final analysis it is the perception that causes the consumer, when choosing between a multitude of similar models, to prefer one above the others; it creates the "must have" feeling (see chapter cover image). Table 16.1 lists some perceptions with their opposites in order to sharpen the meaning. They derive from product reviews and magazines specializing in product design; they are a part of a vocabulary, one that is used to communicate views about product character.

16.4 USING MATERIALS AND PROCESSES TO CREATE PRODUCT PERSONALITY

Do materials, of themselves, have a personality? There is a school of thinking that holds as a central tenet that materials must be used "honestly." This means that deception and disguise are unacceptable—each material must be used in ways that expose its intrinsic qualities and natural appearance. It has its roots in the tradition of craftsmanship—potters' use of clays and glazes, carpenters' use of woods, the skills of silversmiths and glass makers in crafting beautiful objects that exploit the unique qualities of the materials with which they work and the integrity of their craft.

This is a view to be respected. But it is not the only one. Design integrity is a quality that consumers value, but they also value other qualities: humor, sympathy, surprise, provocation, even shock. You don't have to look far to find a product that has one of these, and often that quality is achieved by using materials in ways that deceive. Polymers are frequently used in this way—their adaptability invites it. And, of course, it is partly a question of definition—if you say that a characterizing attribute of polymers is their ability to mimic other materials, then using them in this way *is* honest.

Materials and the senses: Aesthetic attributes

Aesthetic attributes are those that relate to the senses: touch, sight, hearing, taste, and smell (Table 16.2). Almost everyone would agree that metals feel "cold"; that cork feels "warm"; that a wine glass, when struck, "rings"; that a pewter mug sounds "dull," even "dead." A polystyrene water glass can look indistinguishable from one made of glass, but pick it up and it feels lighter, less cold, less rigid; tap it and it does not sound the same. It leaves the impression that is so different from glass that, in an expensive restaurant, it would be completely unacceptable. Materials, then, have certain characterizing aesthetic attributes. Let us see if we can pin these down.

Table 16.2 Some Aesthetic Attributes of Materials

Sense	Attribute	Sense	Attribute
Touch	Warm	Hearing	Muffled
	Cold		Dull
	Soft		Sharp
	Hard		Resonant
	Flexible		Ringing
	Stiff		High-pitched
			Low-pitched
Sight	Optically clear	Taste	Bitter
	Transparent	Smell	Sweet
	Translucent		
	Opaque		
	Reflective		
	Glossy		
	Matte		
	Textured		

Touch: Soft–hard/warm–cold Steel is "hard"; so is glass; diamond is harder than both of them. Hard materials do not scratch easily; indeed they can be used to scratch other materials. They generally accept a high polish, resist wear, and are durable. The impression that a material is hard is directly related to its Vickers hardness H. Here is an example of a sensory attribute that relates directly to a technical one.

"Soft" sounds like the opposite of "hard" but it has more to do with modulus E than with hardness H. A soft material deflects when handled, it gives a little, it is squashy; but when it is released it returns to its original shape. Elastomers (rubbers) feel soft; so do polymer foams. Both have moduli that are 100 to 10,000 lower than ordinary "hard" solids; it is this that makes them feel soft. Soft to hard is used as one axis in Figure 16.4. It uses the quantity $\sqrt{EH}$ as a measure.

A material feels "cold" to the touch if it conducts heat away from the finger quickly; it is "warm" if it does not. This has something to do with its thermal conductivity λ but there is more to it than that—it depends also on its specific heat C_p. A measure of the perceived coldness or warmth of a material is the quantity $\sqrt{\lambda C_p \rho}$. It is shown as the other axis in Figure 16.4, which nicely displays the tactile properties of materials. Polymer foams and low-density woods are warm and soft; so are balsa and cork. Ceramics and metals are cold and hard; so is glass. Polymers and composites lie in between.

Sight: Transparency, color, reflectivity Metals are opaque. Most ceramics, because they are polycrystalline and the crystals scatter light, are either opaque or translucent. Glasses, and single crystals of some ceramics, are transparent. Polymers have the greatest diversity of optical transparency,

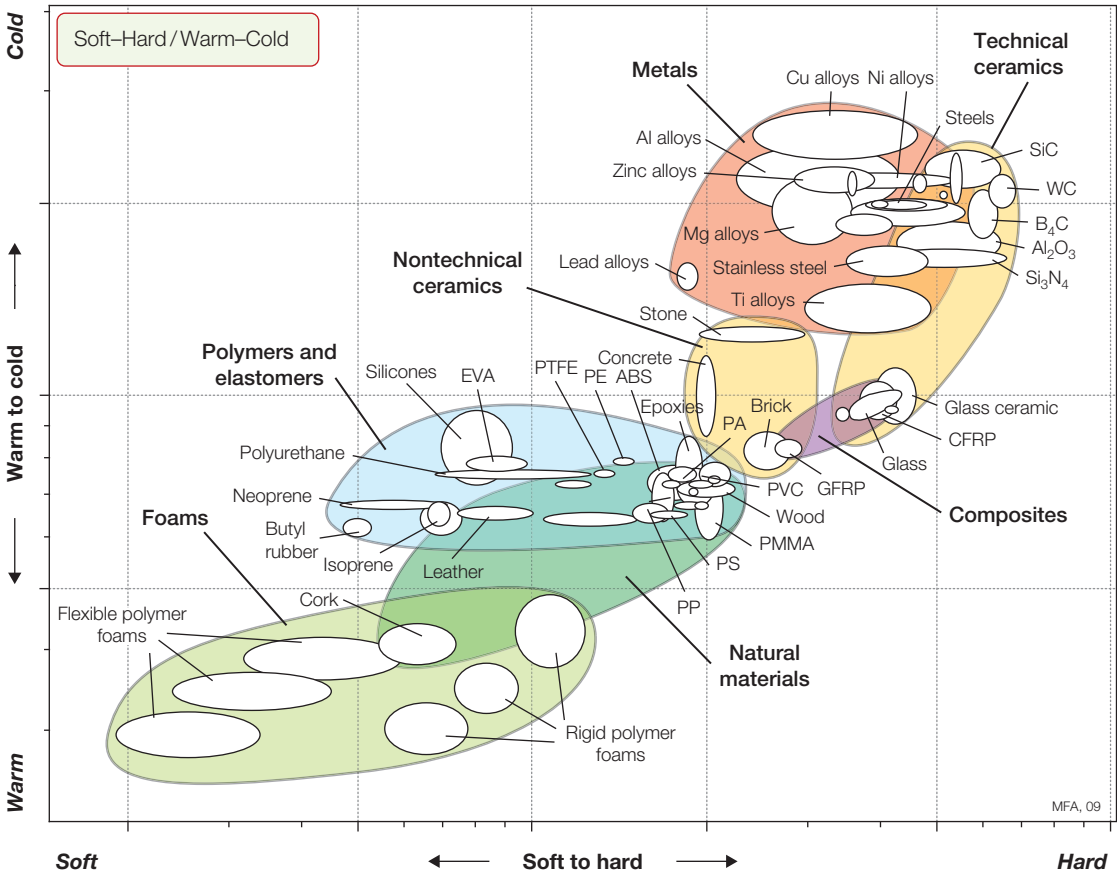


FIGURE 16.4

Tactile qualities of materials. Foams and many natural materials are soft and warm; metals, ceramics, and glasses are hard and cold. Polymers lie in between.

ranging from optical quality to completely opaque. Transparency is commonly described by a four-level ranking that uses easily understood everyday words: "opaque," "translucent," "transparent," and "water-clear." Figure 16.5 ranks the transparency of common materials. So that the data are spread in a useful way—they are plotted against cost. The cheapest materials offering optical-quality transparency ("water clarity") are glass, PS, PET, and PMMA. Epoxies can be transparent but not with water clarity. Nylons are, at best, translucent. All metals, most ceramics, and all carbon-filled or reinforced polymers are opaque.

Color can be quantified by analyzing spectra but this—from a design standpoint—doesn't help much. A more effective method is one of color

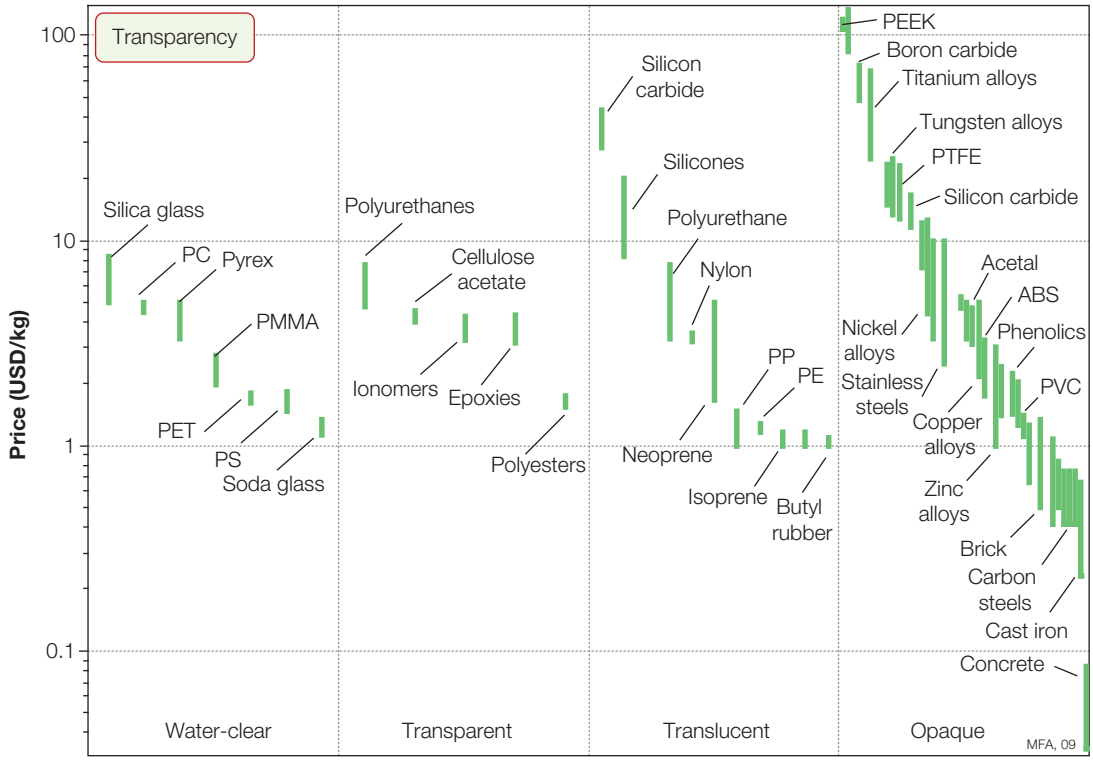


FIGURE 16.5

Here transparency is ranked on a four-point scale, from water-clear to opaque. Water-clear materials are used for windows, display cases, and lenses. Transparent and translucent materials transmit light but diffuse it in doing so. Opaque materials absorb light.

matching, using color charts such as those provided by Pantone[2]; once a match is found it can be described by the code that each color carries. Finally there is reflectivity, an attribute that depends partly on the material and partly on the state of its surface. Like transparency, it is commonly described by a ranking: dead matte, eggshell, semi-gloss, gloss, mirror.

Hearing: Pitch and brightness The frequency of sound (pitch) emitted when an object is struck relates to its material properties. A measure of this pitch $\sqrt{E/\rho}$ is used as one axis in Figure 16.6. Frequency is not the only aspect of acoustic response—the other has to do with the damping or loss coefficient η. A highly damped material sounds dull and muffled; one with low damping rings. Acoustic brightness—the inverse of damping—is used as the other axis in Figure 16.6. It groups materials that have similar acoustic behavior.

[2] Pantone (*www.pantone.com*) provides detailed advice on color selection, including color-matching charts and good descriptions of the associations and perceptions of color.

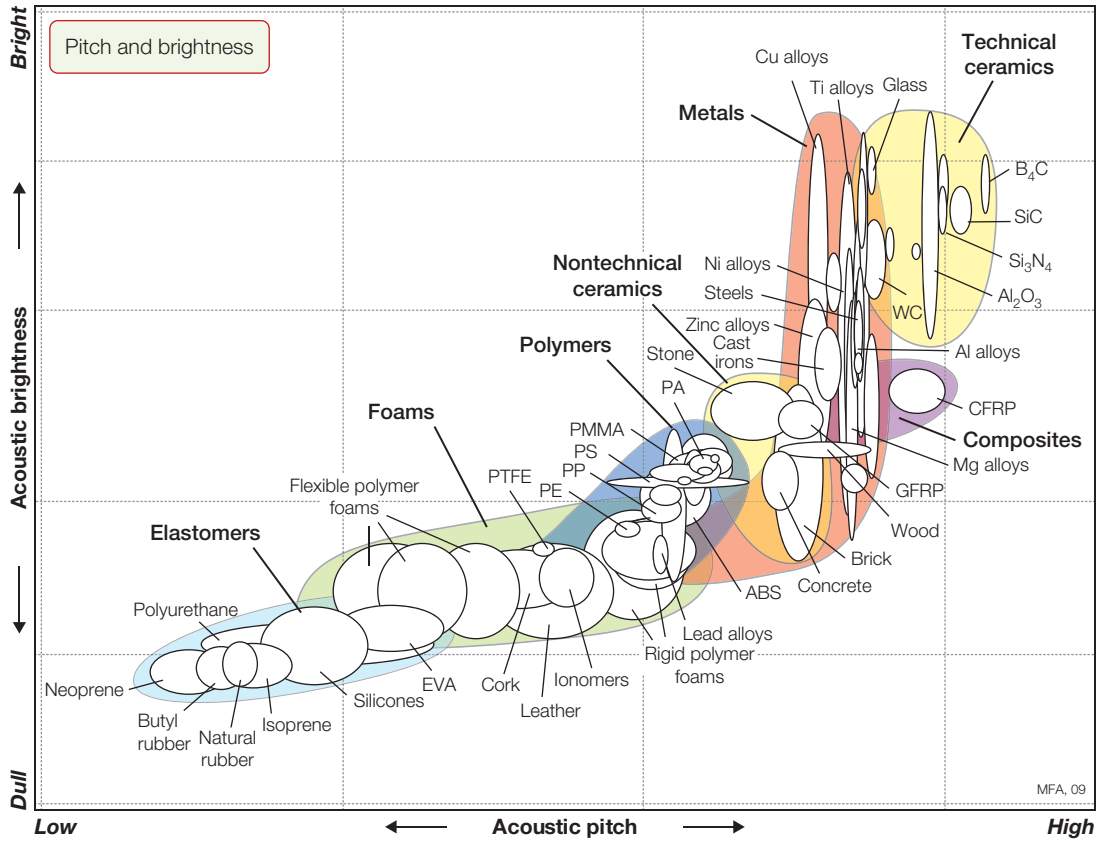


FIGURE 16.6

Acoustic properties of materials. The "ring" of a wine glass occurs because glass is an acoustically bright material with a high natural pitch; the dull "ping" of a plastic glass occurs because polymers are much less bright and—in the same shape—vibrate at a lower frequency. Materials at the top right make good bells; those at the bottom left are good for damping sound.

Bronze, glass, and steel ring when struck, and the sound they emit has—on a relative scale—a high pitch; they are used to make bells. Alumina, on this ranking, has bell-like qualities. Rubber, foams, and many polymers sound dull and, relative to metals, vibrate at low frequencies; they are used for sound damping. Lead, too, is dull and low-pitched; it is used to clad buildings for sound insulation.

The three figures show that each material class has a certain recognizable aesthetic character. Ceramics are hard, cold, high-pitched, and acoustically bright. Metals, too, are relatively hard and cold but although some (e.g., bronze) ring when struck, others (e.g., lead) are dull. Polymers and foams are most nearly like natural materials—warm, soft, low-pitched, and muffled, although some have outstanding optical clarity and almost all can

be colored. But their low hardness means that they scratch easily, losing their gloss.

These qualities of a material contribute to a product's personality. The product acquires some of the attributes of the material from which it is made, an effect that designers recognize and use when seeking to create a personality. A stainless steel fascia, whether it is in a car or on a hi-fi system, has a different personality from one of CFRP, of polished wood, or of leather, and that in part is because the product has acquired some of the aesthetic qualities of the material.

Materials and the mind: Associations and perceptions

So a material certainly has aesthetic qualities—but can it be said to have a personality? At first sight, no—it only acquires one when used in a product. Like an actor, it can assume many different personalities, depending on the role it is asked to play. Wood in fine furniture suggests craftsmanship, but in a packing case, cheap utility. Glass in the lens of a camera has associations of precision engineering, but in a beer bottle, disposable packaging. Even gold, so often associated with wealth and power, has different associations when used in microcircuits: technical functionality.

But wait. The object shown in Figure 16.7 has its own somber association. It appears to be made of polished hardwood—the traditional material for such things. If you had to choose one, you would probably not think polished hardwood inappropriate. But suppose I told you it was made of *polystyrene foam*—would you feel the same? Suddenly it becomes a bin, a wastebasket, inappropriate for its dignified purpose. Materials, it seems, *do* have personality.

Expression through material Think of wood. It is a natural material with a grain that has a surface texture, pattern, color, and feel that other materials do not have. It is tactile—it is perceived as warmer than many other materials and seemingly softer. It is associated with characteristic sounds and smells. It has a tradition; it carries associations of craftsmanship. No two pieces are exactly alike; the woodworker selects the piece on which he will work for its grain and texture. Wood enhances value: The interior of very inexpensive cars is plastic, that of more costly ones is burr-walnut and calf leather. And it ages well, acquiring additional character with time—objects made of wood are more highly valued when they



FIGURE 16.7
A coffin. Wood is perceived to be appropriate for its somber, ceremonial function; plastic is perceived to be inappropriate.

are old than when they are new. There is more to this than just aesthetics; there are the makings of a personality, to be brought out by the designer, certainly, but there nonetheless.

Now consider metal. Metals are cold, clean, precise. They ring when struck. They reflect light—particularly when polished. They are accepted and trusted: Machined metal looks strong—its very nature suggests it has been *engineered*. Metals are associated with robustness, reliability, and permanence. Their strength allows slender structures—the cathedral-like space of railway stations or the span of bridges. They can be worked into flowing forms, such as intricate lace or cast, into solid shapes with elaborate, complex detail. The history of humans and of metals is intertwined—the titles "Bronze age" and "Iron age" tell you how important these metals were—and their qualities are so sharply defined that that they have become ways of describing human qualities—an iron will, a silvery voice, a golden touch, a leaden look. And, like wood, metals can age well, acquiring a patina that makes them more attractive than when newly polished—the bronze of sculptures, the pewter of mugs, the lead of roofs.

Ceramics and glass? They have an exceptionally long tradition—think of Greek pottery and Roman glass. They accept almost any color. Their total resistance to scratching, abrasion, discoloration, and corrosion gives them a certain immortality, threatened only by their brittleness. They are—or were—the materials of great craft-based industries: the glass of Venice, the porcelain of Meissen, the pottery of Wedgwood—valued at certain times more highly than silver. But at the same time ceramics and glass can be robust and functional—think of beer bottles. The transparency of glass gives it an ephemeral quality—sometimes you see it, sometimes you don't. It interacts with light, transmitting it, refracting it, reflecting it. And ceramics today have additional associations—those of advanced technology: kitchen stove-tops, high-pressure/high-temperature valves, space shuttle tiles; all are materials for extreme conditions.

And finally polymers. "A cheap, plastic imitation," it used to be said—and that is a hard reputation to live down. It derives from the early use of plastics to simulate the color and gloss of Japanese handmade pottery, much valued in Europe. Commodity polymers *are* cheap. They are easily colored and molded (that is why they are called "plastic"), making imitation easy. Unlike ceramics, their gloss is easily scratched, and their colors fade—they do not age gracefully. You can see where the reputation came from. But is it justified? No other class of material can take on as many characters as polymers: Colored, they look like ceramics; printed, they can look like wood or textile; metallized, they look exactly like metal. They can be as transparent as glass or as opaque as lead, as flexible as rubber or as stiff—when reinforced—as aluminum.

Plastics emulate precious stones in jewelry, glass in drinking glasses and glazing, wood in counter tops, velvet and fur in clothing, even grass. But despite

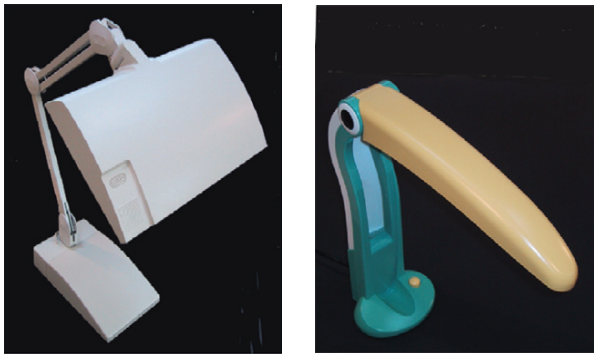


FIGURE 16.8
Lamps. Both have the same technical rating, but differ completely in their personalities. Materials, processes, form, weight, and color all contribute to personality.

this chameleon-like behavior they *do* have a certain personality: They feel warm—much warmer than metal or glass. They are adaptable—that is part of their special character—and they lend themselves particularly to brightly colored, light hearted, even humorous, design. But their very cheapness creates problems as well as benefits: Our streets, countryside, and rivers are littered with discarded plastic bags and packaging that decay only very slowly.

The ways in which material, processes, usability, and personality combine to create a product character tuned to context or "mood" are best illustrated by examples. Figure 16.8 illustrates the first. The lamp shown on the left is designed for the office. It is angular, functional, creamy grey, and heavy. Its form and color echo those of computer consoles and keyboards, creating associations of contemporary office technology. Its form and weight transmit the ideas of stability, robustness, efficiency, and fitness for a task—but for tasks in the workplace, not in the bedroom. Materials and processes have been chosen to reinforce these associations and perceptions. The enameled frame is pressed and folded sheet steel, the base weight is cast iron, and the reflector is stainless steel set in a high-impact ABS enclosure.

The lamp on the right in Figure 16.8 has the same technical rating as the left one, as well as the same functionality and usability. But there the resemblance ends. This product is not designed for the busy executive but for children (and adults that still enjoy being children), to be used in the playroom or bedroom. It has a contoured form and contrasting translucent colors, and it is light. It is made of colored acrylic in translucent and opaque grades so that the outside glows like a neon sign when it is lit. Its form is partly derived from nature, partly from cartoons and comic strips, giving it a light hearted character. I perceive it as playful, funny, cheerful, and clever—but also as eccentric and easily damaged. You may perceive it in other ways—perception is a personal thing; it depends where you are coming from. Skilled designers manipulate perception to appeal to the user group they wish to attract.

Figure 16.9 shows a second example. Here are two contrasting ways of presenting home entertainment systems. On the left: a music center aimed at successful professionals with disposable income, who are comfortable with (or addicted to) advanced technology, and for whom only the best is good enough. The linear form, the use of primitives (rectangles, circles, cylinders, cones), and the matt silver and black proclaim that this product has not just been *made*, it has been *Designed* (big *D*). The formal geometry and finish suggest precision instruments, telescopes, electron microscopes; the shapes resemble those of organ pipes (hence associations of music, of culture). The

perception is that of quality cutting-edge technology, a symbol of discriminating taste. The form has much to do with these associations and perceptions, but so too do the materials: brushed aluminum, stainless steel, and black enamel—these are not materials you choose for a cuddly toy.





FIGURE 16.9
Consumer electronics. The system on the left is aimed at a different consumer group than the radios on the right. The personalities of each (meaning the combination of aesthetics, associations, and perceptions) have been constructed to appeal to the target group. Materials play a central role here in creating personality. (Image on left courtesy of Bang & Olufsen.)

On the right, electronics are presented in another way. This is a company that has retained market share, even increased it, by *not* changing, at least as far as appearance is concerned. (Fifty years ago I had a radio that looked exactly like this.) The context? Clearly, the home, perhaps aimed at consumers who are uncomfortable with modern technology (though the electronics in these radios is modern enough) or who simply feel that it clashes with the home environment. Each radio has a simple form, it is pastel-colored, it is soft and warm to touch. It is the materials that make the difference: These products are clothed in suede or leather in six or more colors. The combination of form and material creates associations of comfortable furniture, leather purses and handbags (hence luxury, comfort, style), the past (hence stability), and perceptions of solid craftsmanship, reliability, retro-appeal, and traditional but durable design.

So there is a character hidden in a material even before it has been made into a recognizable form. It is a sort of embedded personality, a shy one, not always obvious, easily concealed or disguised, but one that, when appropriately manipulated, imparts its qualities to the design. It is for this reason that certain materials are so closely linked to certain *design styles*. *Style* is shorthand for a manner of design with a shared set of aesthetics, associations, and perceptions. The Early Industrial style (1800–1890)[3] embraced the technologies of the Industrial Revolution, using cast iron and steel, often elaborately decorated to give it a historical façade. The Arts and Crafts Movement from 1860 to 1910 rejected this, choosing instead natural materials and fabrics to create products with the character of traditional handcrafted quality. Art Nouveau (1890–1918), by contrast, exploited the fluid shapes and durability made possible by wrought iron and cast bronze, the warmth and textures of hardwood, and the transparency of glass to create products of flowing, organic character. Art Deco (1918–1935) extended the range of materials to include for the first time plastics (Bakelite and Catalin), allowing production of both luxury products for the rich and mass-produced products for a wider market. The simplicity and explicit character of Bauhaus

[3] The dates are, of course, approximate. Design styles do not switch on and off on specific dates; they emerge as a development of, or a reaction to, earlier styles with which they often coexist, and they merge into the styles that follow.

designs (1919–1933) are most clearly expressed by the use of chromed steel tubing, glass, and molded plywood. Plastics first reached maturity in product design in the cheeky iconoclastic character of the Pop Art style (1940–1960). Since then the range of materials has continued to increase, and their role in helping to mold product character remains.

16.5 SUMMARY AND CONCLUSIONS

What do we learn? The element of satisfaction is central to contemporary product design. It is achieved through an integration of good technical design to provide functionality, proper consideration of the needs of the user in the design of the interface, and imaginative industrial design to create a product that will appeal to the consumers at whom it is aimed.

Materials play a central role in this. Functionality is dependent on the choice of proper material and process to meet the technical requirements of the design safely and economically. Usability depends on the visual and tactile properties of materials to convey information and respond to user actions. Above all, the aesthetics, associations, and perceptions of the product are strongly influenced by the choice of the material and its processing, imbuing the product with a personality that, to a greater or lesser extent, reflects that of the material itself.

Consumers look for more than functionality in the products they purchase. In the sophisticated markets of developed nations, the "consumer durable" is a thing of the past. The challenge for the designer no longer lies in meeting functional requirements alone but in doing so in a way that also satisfies aesthetic and emotional needs. The product must carry the image and convey the meaning that the consumer seeks: timeless elegance, perhaps, or racy newness. One Japanese manufacturer goes so far as to say: "*Desire* replaces *need* as the engine of design."

Not everyone, perhaps, would wish to accept that idea. So we end with simpler words—the same ones with which we started. Good design works. Excellent design also gives pleasure. The imaginative use of materials provides it.

If you found this chapter interesting and would like to read more, you will find the ideas it contains developed more fully in the first book listed in the next section.

16.6 FURTHER READING

Ashby, M.F., & Johnson, K. (2010). *Materials and design—the art and science of materials selection in product design* (2nd ed.). Butterworth-Heinemann, ISBN 0-7506-5554-2.
A text that develops further the ideas outlined in this chapter.

Clark, P., & Freeman, J. (2000). *Design, a crash course*. The Ivy Press Ltd, Watson-Guptil Publications, BPI Communications, ISBN 0-8230-0983-1.
An entertainingly written scoot through the history of product design from 5000 BC to the present day.

Dormer, P. (1993). *Design since 1945*. Thames and Hudson, ISBN 0-500-20269-9.
A well-illustrated and inexpensive paperback documenting the influence of industrial design in furniture, appliances, and textiles—a history of contemporary design that complements the wider-ranging history of Haufe (1998), q.v.

Forty, A. (1986). *Objects of desire—design in society since 1750*. Thames and Hudson, ISBN 0-500-27412-6.
A refreshing survey of the design history of printed fabrics, domestic products, office equipment, and transport systems. The book is mercifully free of eulogies about designers and focuses on what industrial design does rather than who did it. The black-and-white illustrations are disappointing, mostly drawn from the late nineteenth or early twentieth centuries, with few examples of contemporary design.

Haufe, T. (1998). *Design, a concise history*. Laurence King Publishing (originally in German), ISBN 1-85669-134-9.
An inexpensive paperback. Probably the best introduction to industrial design for students (and anyone else). Concise, comprehensive, clear, and with intelligible layout and good, if small, color illustrations.

Jordan, P.S. (2000). *Designing pleasurable products*. Taylor and Francis, ISBN 0-748-40844-4.
Jordan, Manager of Aesthetic Research at Philips Design, argues that products today must function properly, must be usable, and must also give pleasure. Much of the book is a description of market research methods for eliciting user reactions to products.

Julier, G. (1993). *Encyclopedia of 20th-century design and designers*. Thames and Hudson, ISBN 0-500-20261-3.
A brief summary of design history with good pictures and discussions of the evolution of product designs.

Manzini, E. (1989). *The material of invention*. The Design Council, ISBN 0-85072-247-0.
Intriguing descriptions of the role of material in design and in inventions. The translation from Italian provides interesting—and often inspiring—commentary and vocabulary that is rarely used in traditional writings about materials.

McDermott, C. (1999). *The product book*. D & AD in association with Rotovison.
Fifty essays by respected designers who describe their definition of design, the role of their respective companies, and their approach to product design.

Norman, D.A. (1988). *The design of everyday things*. Doubleday, ISBN 0-385-26774-6.
A book that provides insight into the design of products with particular emphasis on ergonomics and ease of use.

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CHAPTER 17

Forces for Change



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17.1 INTRODUCTION AND SYNOPSIS

If there were no forces for change, everything would stay the same. The clear message of Figure 1.1, at the start of this book, is that exactly the opposite is true: Things are changing faster now than ever before. The evolving circumstances of the world in which we live change the boundary conditions for design, and with them those for selecting materials and processes.

These changes are driven by a number of forces. First, there is the *market pull*: the demand from industry for materials that are lighter, stiffer, stronger, tougher, cheaper, and more tolerant of extremes of temperature and environment, and that offer greater functionality. Then there is the *science push*: the curiosity-driven research of materials experts in the laboratories of universities, industries, and government agencies. There is the driving force of what might be called *mega-projects*: historically, the Manhattan Project, the development of nuclear power, the space race, and various defense programs; today, one might think of alternative energy technologies, the problems of maintaining an aging infrastructure of drainage, roads, bridges, and aircraft, and the threat of terrorism. There is the trend to miniaturization while at the same time increasing the functionality of products. There is legislation regulating product safety, and there is the increased emphasis on liability established by recent legal precedent.

This chapter examines forces for change and the directions in which they push materials and their deployment. Figure 17.1 sets the scene.

17.2 MARKET PULL AND SCIENCE PUSH

Market forces and competition

The end users of materials are the manufacturing industries. They decide which materials they will purchase, and adapt their designs to make the best use of them. Their decisions are based on the nature of their products. Materials for large civil structures (which might weigh 100,000 tonnes or more) must be cheap; economy is the overriding consideration. By contrast, the cost of materials for high-tech products (sports equipment, military hardware, space projects, biomedical applications) plays a less important role: For an artificial heart valve, for instance, material cost is almost irrelevant. Performance, not economy, dictates the choice.

The market price of a product has several contributions. One is the cost of the materials of which the product is made, but there is also the cost of the research and development that went into its design, the cost of manufacture and marketing, and the perceived value associated with fashion, scarcity, lack of competition, and such like—as described in Chapter 13. When the

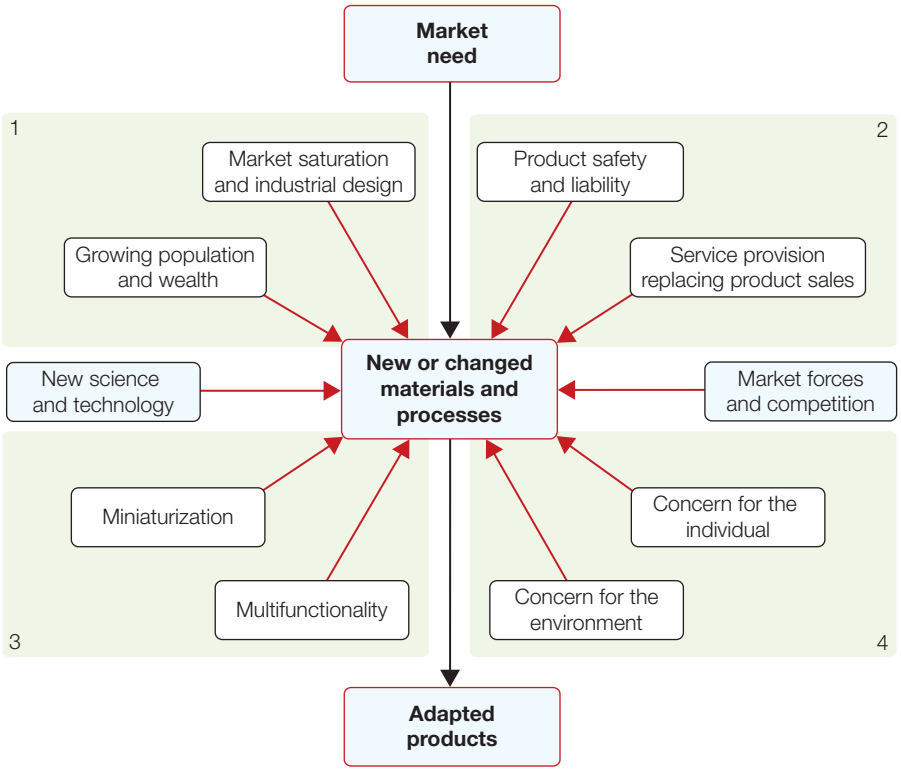


FIGURE 17.1
Forces for change. Each of the influences exerts pressure to change the choice of material and process, and stimulates efforts to develop new ones. Market pull and science push are shown on the extreme *right* and *left* of the figure.

material costs are a large part of the market value (50%, say)—that is, when the value added to the material is small—the manufacturer seeks to economize on material usage to increase profit or market share. When, by contrast, material costs are a tiny fraction of the market value (1%, say), the manufacturer seeks the materials that will most improve the performance of the product, with little concern for their cost.

With this background, examine Figures 17.2 and 17.3. The vertical axis in both is the price per unit weight ($/kg) of materials and products: It gives a common measure by which materials and products can be compared. The measure is a crude one but its great merit is that it is unambiguous, easily determined, and bears some relationship to value added. A product with a price/kg that is only two or three times that of the materials of which it is made is material-intensive and is sensitive to material costs; one with a price/kg that is 100 times that of its materials is insensitive to material

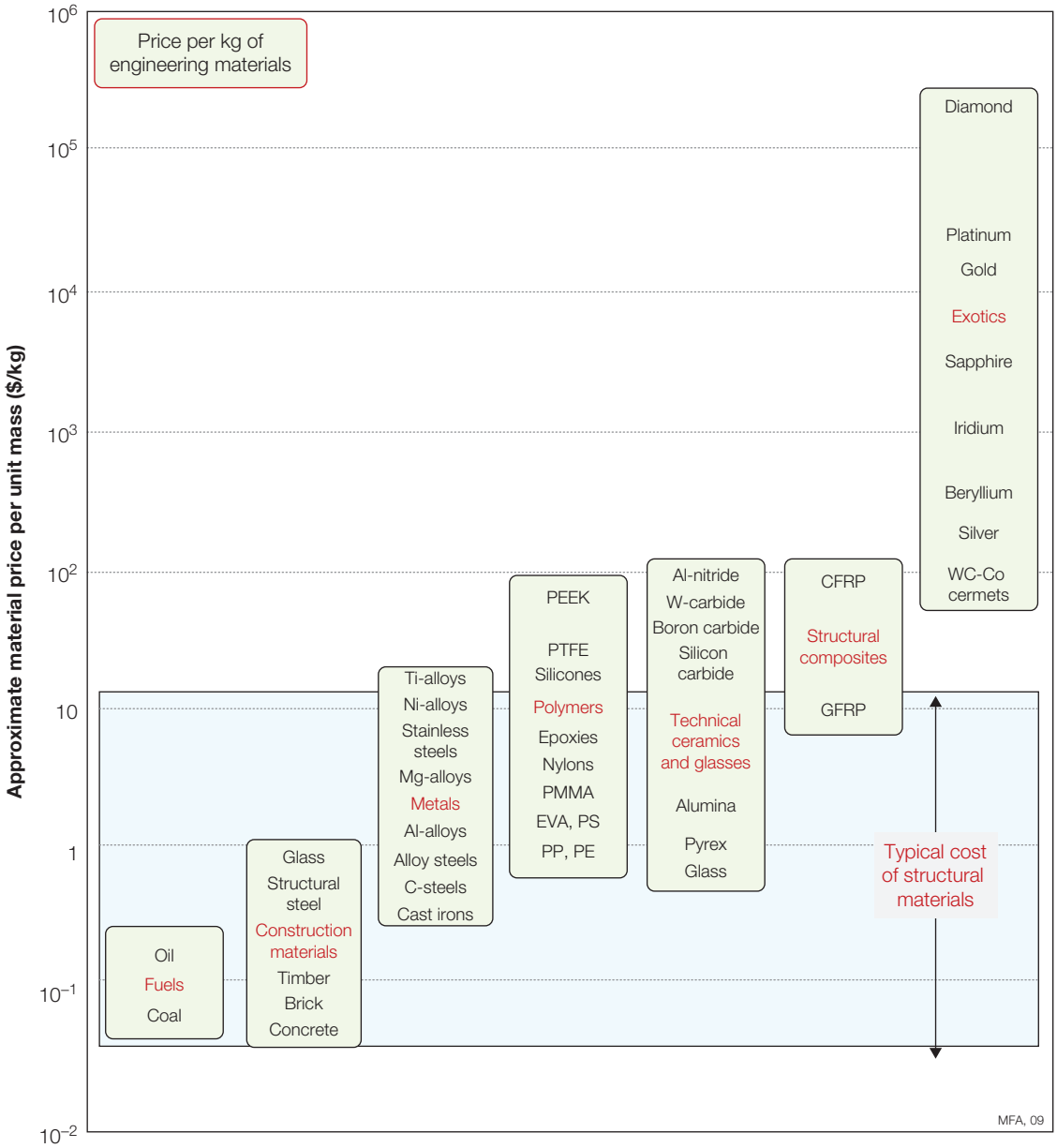


FIGURE 17.2

The price per unit weight for materials. The shaded band spans the range where the most widely used commodity materials of manufacture and construction lie.

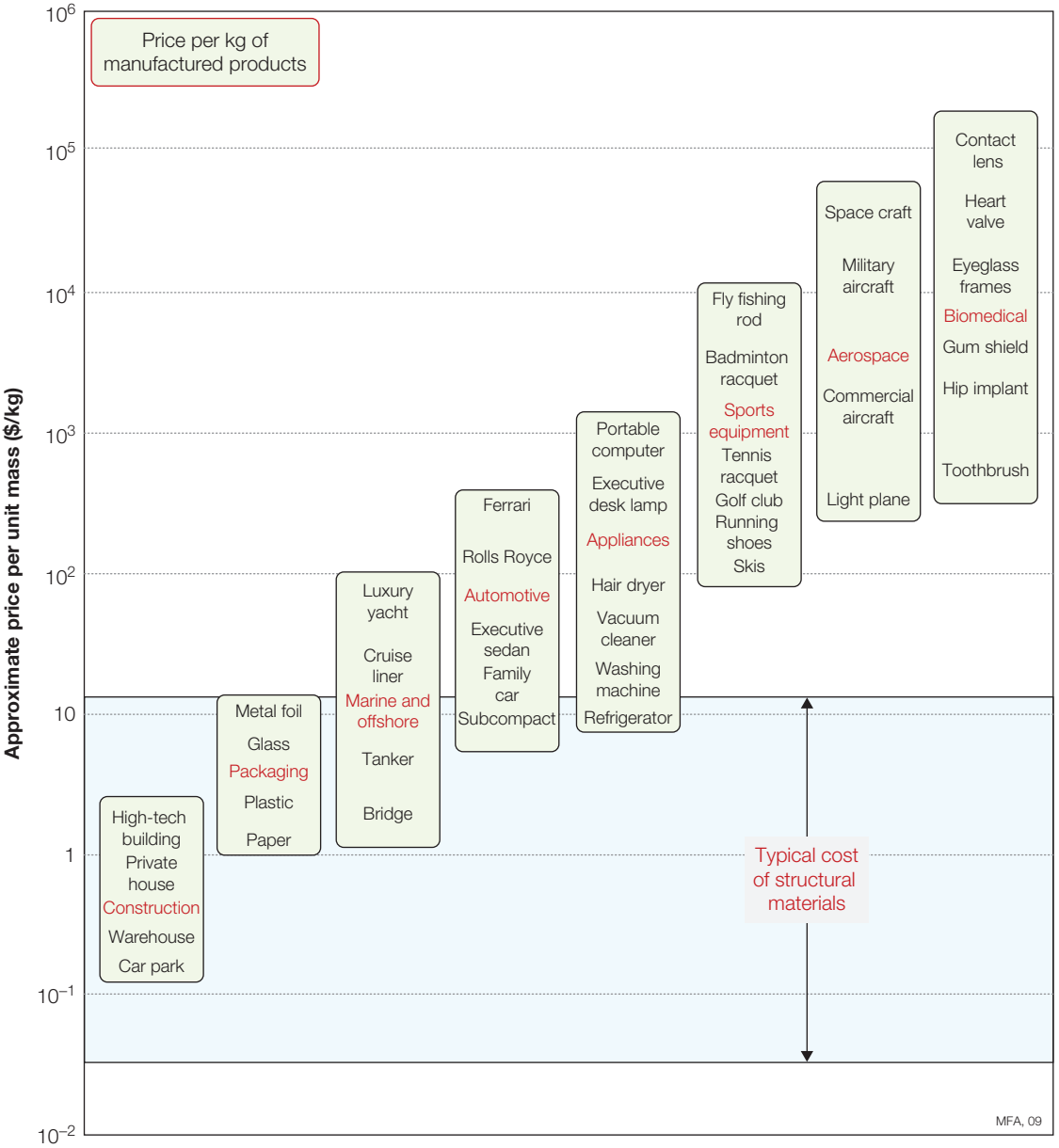


FIGURE 17.3

The price per unit weight for products. The shaded band spans the range in which lies most of the material of which they are made. Products in the shaded band are material-intensive; those above it are not.